

ERECTIELE EN SEKSUELE DYSFUNCTIE BIJ DE MAN: DIAGNOSE EN BEHANDELING

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DE MAN EN ZIJN KRUIS



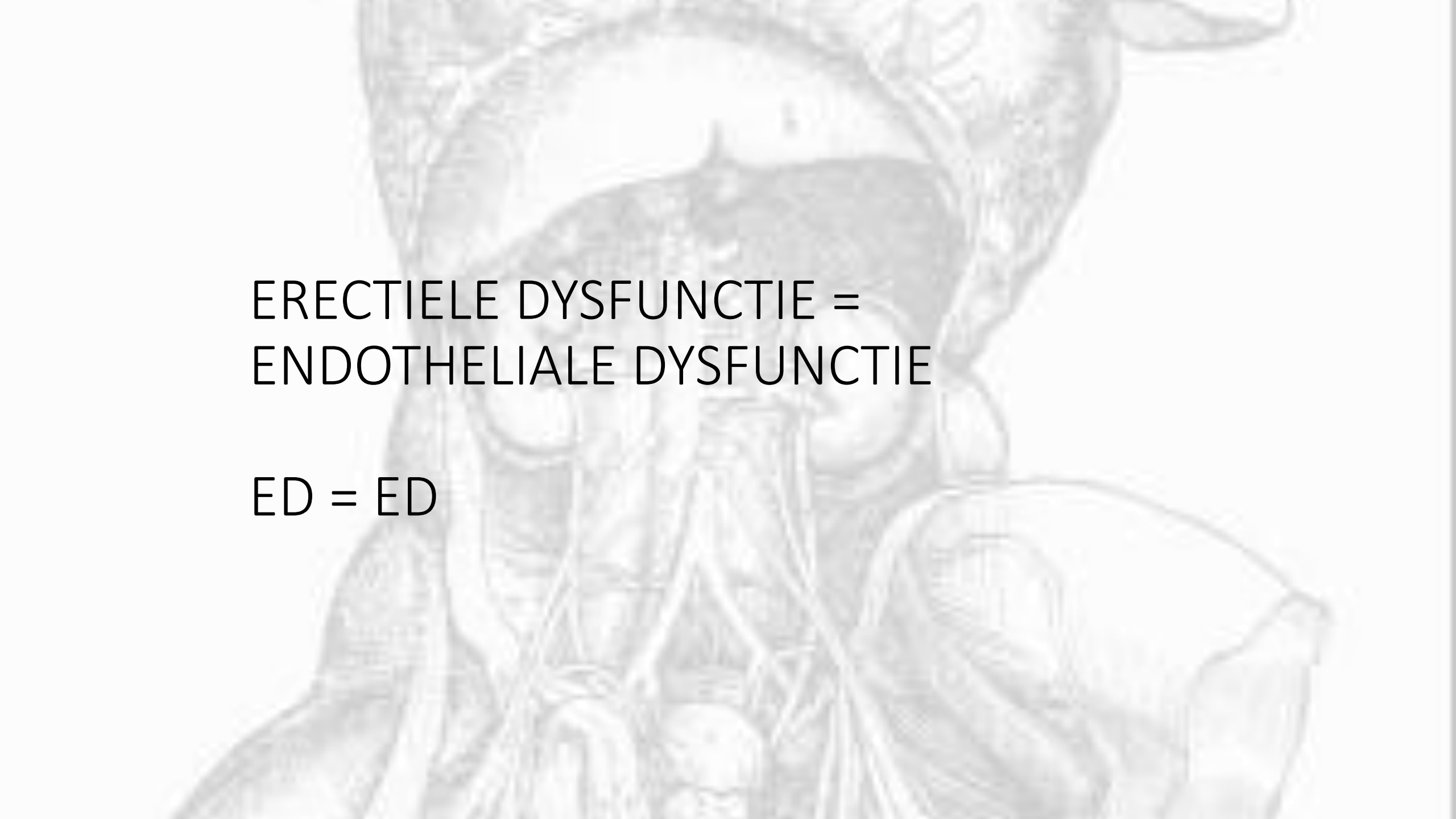


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ERECTIELE DYSFUNCTIE =
ENDOTHELIALE DYSFUNCTIE

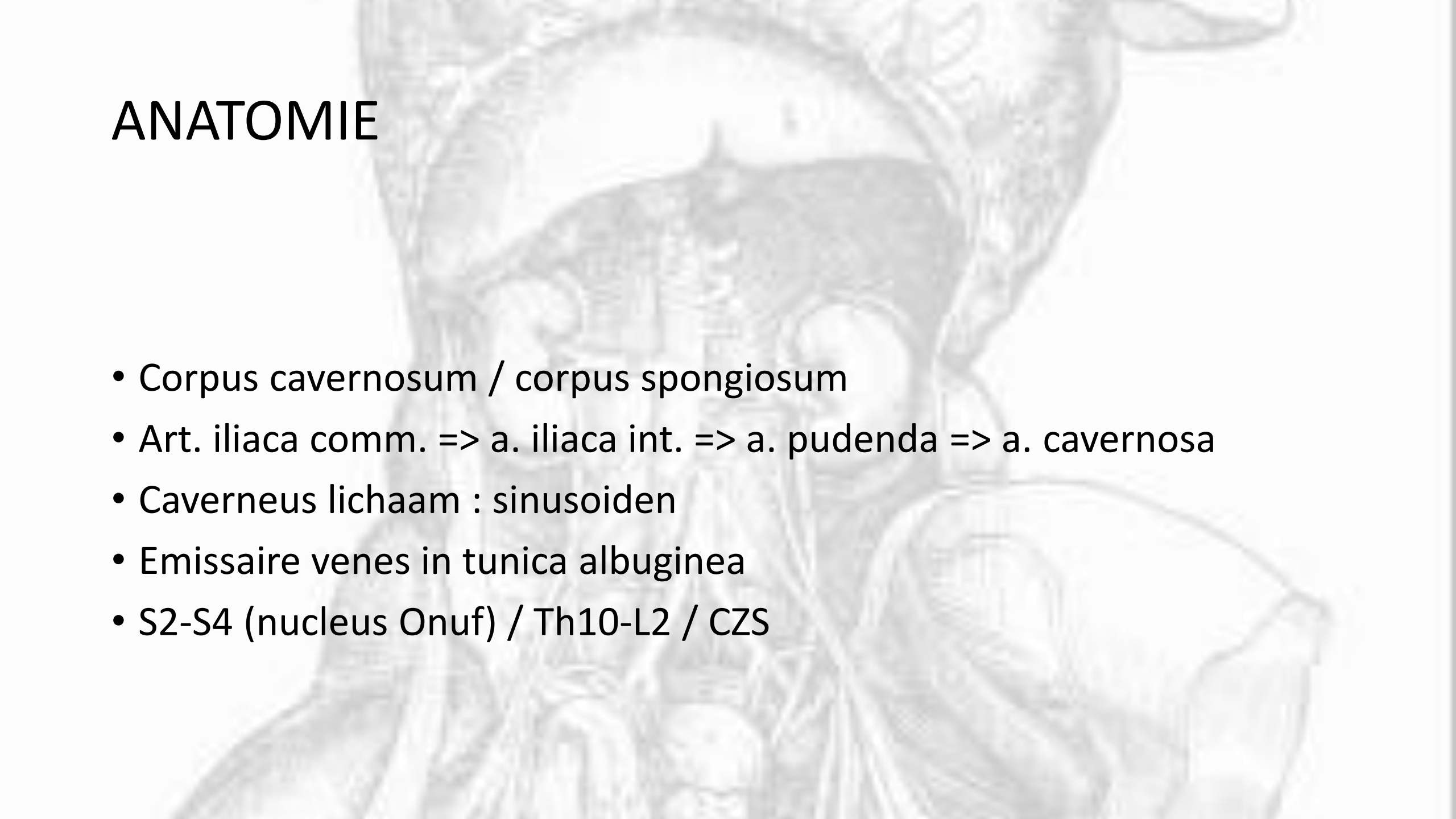
ED = ED

A detailed anatomical illustration of the male reproductive system, showing the testes, vas deferens, ureters, and associated structures. The illustration is in a classic medical style, with clear labels and shading to indicate depth and structure. It serves as a background for the text.

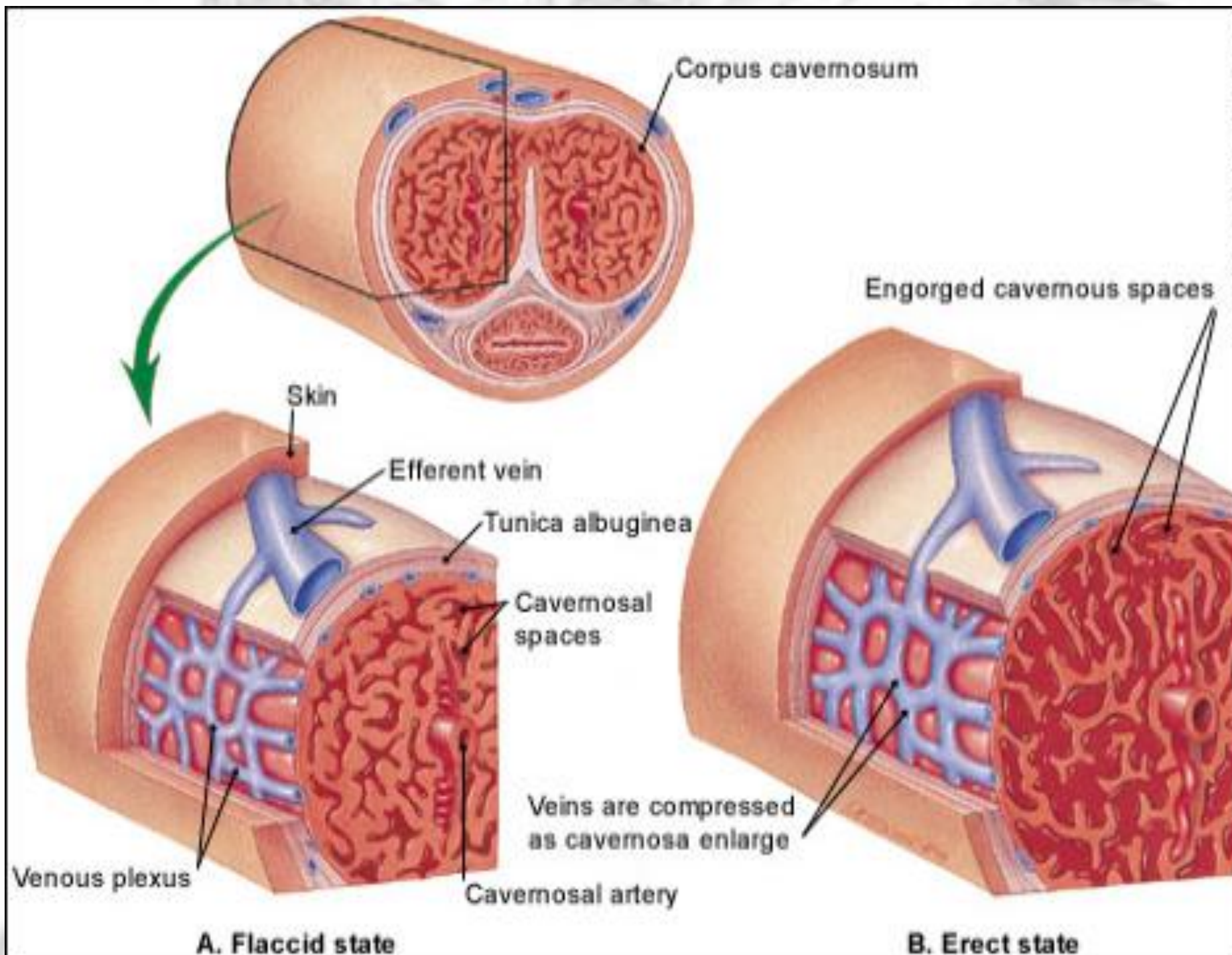
- DYSFUNCTIE VAN

- Erectie
- Libido
- Ejaculatie
- Orgasme

ANATOMIE



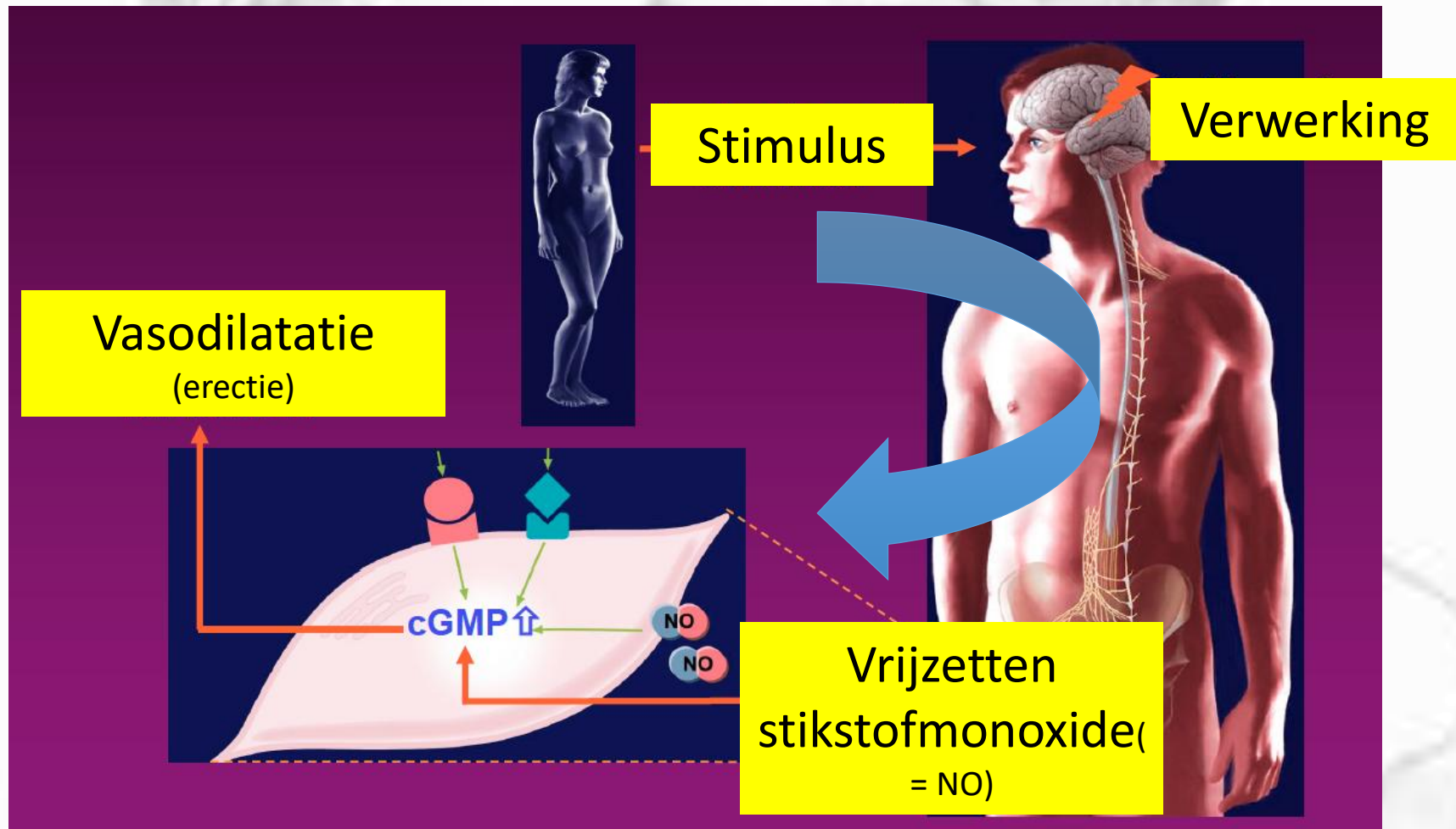
- Corpus cavernosum / corpus spongiosum
- Art. iliaca comm. => a. iliaca int. => a. pudenda => a. cavernosa
- Caverneus lichaam : sinusoiden
- Emissaire venes in tunica albuginea
- S2-S4 (nucleus Onuf) / Th10-L2 / CZS

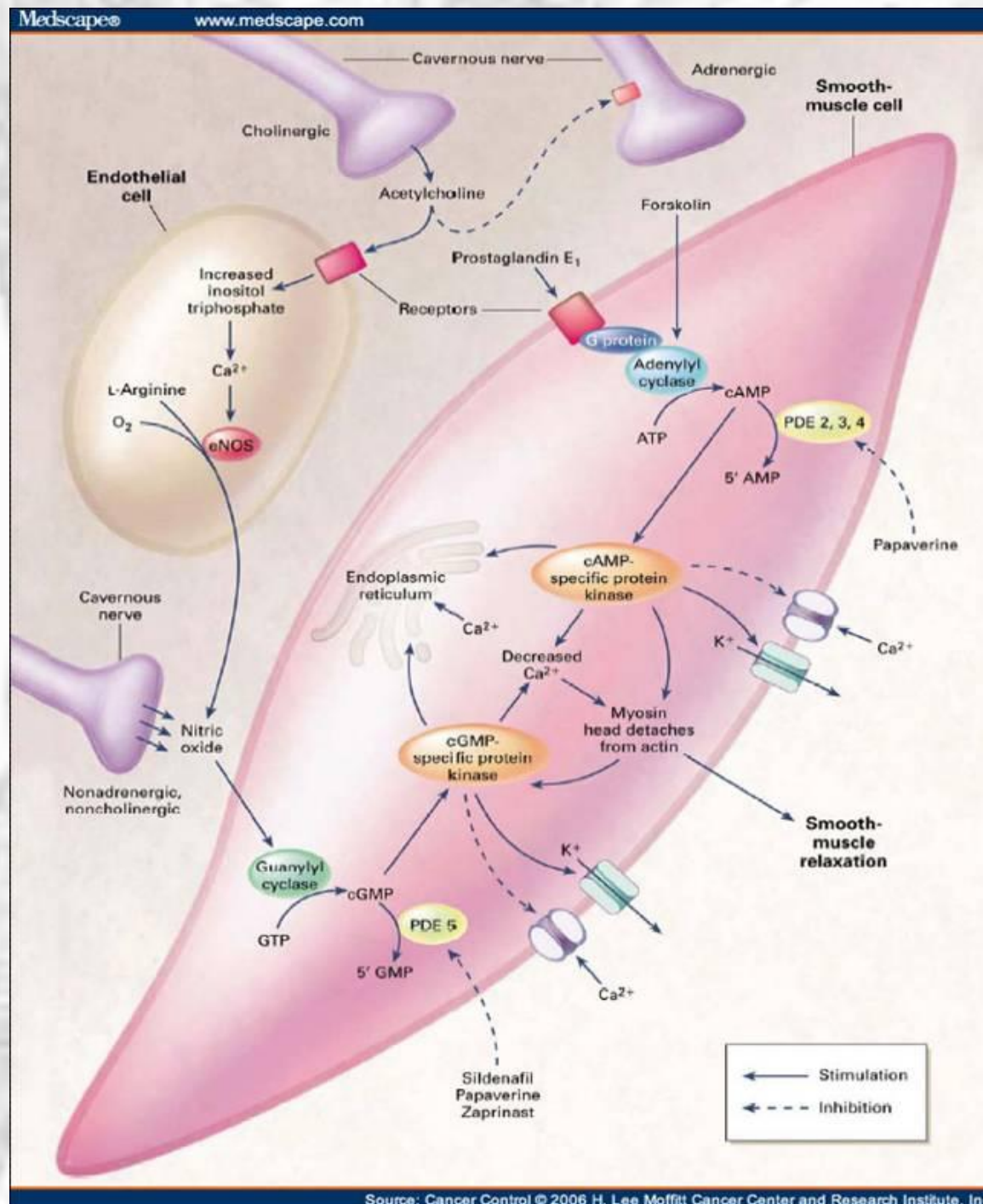


FYSIOLOGIE

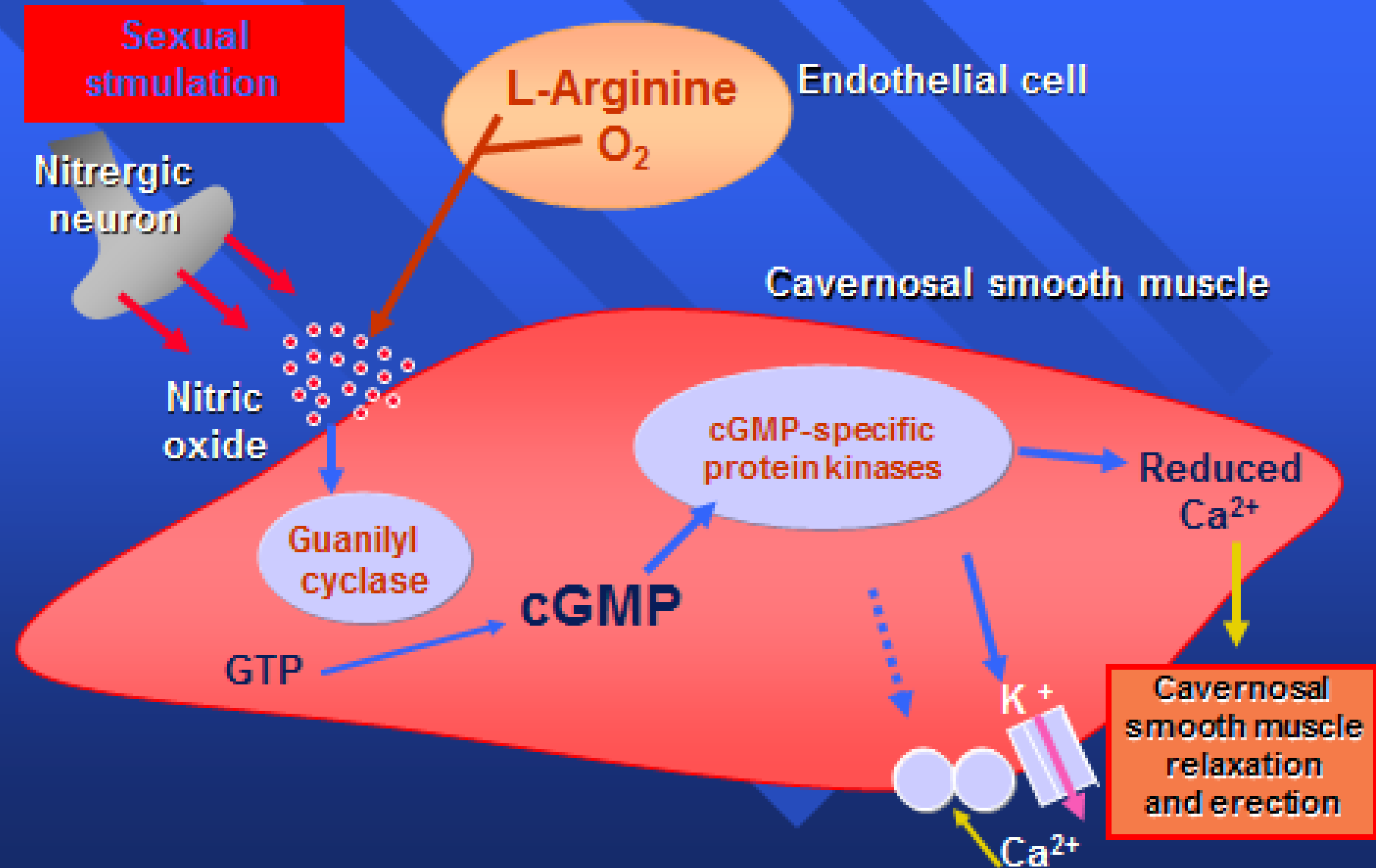
A detailed anatomical illustration of the human torso, showing the internal organs. The heart is centrally located, with the lungs on either side. The kidneys are visible below the lungs, and the liver is on the right side. The illustration is in a classic, detailed style, showing the complex network of blood vessels and other structures within the body.

- Vasodilatatie
- Relaxatie sinusoiden
- Toename veneuze weerstand





Mechanism of erection



EPIDEMIOLOGIE

- **Massachusetts Male Aging Study (MMAS)**

- 40-70 jr. 52%

- **Olmstead County Study**

- 40-49: 12,6%
- 50-59: 13,3%
- 60-69: 20,4%
- 70-79: 25,2%

- **Krimpen Study**

- 50-54: 3%
- 55-59: 5%
- 60-64: 11%
- 65-69: 18%
- 70-78: 26%

- **Cologne Male Survey**

- 30-39: 2,3%
- 40-49: 9,5%
- 50-59: 15,7%
- 60-69: 34,4%
- 70-80: 53,4%

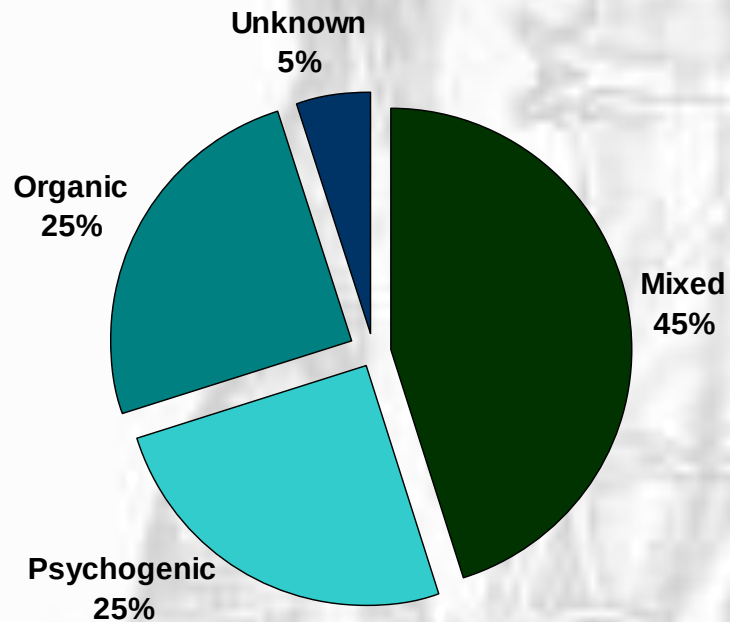
RISICOFACTOREN



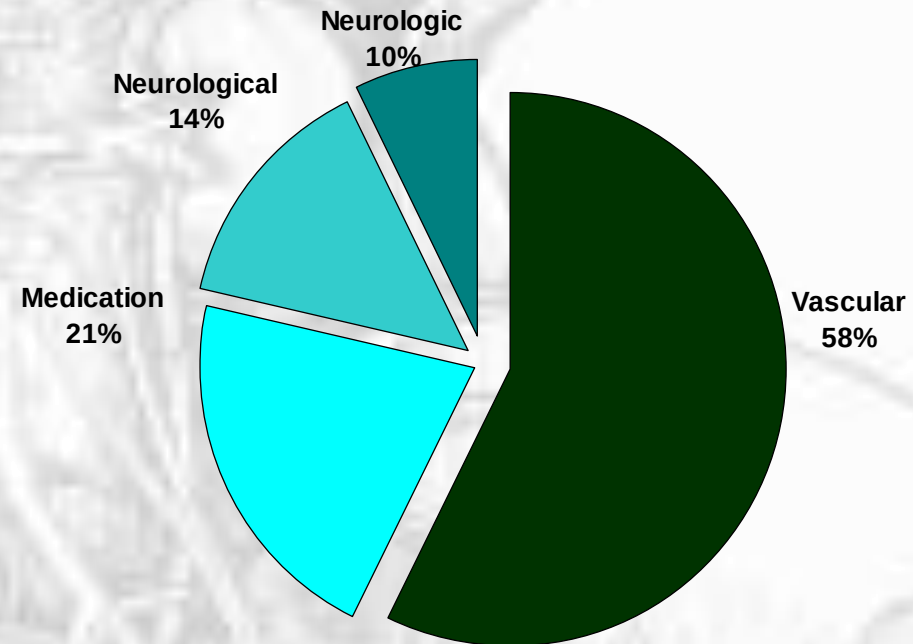
- Leeftijd
- DM
- AHT
- Hypercholesterolemie
- Roken
- Ethyl
- Vroegere heekunde klein bekken
- Schildklier lijden
- Nierinsufficiëntie
- Trauma
- Psychische problemen

Pathophysiology of ED

ED by Classification



Causes of Organic ED



Other includes surgery, trauma, radiation and endocrine

Clinical Manual of Sexual Medicine Sexual Dysfunctions in Men.

Based on the Reports of the 2nd International Consultation on Sexual Dysfunctions, 2003.

Effect of lifestyle changes on erectile dysfunction in obese men: a randomized controlled trial

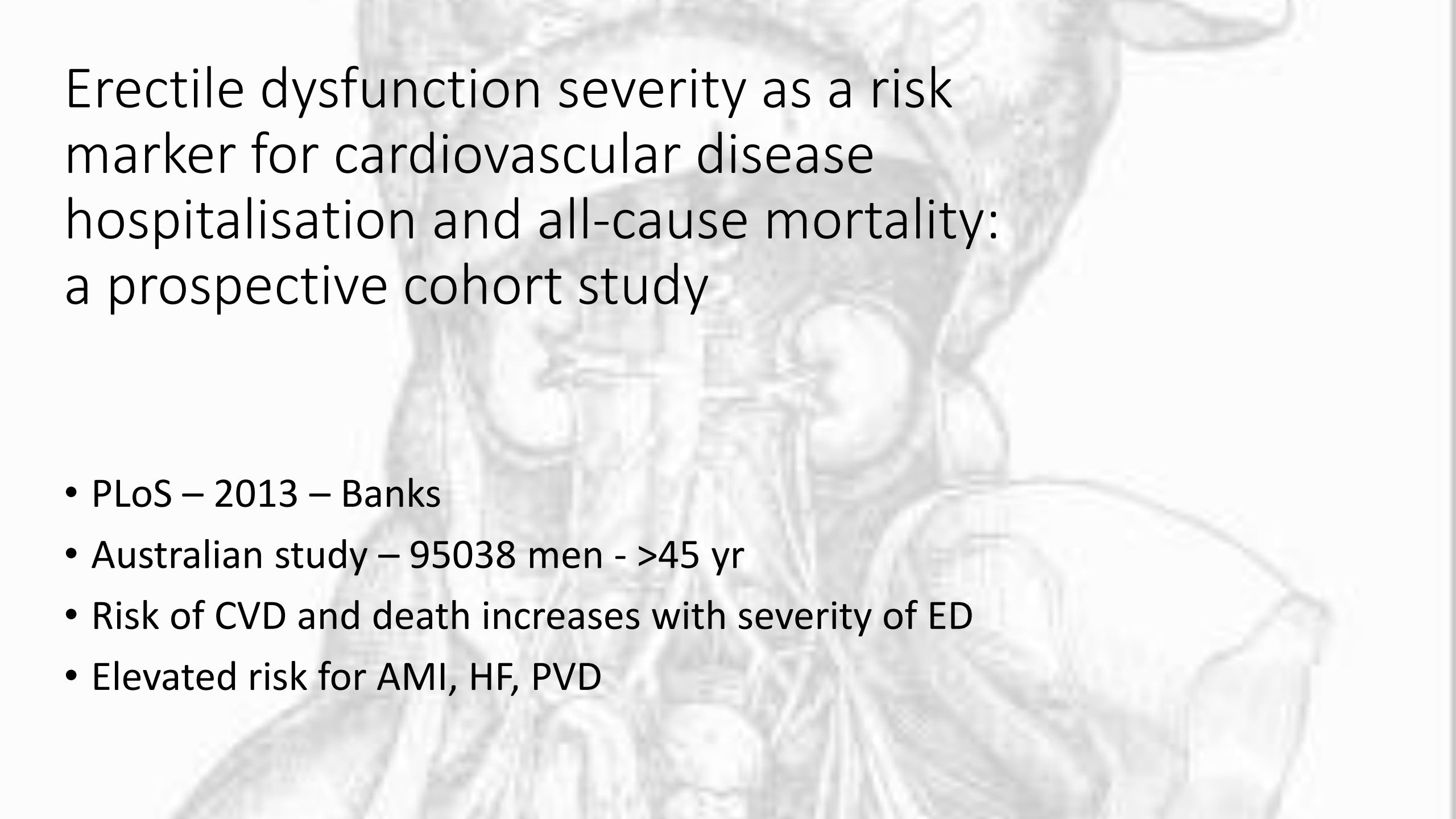
- JAMA – 2004 – Esposito
- 110 obese men with ED
- 55 men intervention group 10% weight loss
- Evaluation after 2 years
- Intervention group BMI decreases
- Intervention group better IIEF

Erectile dysfunction diagnosis and treatment as a means to improve medication adherence and optimize comorbidity management

- J Sex Med – 2013 – Scranton
- Appropriate management of ED and its risk factors may have beneficial effects on diseases that are comorbid with ED and v.v.
- Early diagnosis of ED may provide an opportunity to prevent future CV events

A systematic review of the association between erectile dysfunction and cardiovascular disease

- Eur Urol – 2013 – Gandaglia
- ED and CVD should be regarded as two different manifestations of the same systemic disorder
- ED usually precedes CVD onset

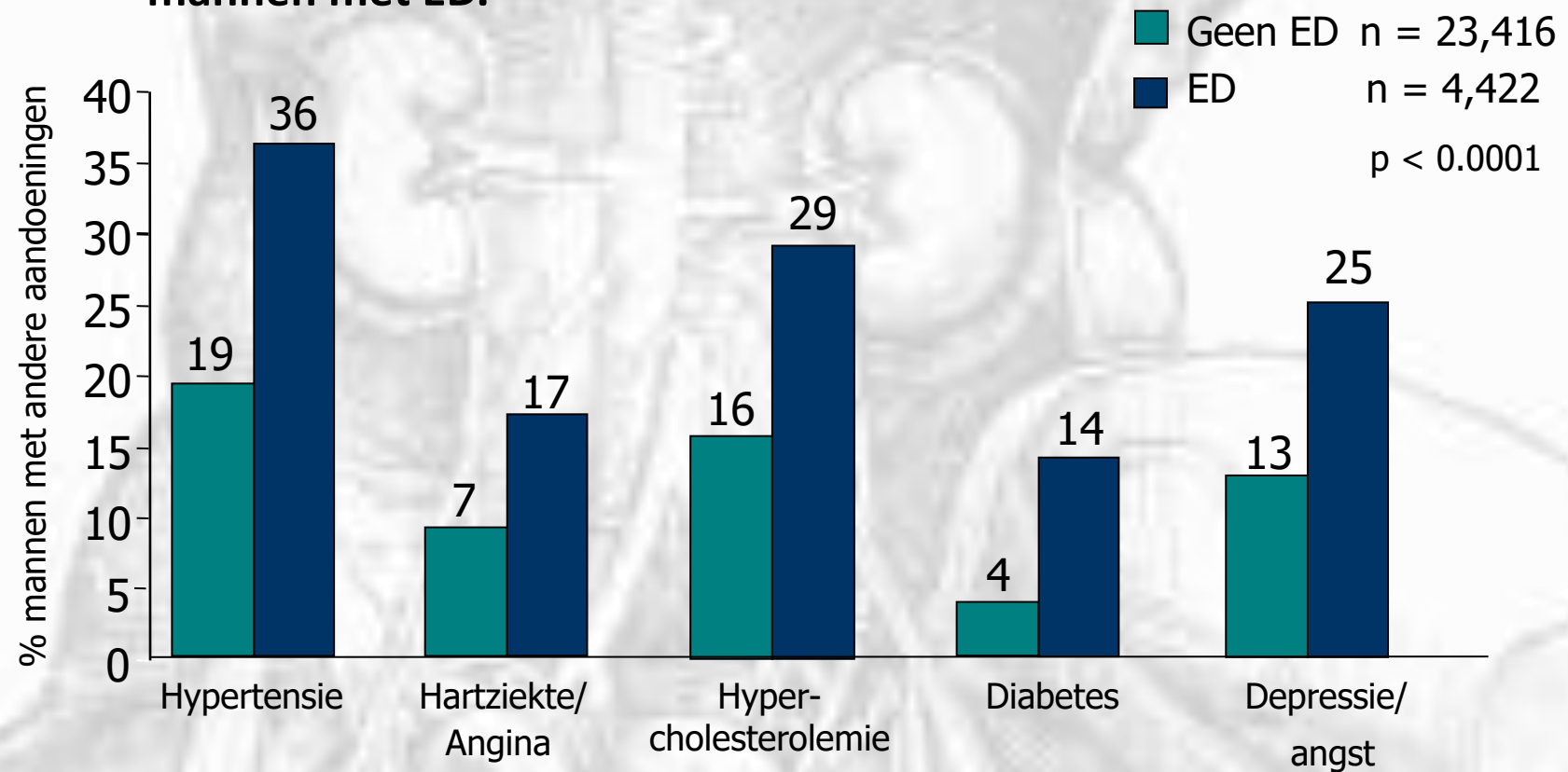


Erectile dysfunction severity as a risk marker for cardiovascular disease hospitalisation and all-cause mortality: a prospective cohort study

- PLoS – 2013 – Banks
- Australian study – 95038 men - >45 yr
- Risk of CVD and death increases with severity of ED
- Elevated risk for AMI, HF, PVD

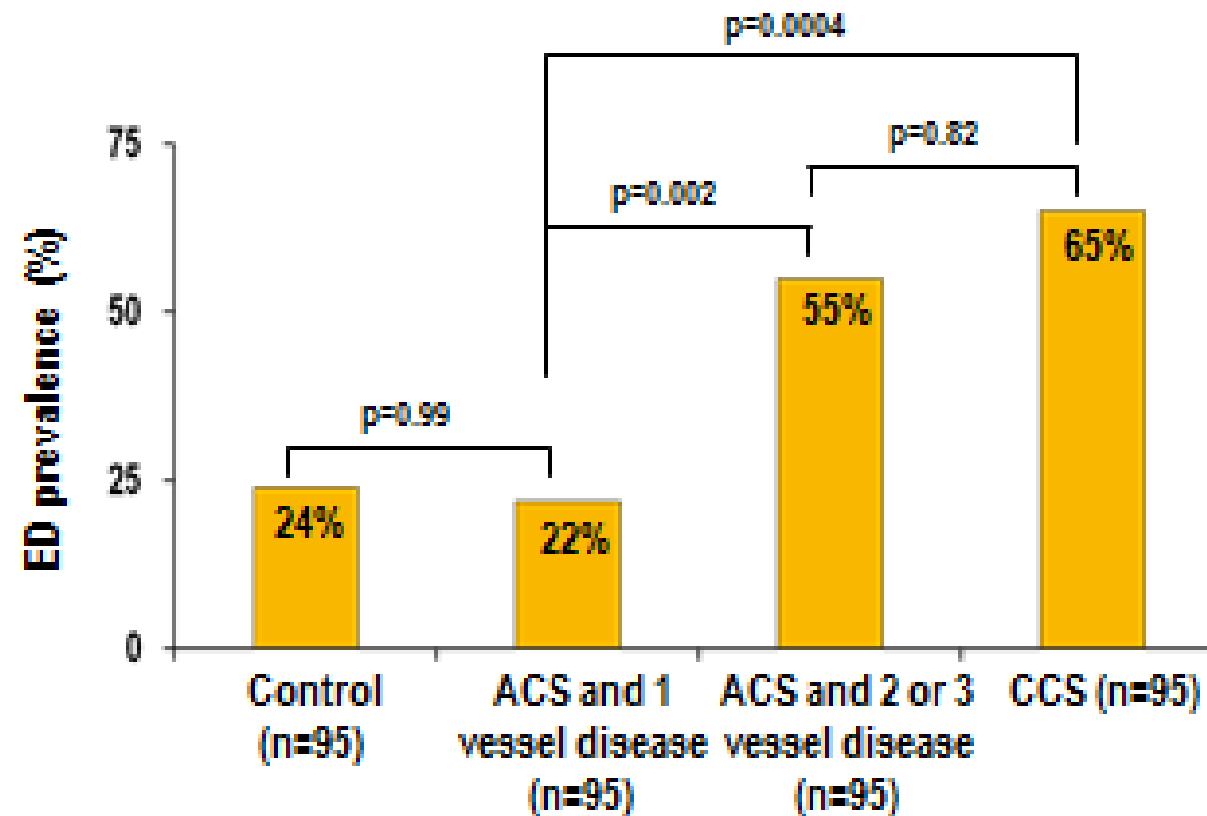
Een aantal aandoeningen zijn geassocieerd met ED

Er is een hogere prevalentie van comorbiditeiten bij mannen met ED.



Opm: 64% van de mannen met ED rapporteerden minstens 1 of meer van deze aandoeningen

PREVALENCE OF ED AMONG PATIENTS WITH ACUTE CORONARY SYNDROME (ACS) OR CHRONIC CORONARY SYNDROME (CCS)



Erectiele dysfunctie...



Circulation

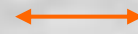
JOURNAL OF THE AMERICAN HEART ASSOCIATION

Cardiovascular Implications of Erectile Dysfunction
Bryan G. Schwartz and Robert A. Kloner

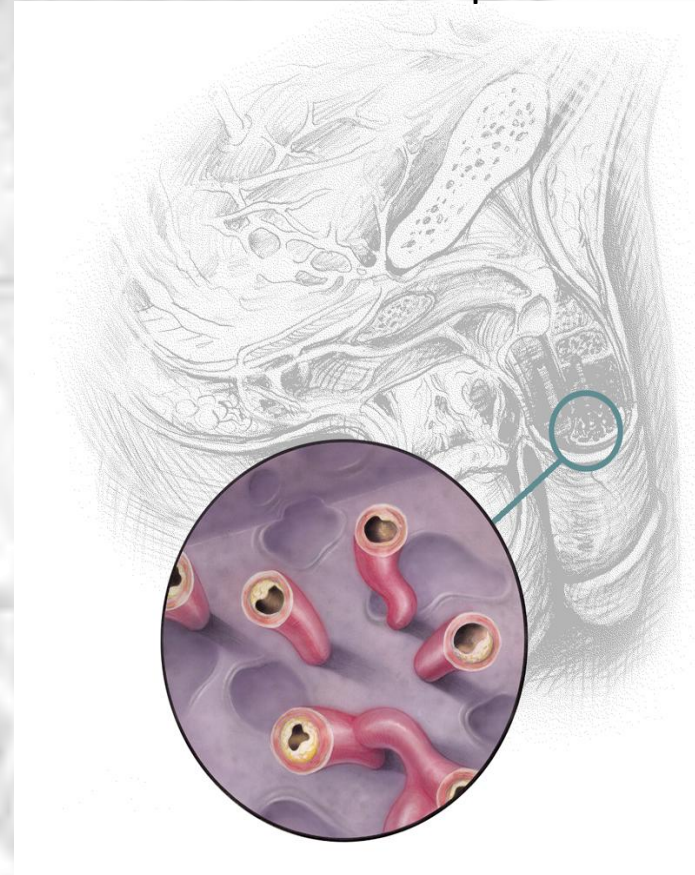
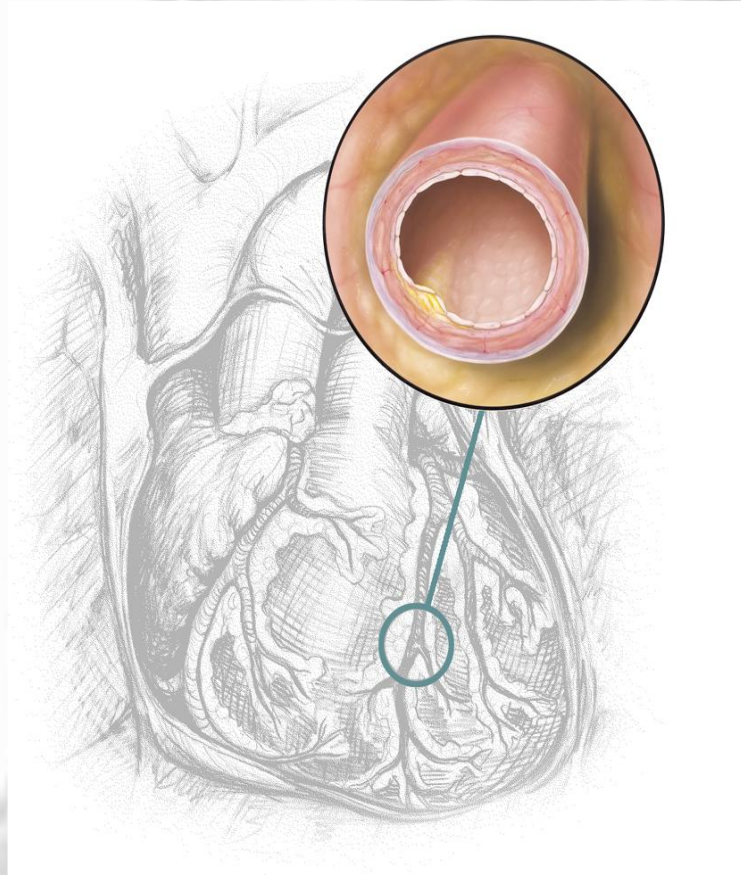
the sentinel symptom of vascular
pathology

Doormeter van de bloedvaten is belangrijk

Atherosclerose in coronaire bloedvaten

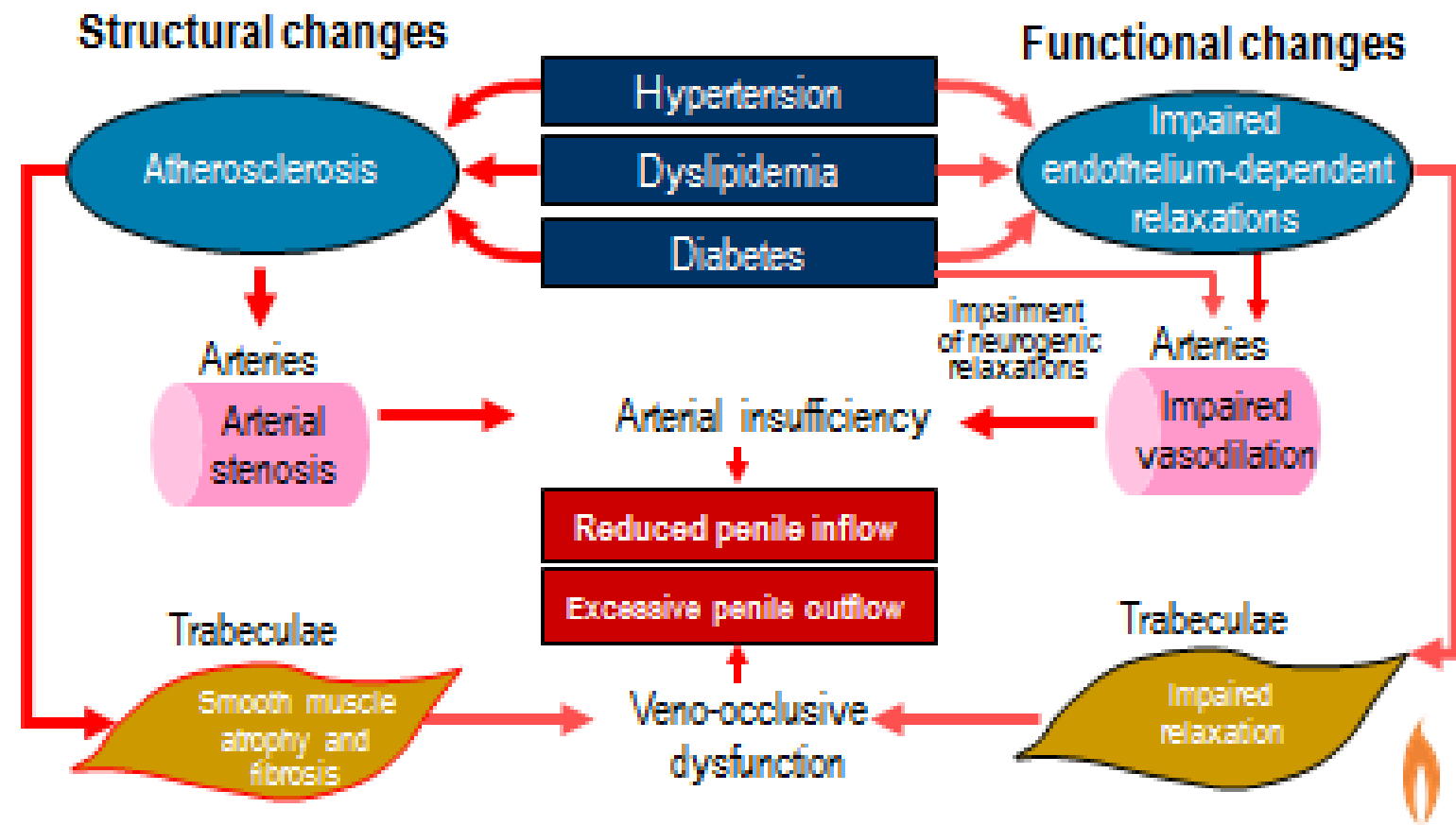


Atherosclerose in de bloedvaten
van de penis



Waarom mannen in een vroeg stadium van ED zelden te maken hebben met CAD maar waarom mannen met CAD frequent lijden aan ED...

THE INTERPLAY BETWEEN UNDERLYING CONDITIONS AND IMPAIRED ENDOTHELIAL FUNCTION IN ED



ED is de barometer van de gezondheid van de verouderende man

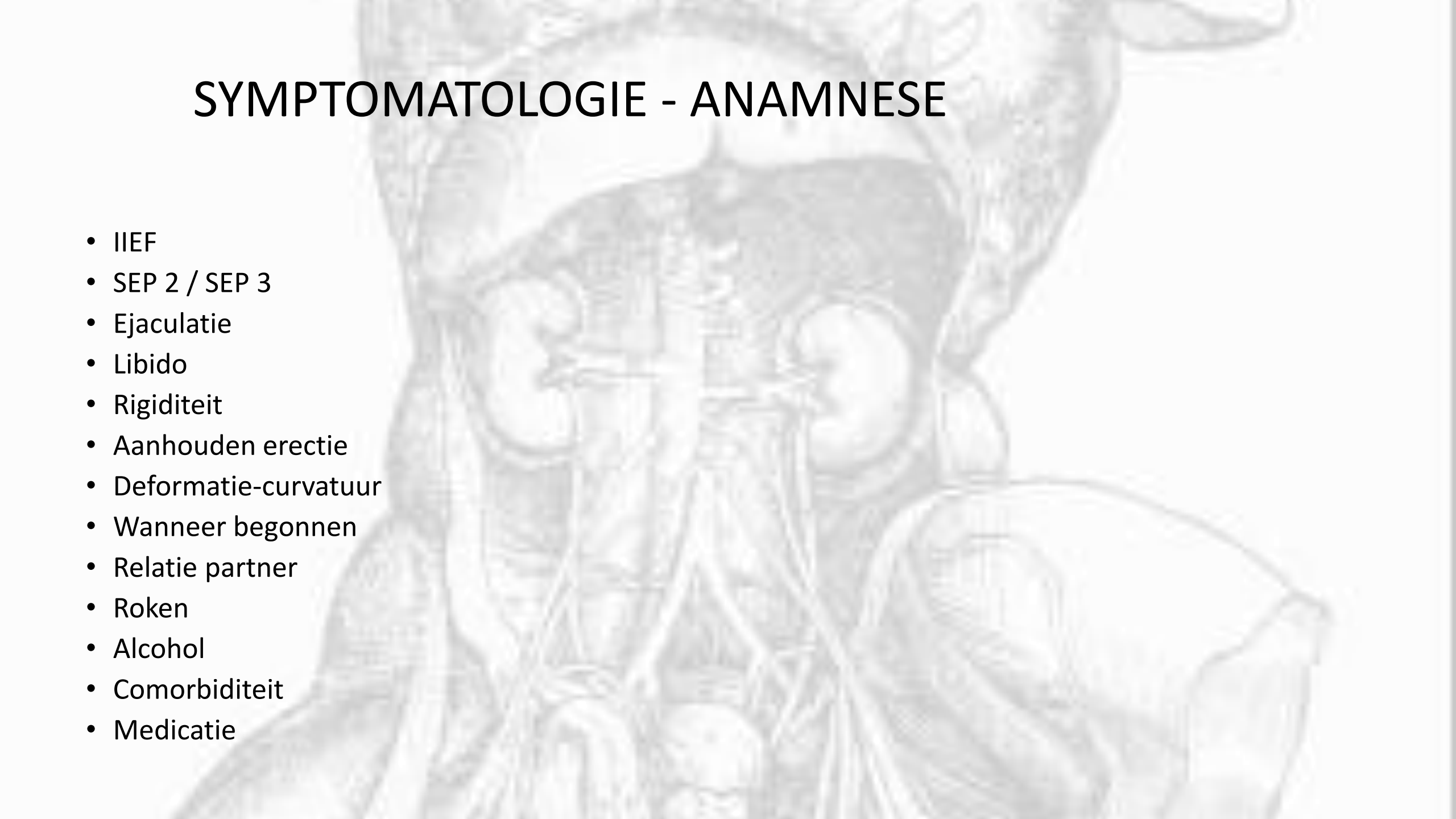
- ED is geassocieerd met het voorkomen van een groter aantal comorbiditeiten (hypertensie, hartkwalen, diabetes)
- Patiënten met een acuut MI en die ED hebben, zullen een 6 keer hoger risico hebben op een “multivessel disease” en dit onafhankelijk van andere risicofactoren (Cobra trial)
- Elke mmol/liter toename in totaal cholesterol = 1.32 keer verhoogd ED risico
- Insuline resistentie komt in 79.2% van de ED populatie voor vs. 25% in de algemene populatie
- ED kan beschouwd worden als een merker voor atherosclerosis
- Hou rekening met deze comorbiditeiten bij patiënten die consulteren met ED.
- Hou rekening met ED bij patiënten die consulteren voor deze comorbiditeiten Moeten ze behandeld worden?

- 
- E.D. = Erectiele Dysfunctie = Endotheliale dysfunctie
 - E.D. is veel voorkomend bij mannen met chronisch stabiel coronair lijden
 - E.D. is een vroegtijdige marker van vasculair lijden

DIAGNOSTIEK

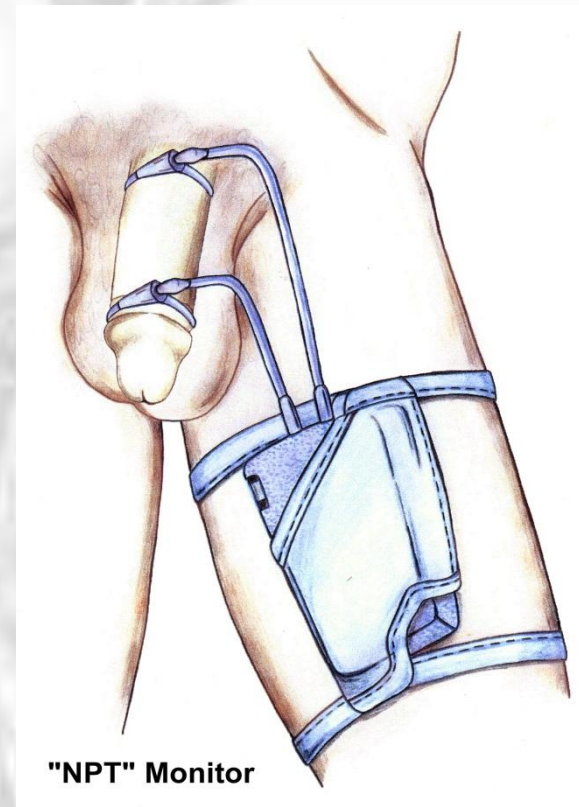
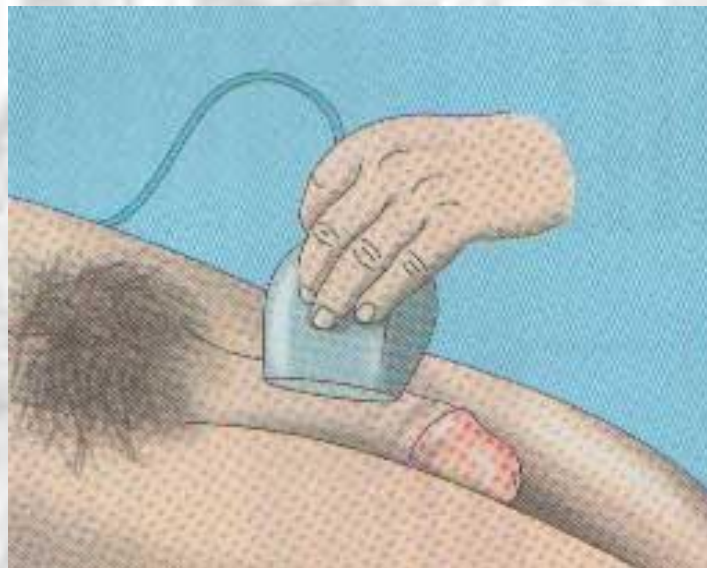
A detailed anatomical illustration of the human urinary system, showing the kidneys, ureters, bladder, and associated blood vessels and nerves. The illustration is rendered in a classic, detailed style, likely from a medical textbook. The kidneys are shown as bean-shaped organs, with the ureters leading down to the bladder. The surrounding structures, including the adrenal glands and the network of blood vessels and nerves, are also depicted with precision.

SYMPTOMATOLOGIE - ANAMNESE

A detailed anatomical illustration of the male reproductive system, showing the testes, epididymis, vas deferens, ureters, and associated structures. The illustration is rendered in a light, sketchy style, with the central structures in a slightly darker shade than the surrounding tissues.

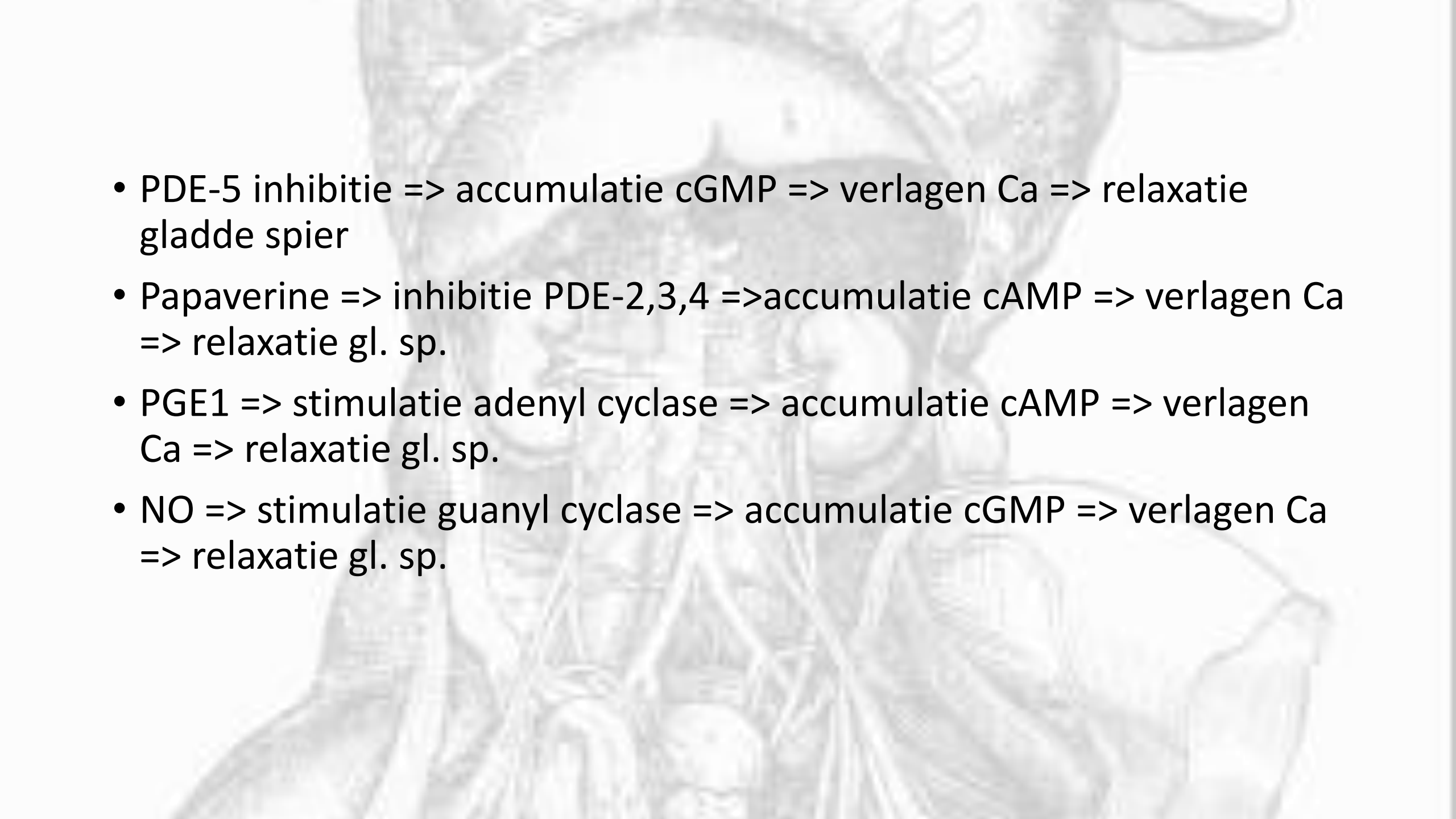
- IIEF
- SEP 2 / SEP 3
- Ejaculatie
- Libido
- Rigiditeit
- Aanhouden erectie
- Deformatie-curvatuur
- Wanneer begonnen
- Relatie partner
- Roken
- Alcohol
- Comorbiditeit
- Medicatie

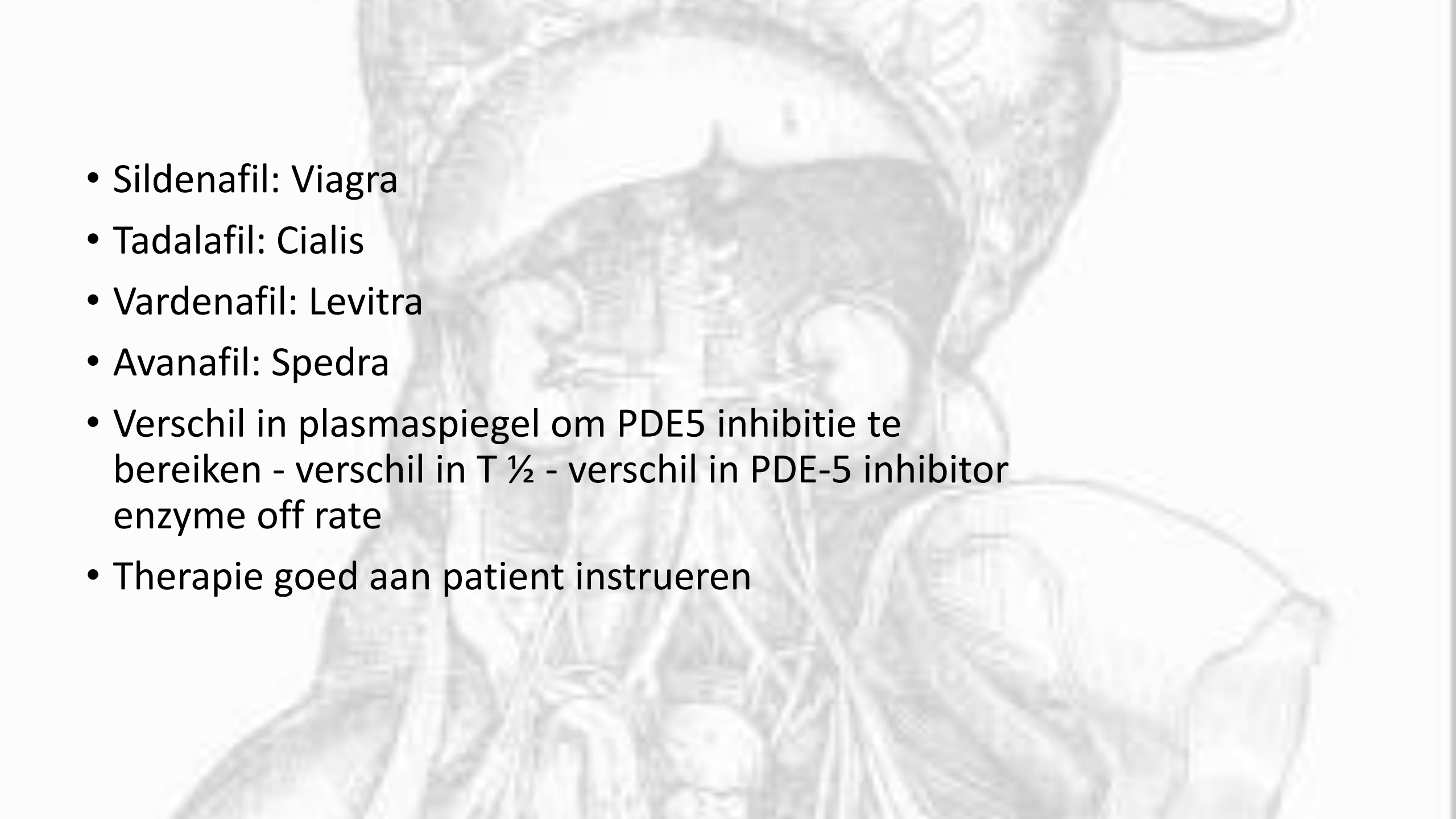
- Klinisch onderzoek
- Labo testen (testosterone)
- Doppler
- NPT
- NUFO



THERAPIE



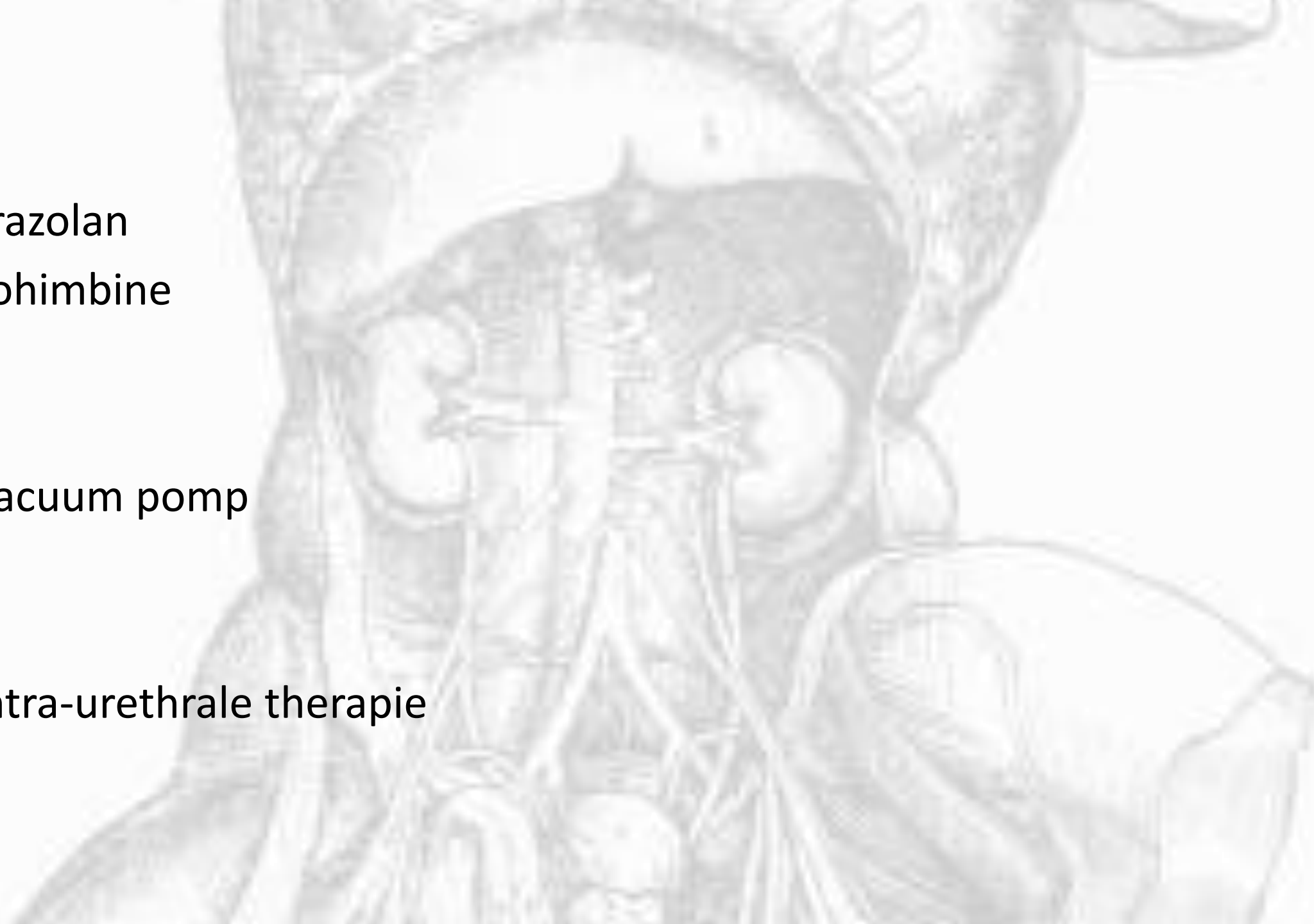
- 
- PDE-5 inhibitie => accumulatie cGMP => verlagen Ca => relaxatie gladde spier
 - Papaverine => inhibitie PDE-2,3,4 => accumulatie cAMP => verlagen Ca => relaxatie gl. sp.
 - PGE1 => stimulatie adenyl cyclase => accumulatie cAMP => verlagen Ca => relaxatie gl. sp.
 - NO => stimulatie guanyl cyclase => accumulatie cGMP => verlagen Ca => relaxatie gl. sp.

- 
- Sildenafil: Viagra
 - Tadalafil: Cialis
 - Vardenafil: Levitra
 - Avanafil: Spedra
 - Verschil in plasmaspiegel om PDE5 inhibitie te bereiken - verschil in $T_{1/2}$ - verschil in PDE-5 inhibitor enzyme off rate
 - Therapie goed aan patient instrueren

INTRA CAVERNEUZE THERAPIE

An anatomical illustration of the male reproductive system, showing the penis, testis, and associated vasculature and nerves. The illustration is in a light, faded style, serving as a background for the text.

- Alprostadil (Caverject)
- Papaverine / fentolamine (niet meer) / PGE1
- Instructie !!!
- Priapisme

- 
- A detailed anatomical illustration of the male reproductive system, showing the testes, vas deferens, ureters, and associated structures. The illustration is in a light gray tone, serving as a background for the text.
- Trazolan
 - Yohimbine
 - Vacuum pomp
 - Intra-urethrale therapie

VACUUMPOMP

V.S.

1917

OMSLACHTIG

Which route would you take



LI-ESWT: clinical data

available at www.sciencedirect.com
journal homepage: www.europeanurology.com



European Association of Urology



Platinum Priority – Sexual Medicine

Editorial by Konstantinos Hatzimouratidis on pp. 249–250 of this issue

Can Low-Intensity Extracorporeal Shockwave Therapy Improve Erectile Function? A 6-Month Follow-up Pilot Study in Patients with Organic Erectile Dysfunction

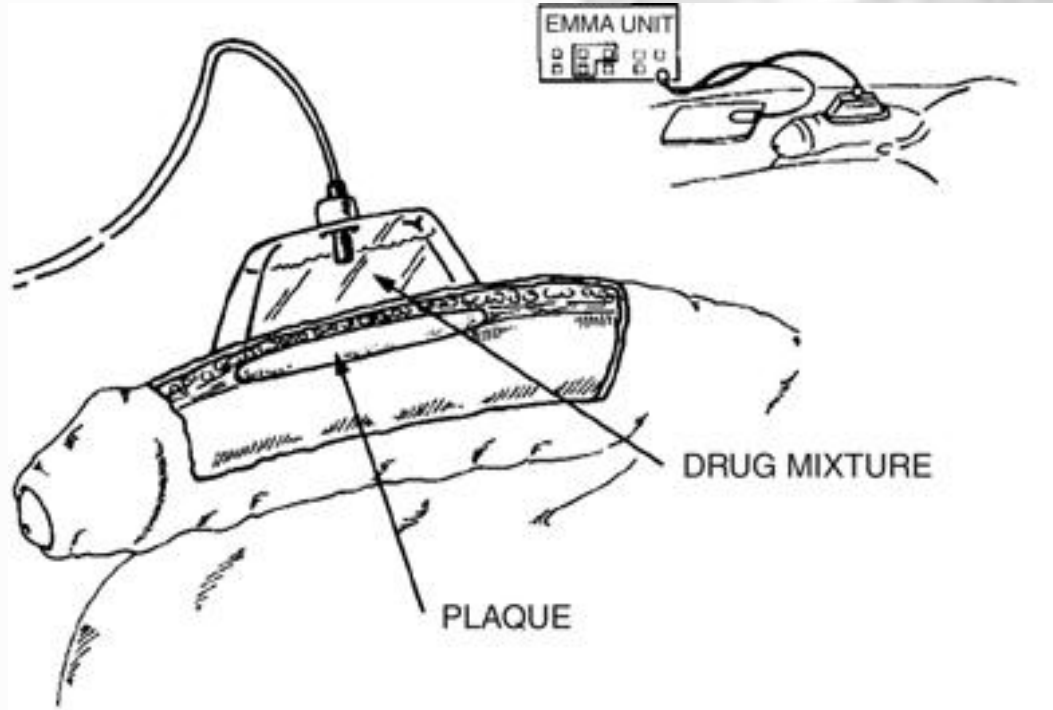
Yoram Vardi^{}, Boaz Appel, Giris Jacob, Omar Massarwi, Ilan Gruenwald*

Neuro-Urology Unit, Rambam Healthcare Campus and the Technion, Haifa, Israel

Iontophoresis (EMDA)

- Iontophoresis (also known as transdermal electromotive drug administration or electromotive drug administration [EMDA]) has been introduced to try and overcome limitations on the local uptake of the drugs themselves.
- Uncontrolled studies showed promising results in terms of improvement in penile curvature, plaque size and penile pain during erection (Di Stasi;BJU Int;2003).
- In a **randomised, double-blind, controlled study**, iontophoresis with verapamil 5 mg and dexamethasone 8 mg resulted in a statistically significant **improvement in penile curvature and plaque size** (Di Stasi;J Urol;2004).
- However, in **another randomised, double-blind, placebo-controlled** study, penile curvature was **not statistically improved** after iontophoresis with verapamil 10 mg (Greenfield;J Urol;2007).
- The method is not associated with any significant adverse event.

Iontophoresis (EMDA)



The active electrode receptacle containing the drug mixture is positioned at the level of the penile plaque. The dispersive electrode is placed on the patient's leg.



DIFFICULT TO TREAT

RADICAL PELVIC SURGERY

end-organ
damaged



fibrosis, smooth muscle
/ endothelial apoptosis

CVOD

SEVERE DIABETES



Lack of
neuronal /
endothelial

NO

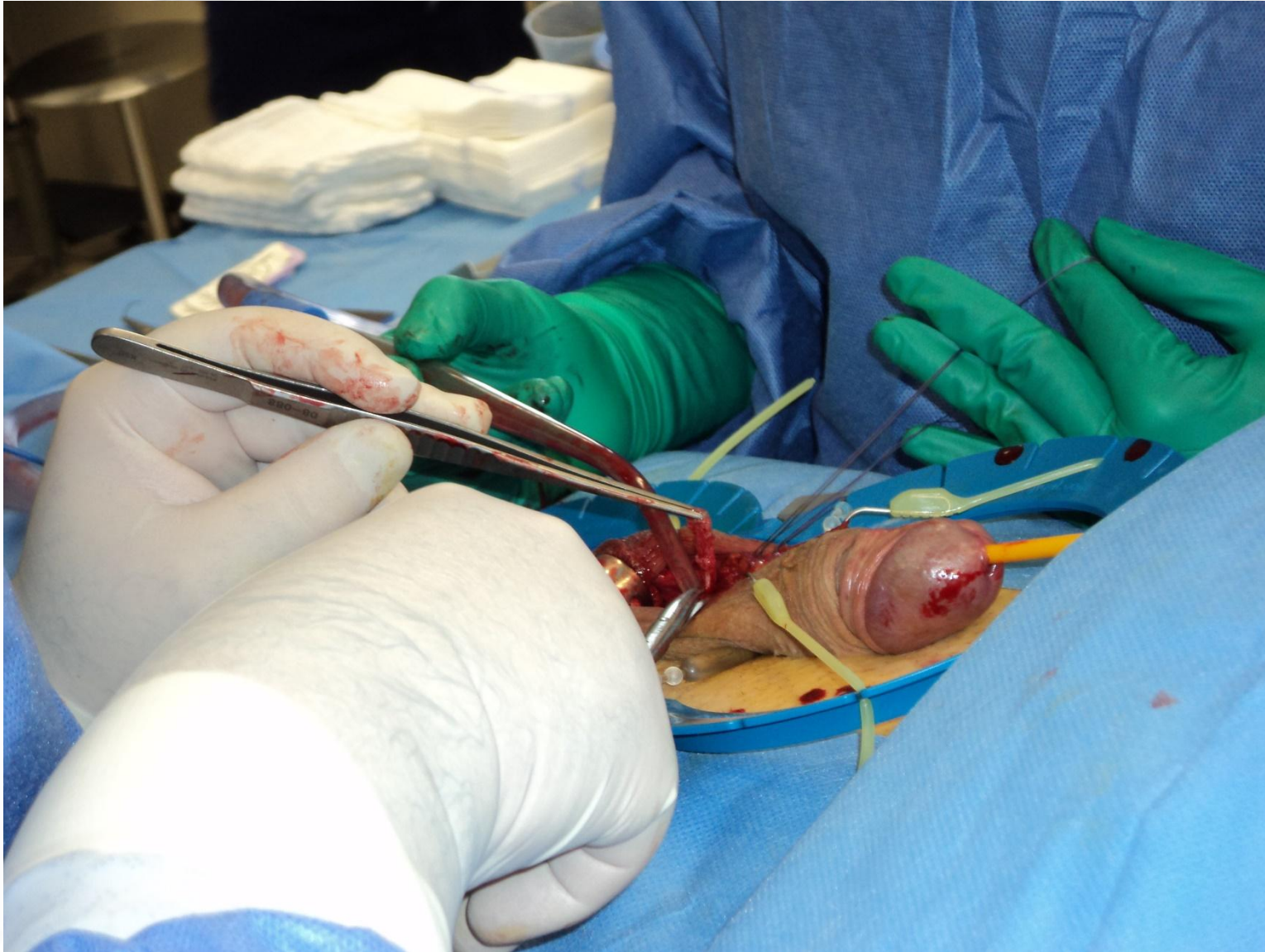
end-organ
ok

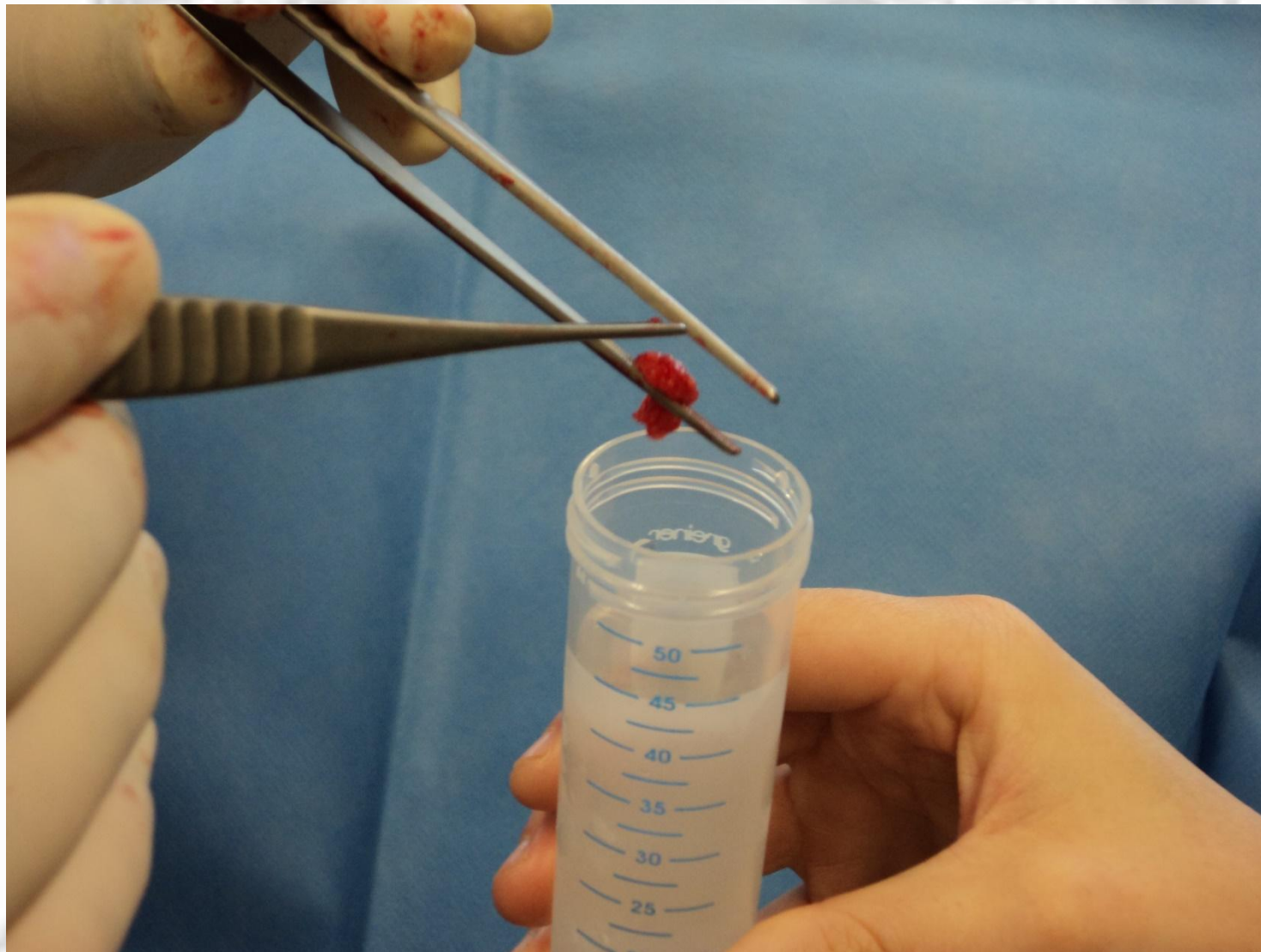


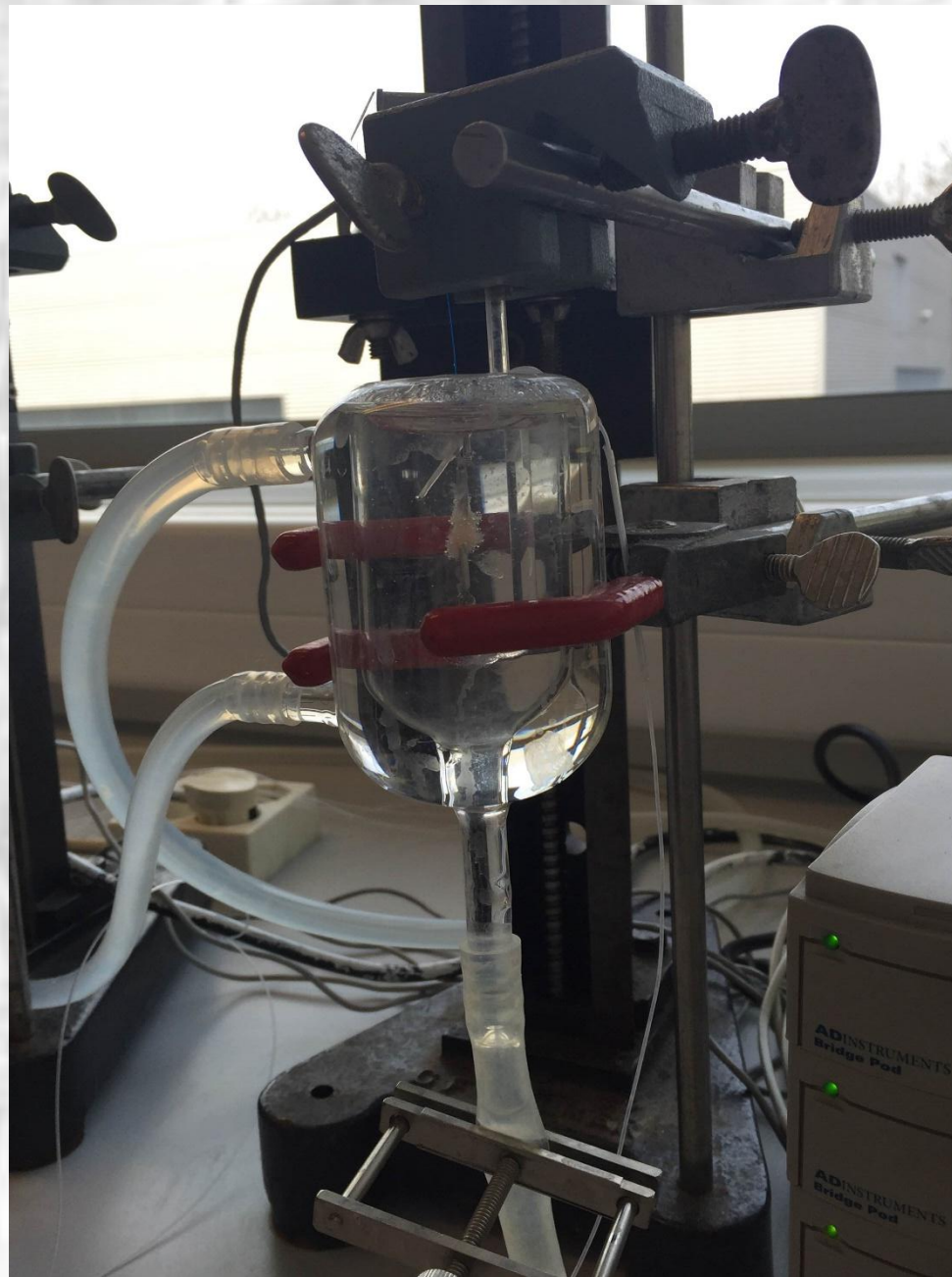
PGE1-non-responders

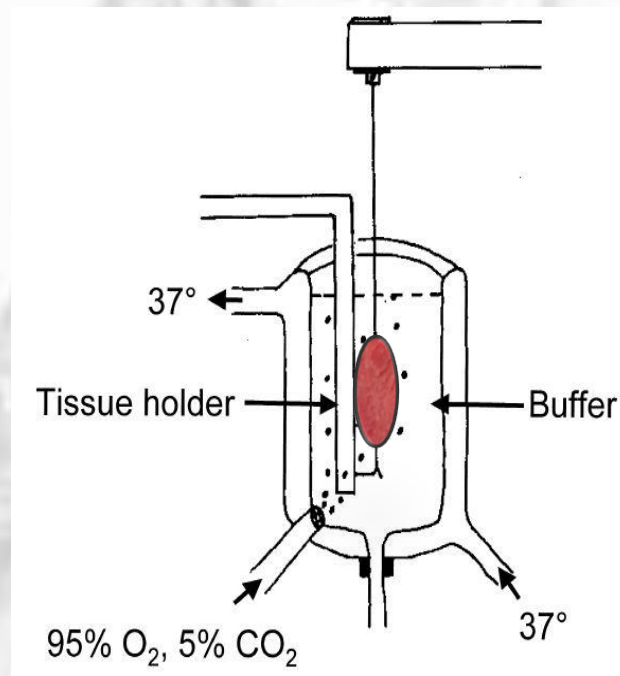
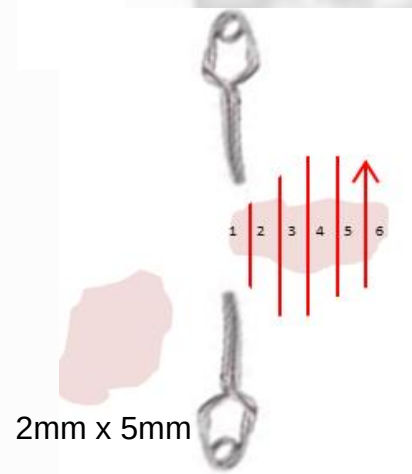
PDE5-i non-responders

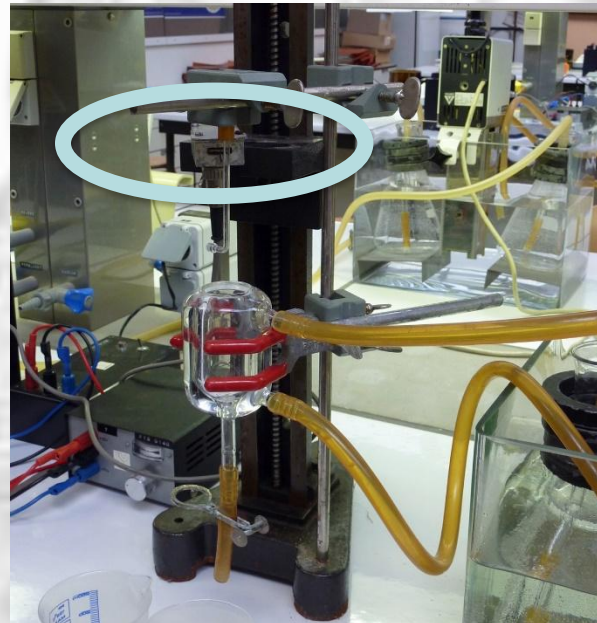
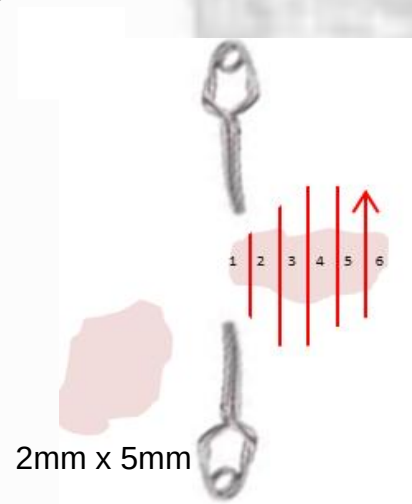


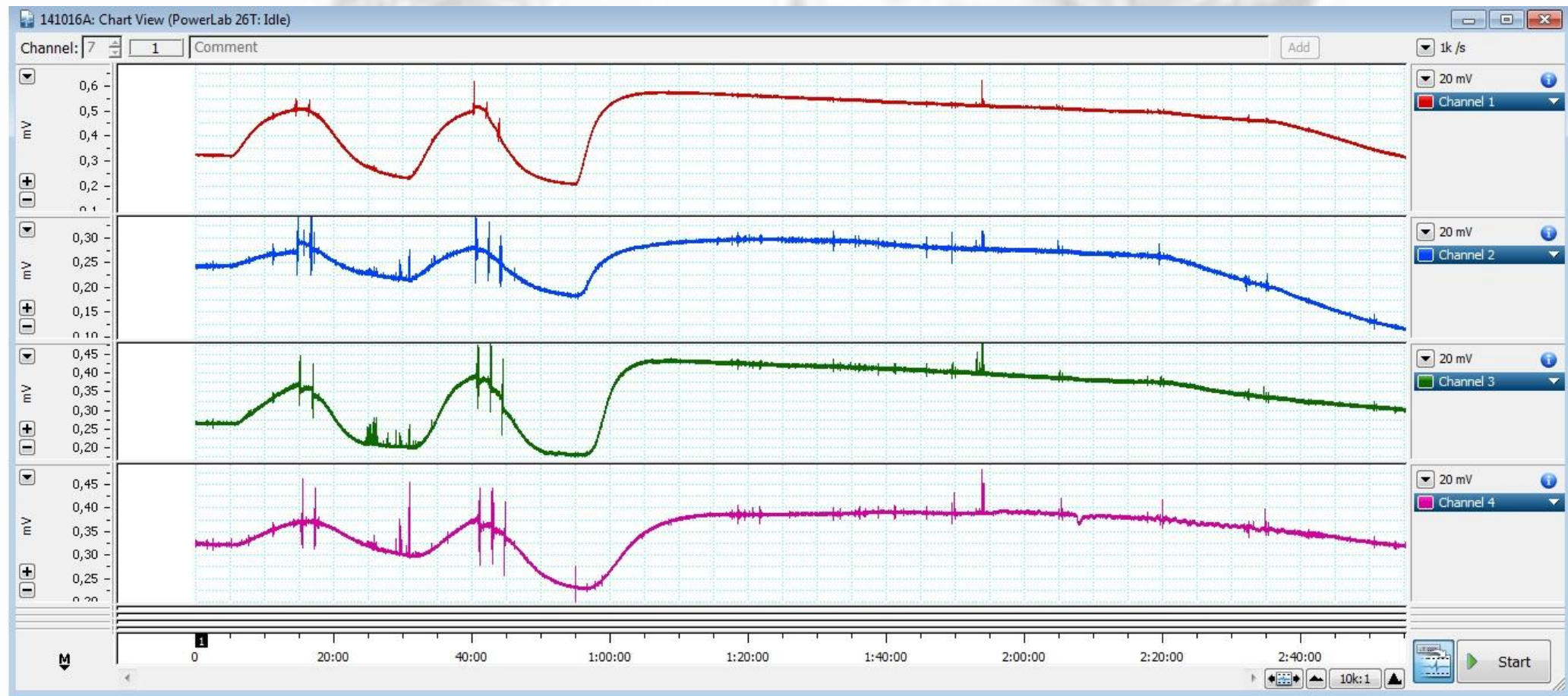


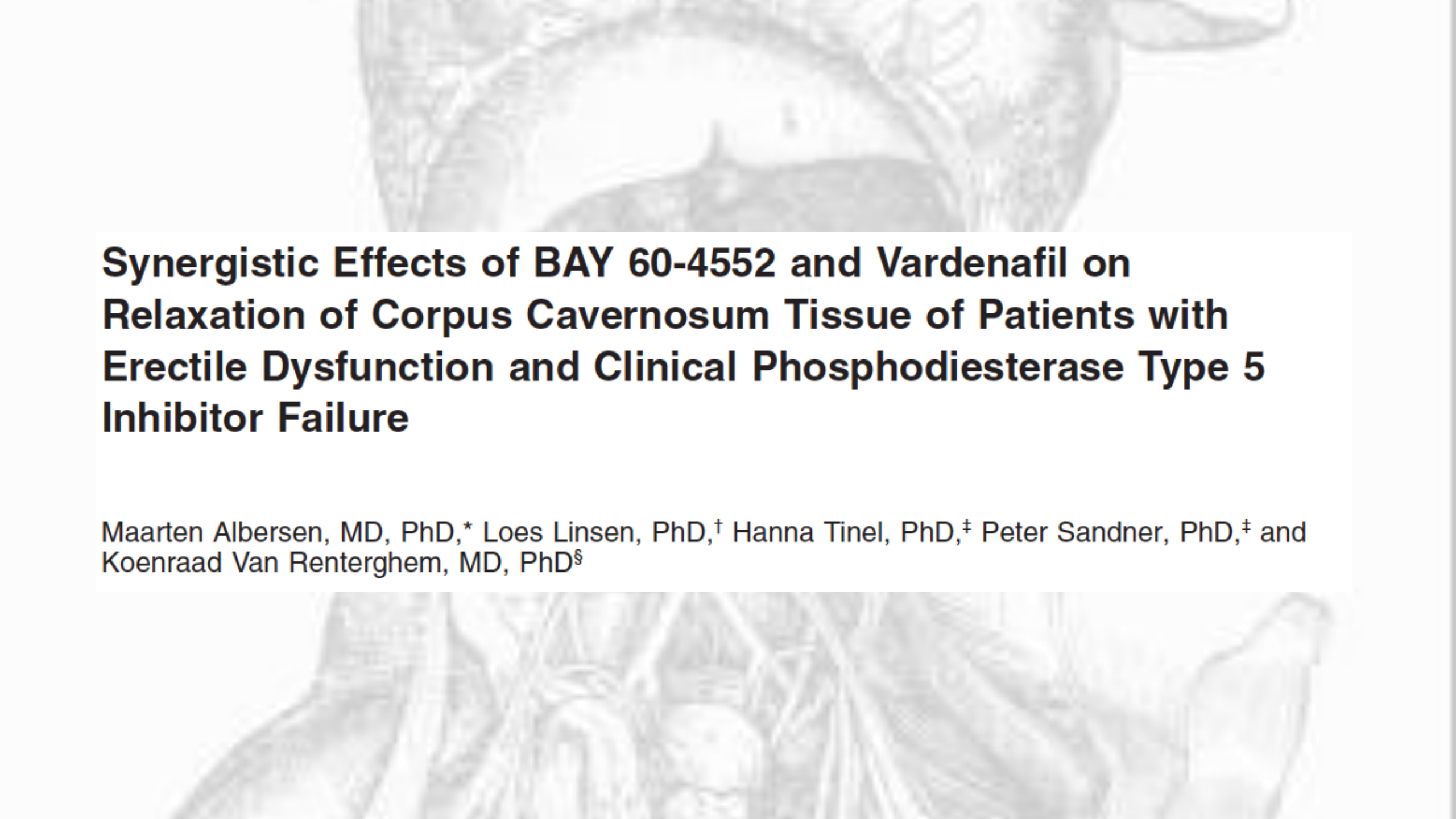








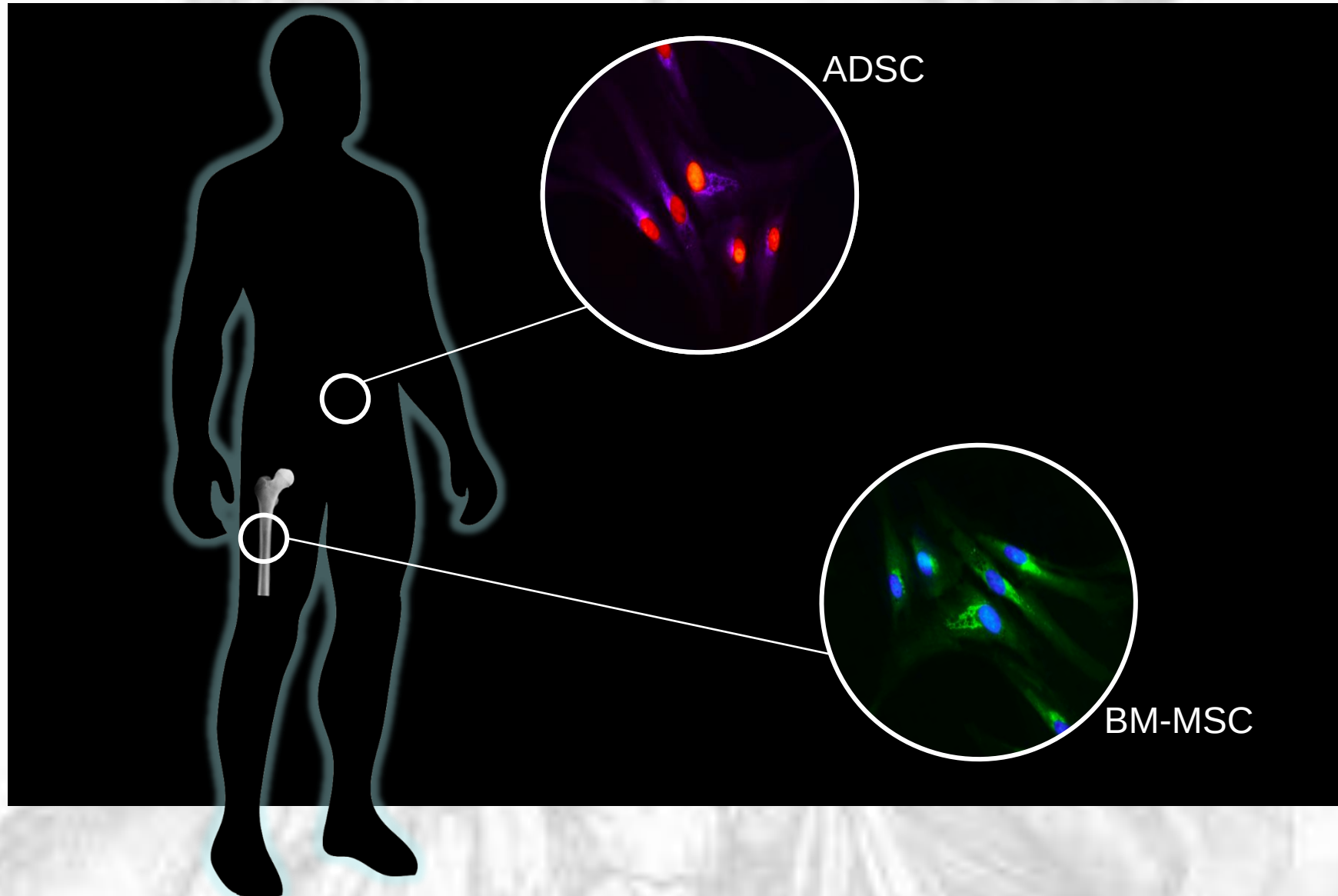


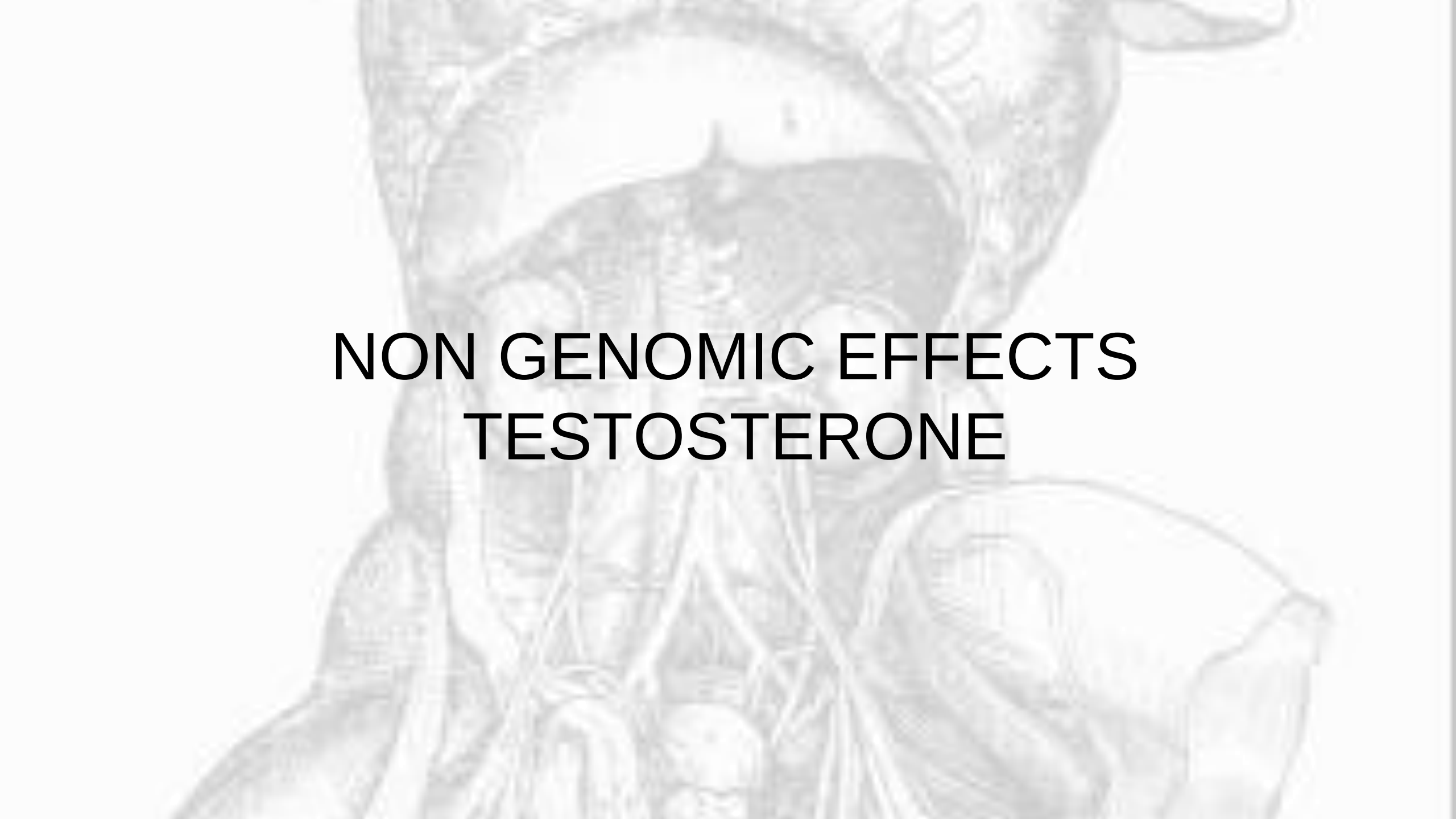


Synergistic Effects of BAY 60-4552 and Vardenafil on Relaxation of Corpus Cavernosum Tissue of Patients with Erectile Dysfunction and Clinical Phosphodiesterase Type 5 Inhibitor Failure

Maarten Albersen, MD, PhD,* Loes Linsen, PhD,[†] Hanna Tinel, PhD,[‡] Peter Sandner, PhD,[‡] and Koenraad Van Renterghem, MD, PhD[§]

Stem cells?





NON GENOMIC EFFECTS TESTOSTERONE



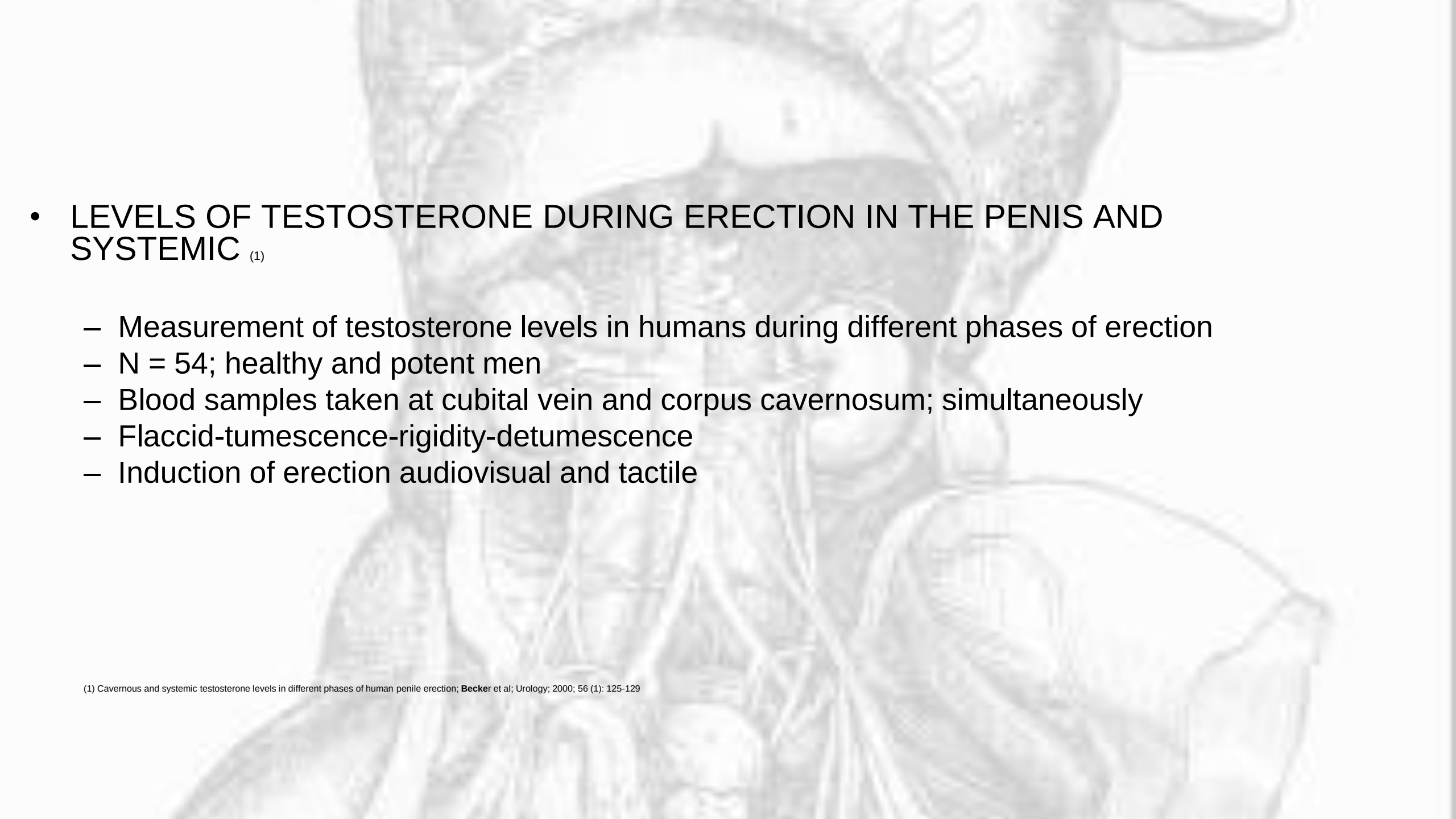
- Acute effect on myocardial ischemia ^{(1) (2) (3)}:
 - Men 35-75 yrs; stable exertional angina; angiographic proven CAD; total testosterone < 11 nmol/l; PSA normal
 - Screening treadmill test/ patients with positive test for myocardial ischemia were randomised one week later between T versus placebo/ one week later opposite treatment with new treadmill test
 - Intracoronary administration
- Testosterone delayed onset of myocardial ischemia
 - Effect was seen at physiological concentrations
 - No effect of inhibition AR or inhibition aromatase
 - Independent of endothelium and nuclear AR; initiated at smooth muscle cell membrane; direct calcium antagonistic action? (rat endothelial denudated coronary art.) ⁽⁴⁾

(1) Effect of acute testosterone on myocardial ischemia in men with coronary artery disease; **Webb** et al; American J of Cardiol; vol 83; Feb 1; 1999

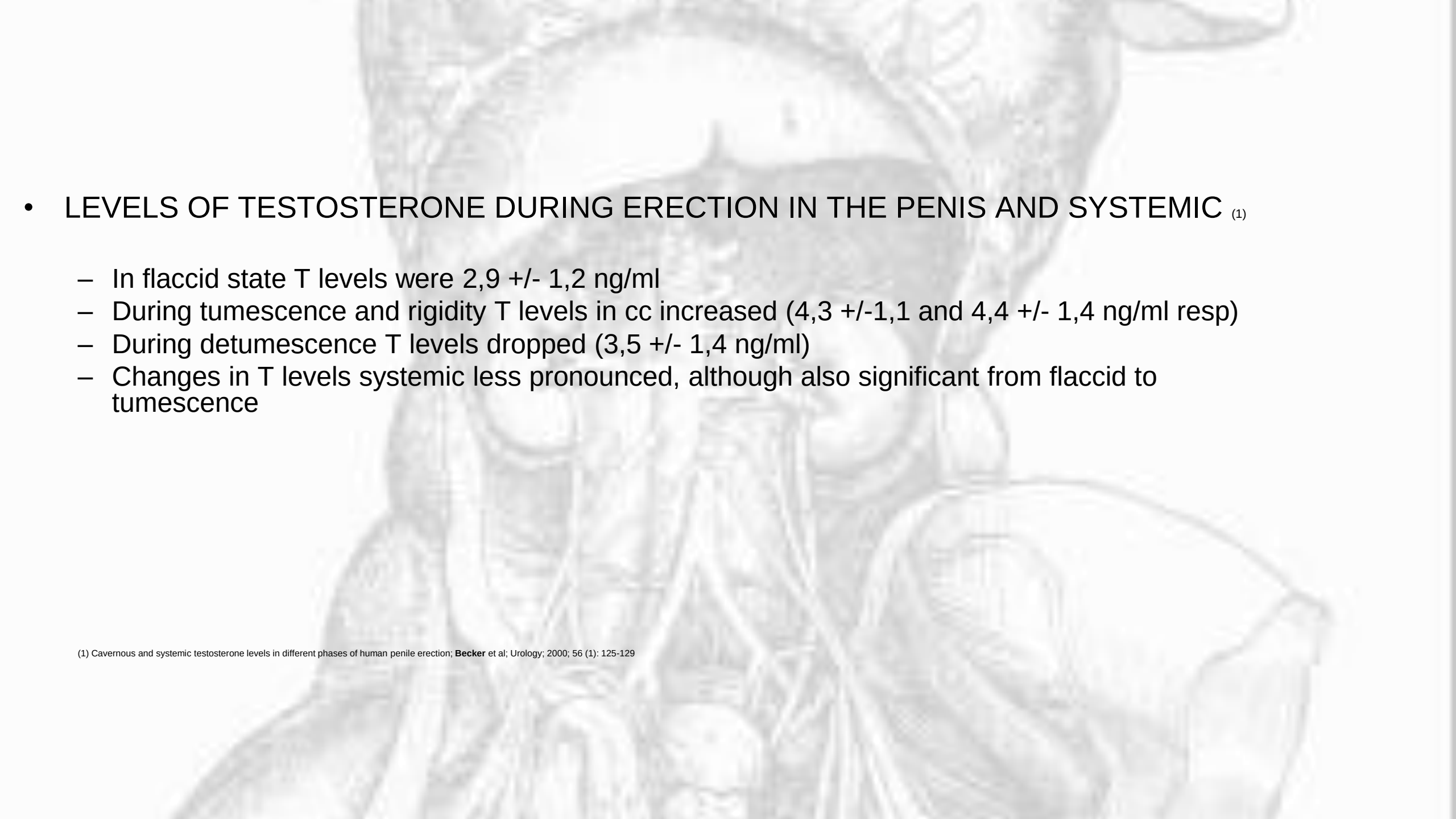
(2) Effects of testosterone on coronary vasomotor regulation in men with coronary heart disease; **Webb** et al; Circulation; 1999; 100: 1690-96

(3) Acute anti-ischemic effect of testosterone in men with coronary artery disease; **Rosano** et al; Circulation; 1999; 99: 1666-1670

(4) Testosterone-induced coronary vasodilatation occurs via a non-genomic mechanism: evidence of a direct calcium antagonistic action; **Jones** et al; Clin Sci (Lond); 2004 Aug; 107(2): 149-58

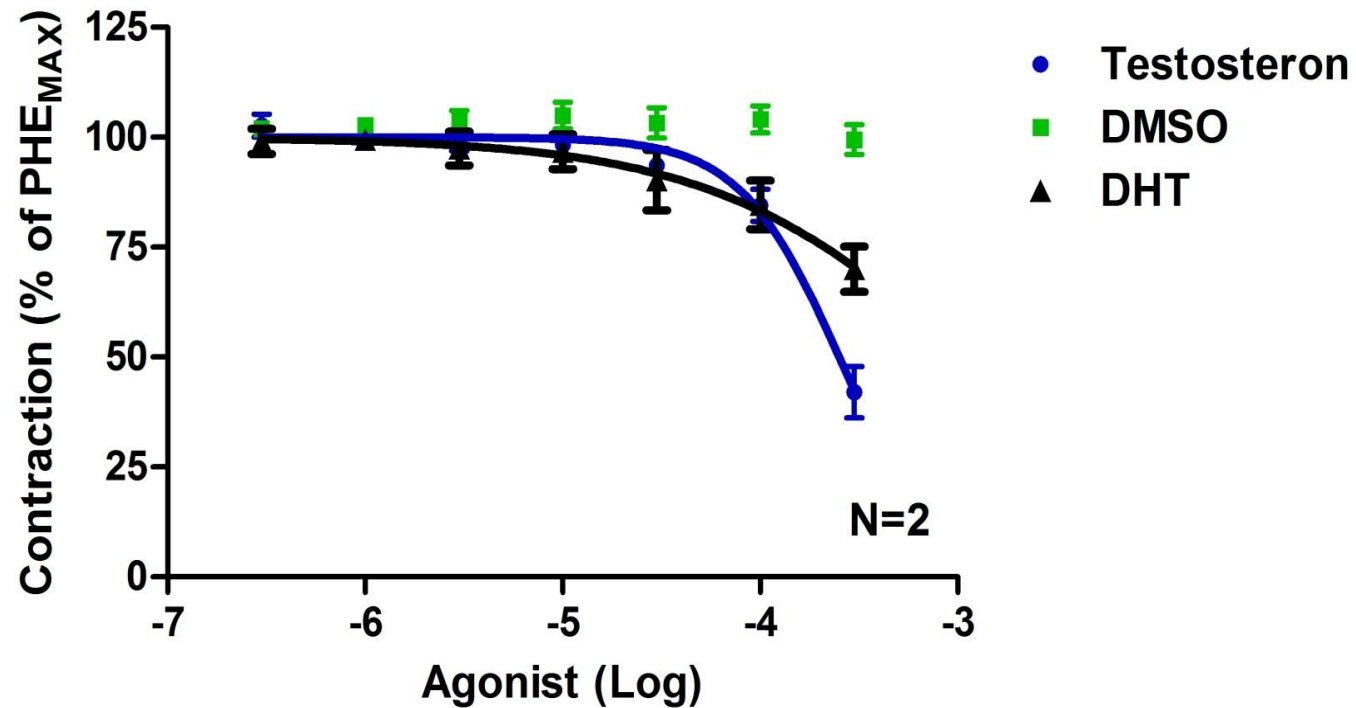
- 
- LEVELS OF TESTOSTERONE DURING ERECTION IN THE PENIS AND SYSTEMIC ⁽¹⁾
 - Measurement of testosterone levels in humans during different phases of erection
 - N = 54; healthy and potent men
 - Blood samples taken at cubital vein and corpus cavernosum; simultaneously
 - Flaccid-tumescence-rigidity-detumescence
 - Induction of erection audiovisual and tactile

(1) Cavernous and systemic testosterone levels in different phases of human penile erection; **Becker** et al; Urology; 2000; 56 (1): 125-129

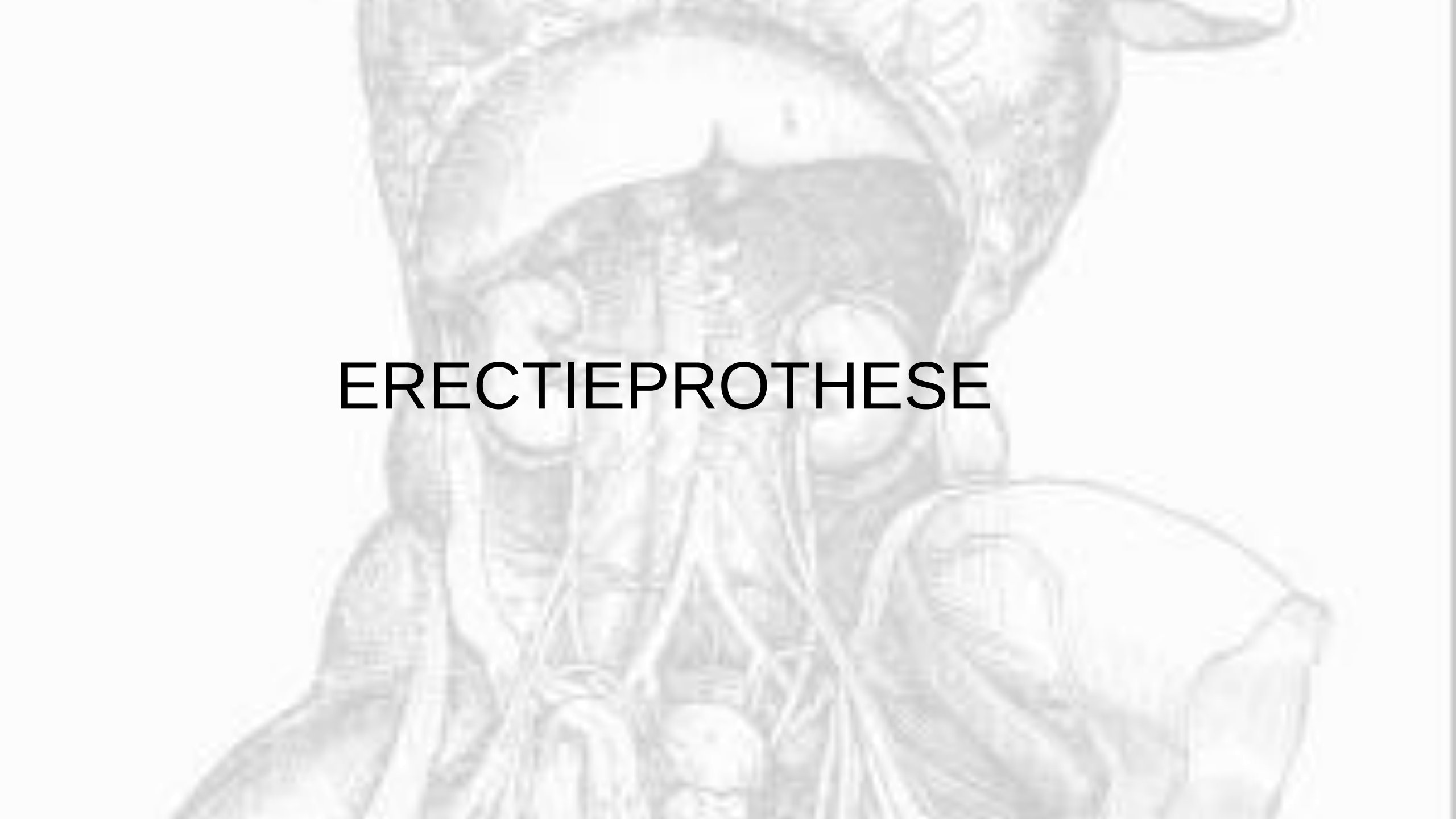
- 
- LEVELS OF TESTOSTERONE DURING ERECTION IN THE PENIS AND SYSTEMIC ⁽¹⁾
 - In flaccid state T levels were 2,9 +/- 1,2 ng/ml
 - During tumescence and rigidity T levels in cc increased (4,3 +/-1,1 and 4,4 +/- 1,4 ng/ml resp)
 - During detumescence T levels dropped (3,5 +/- 1,4 ng/ml)
 - Changes in T levels systemic less pronounced, although also significant from flaccid to tumescence

(1) Cavernous and systemic testosterone levels in different phases of human penile erection; **Becker et al**; Urology; 2000; 56 (1): 125-129

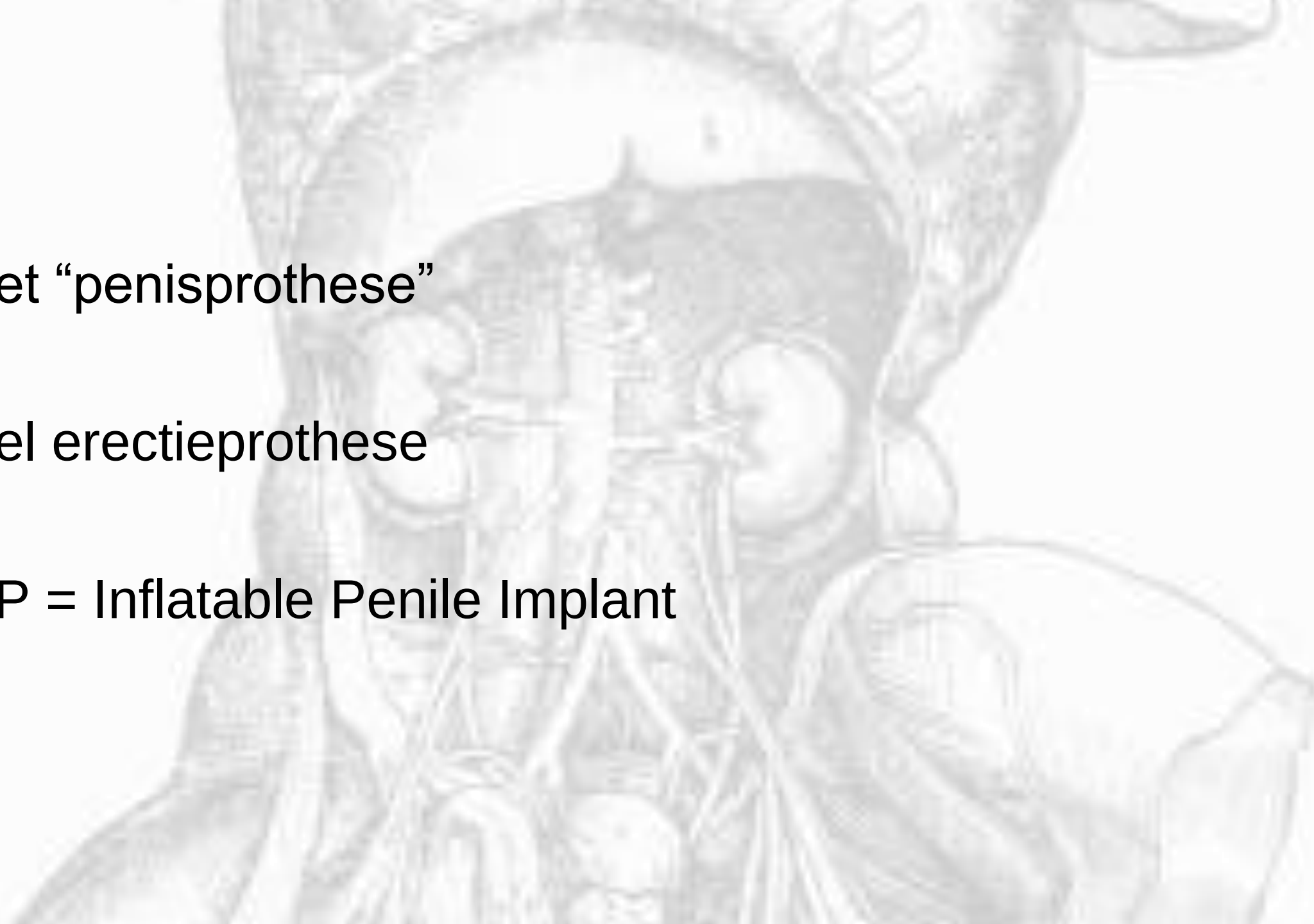
Dose response 5min interval



LogIC50	T	DHT	DMSO
0			
T		0.0130	/
DHT	0.0130		/
DMSO	/		/
DMSO: N=1			

An anatomical illustration of the male reproductive system, showing the penis, testicles, and associated structures. The illustration is in a light, faded style, with the central text overlaid. The text "ERECTIEPROTHESE" is written in a bold, black, sans-serif font, centered horizontally and vertically across the image. The background shows the internal organs of the male reproductive system, including the penis, testicles, and the associated ducts and glands. The illustration is detailed, showing the texture of the organs and the complex network of ducts and glands. The overall tone is educational and clinical.

ERECTIEPROTHESE

- 
- Niet “penisprothese”
 - Wel erectieprothese
 - IPP = Inflatable Penile Implant

HISTORICAL OVERVIEW



18^{DE} EEUW **ELECTROTHERAPY - COLD THERAPY**

1889 **BROWN SEQUARD - INJECTION TESTIC.EXTRACT DOGS**

1892 **STANLEY - IDEM EXP. ON PRISONERS**

1900 **VASECTOMY**

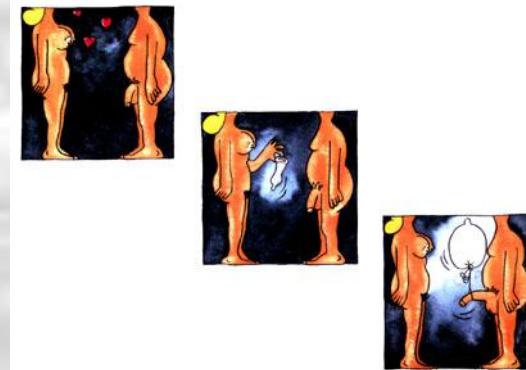
1902 **WOOTEN C.C.**

1908 **LYDSSTON IDEM**

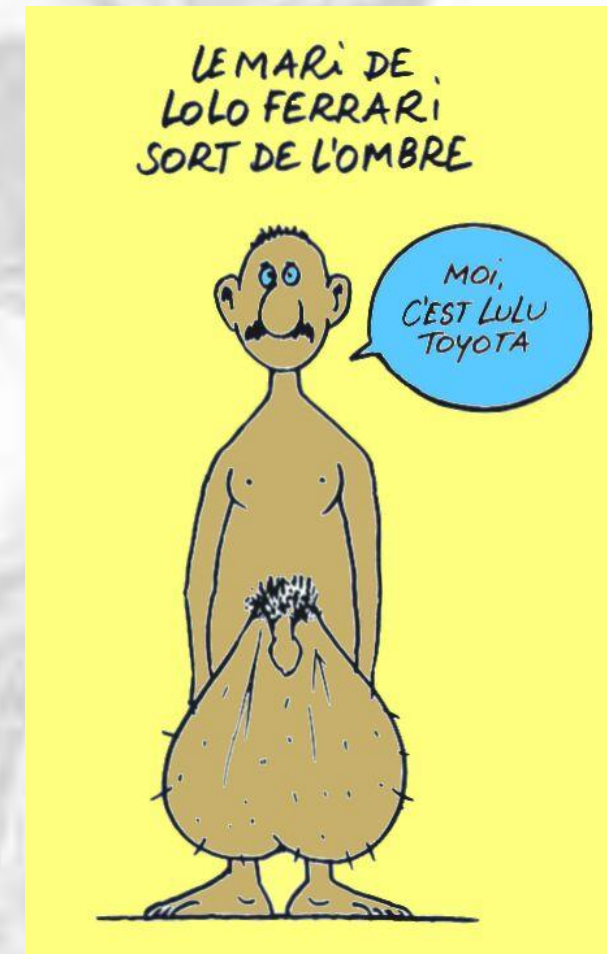
1918 **LESPINASSE**
 ■ **TESTIC.EXTRACT IN RECTUS ABDOMINUS**

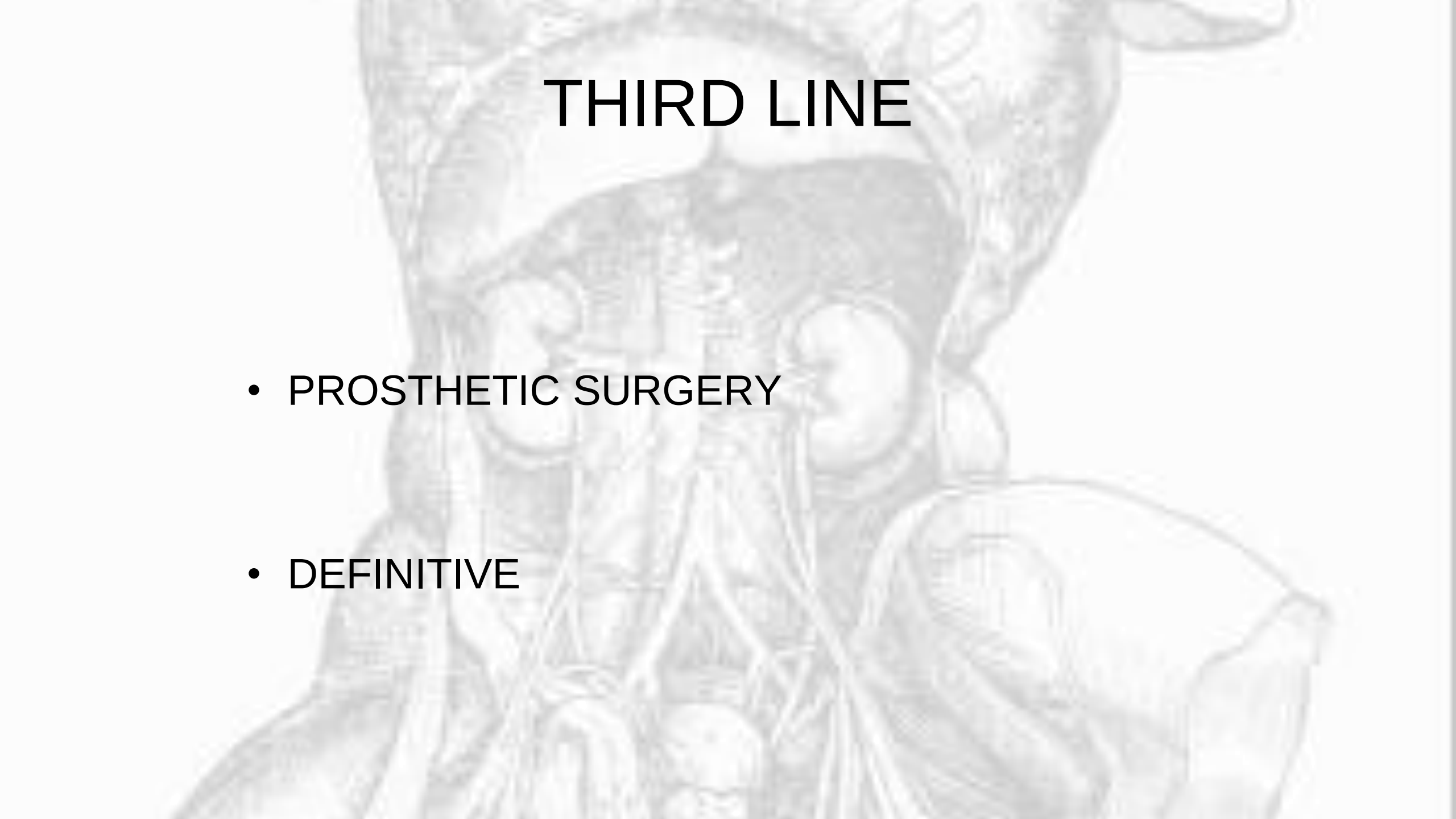
1935 **LOWSLEY**
 ■ **PLICATURATION BULBO-ISCHIO CAVERNOSUS**

1936 **BOGORAS RIB CARTILAGUE**



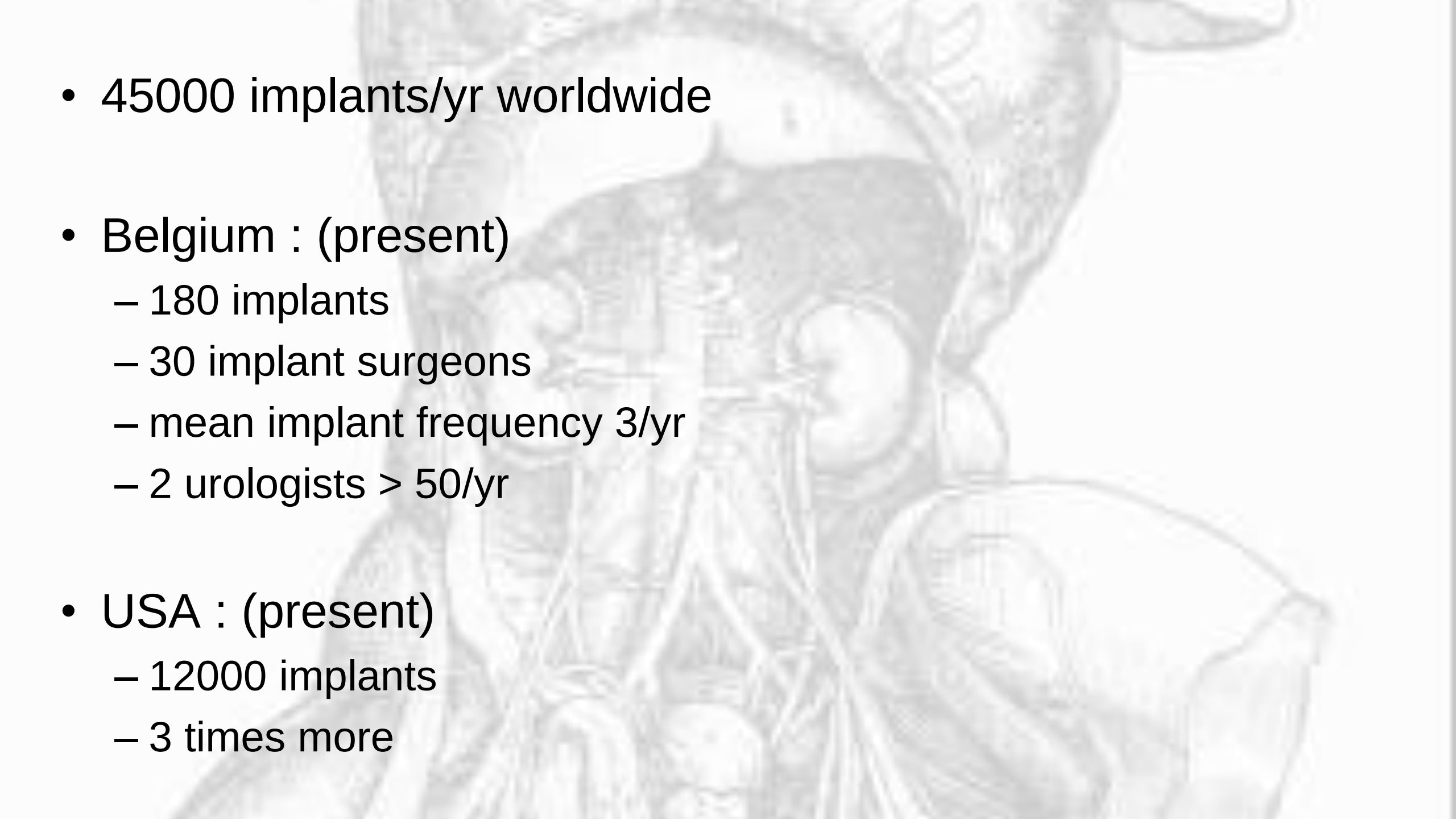
1950	SCARDINO	FIRST SYNTHET. IMPLANT
1952	GOODWIN / SCOTT	5 PAT.
1960		INTRACAVERNEOUS IMPLANT
1968	PEARLMAN	SILICONE
1970		MULTICOMPONENT PROTHESIS
1973	SCOTT – BRADLEY – TIMM	
	■	FIRST INFLAT. PROTH
	■	TWO PUMPS
1974	AMS	3 PIECE
	700	
1980	JONAS	SEMI-RIGID
1983	AMS	REAR TIP EXTENDER
	COLOPL	POLY-URETHANE





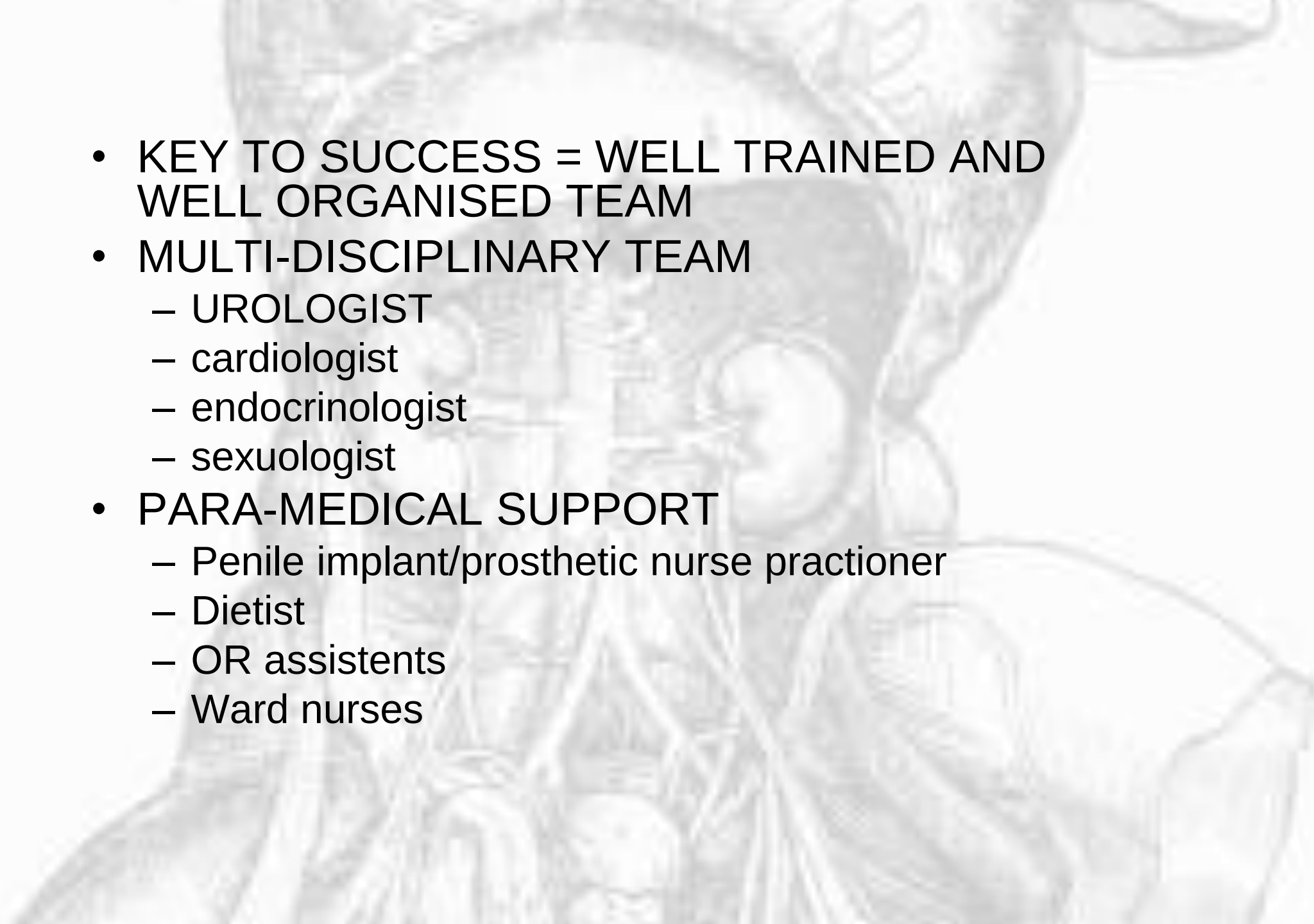
THIRD LINE

- PROSTHETIC SURGERY
- DEFINITIVE

- 
- 45000 implants/yr worldwide
 - Belgium : (present)
 - 180 implants
 - 30 implant surgeons
 - mean implant frequency 3/yr
 - 2 urologists > 50/yr
 - USA : (present)
 - 12000 implants
 - 3 times more

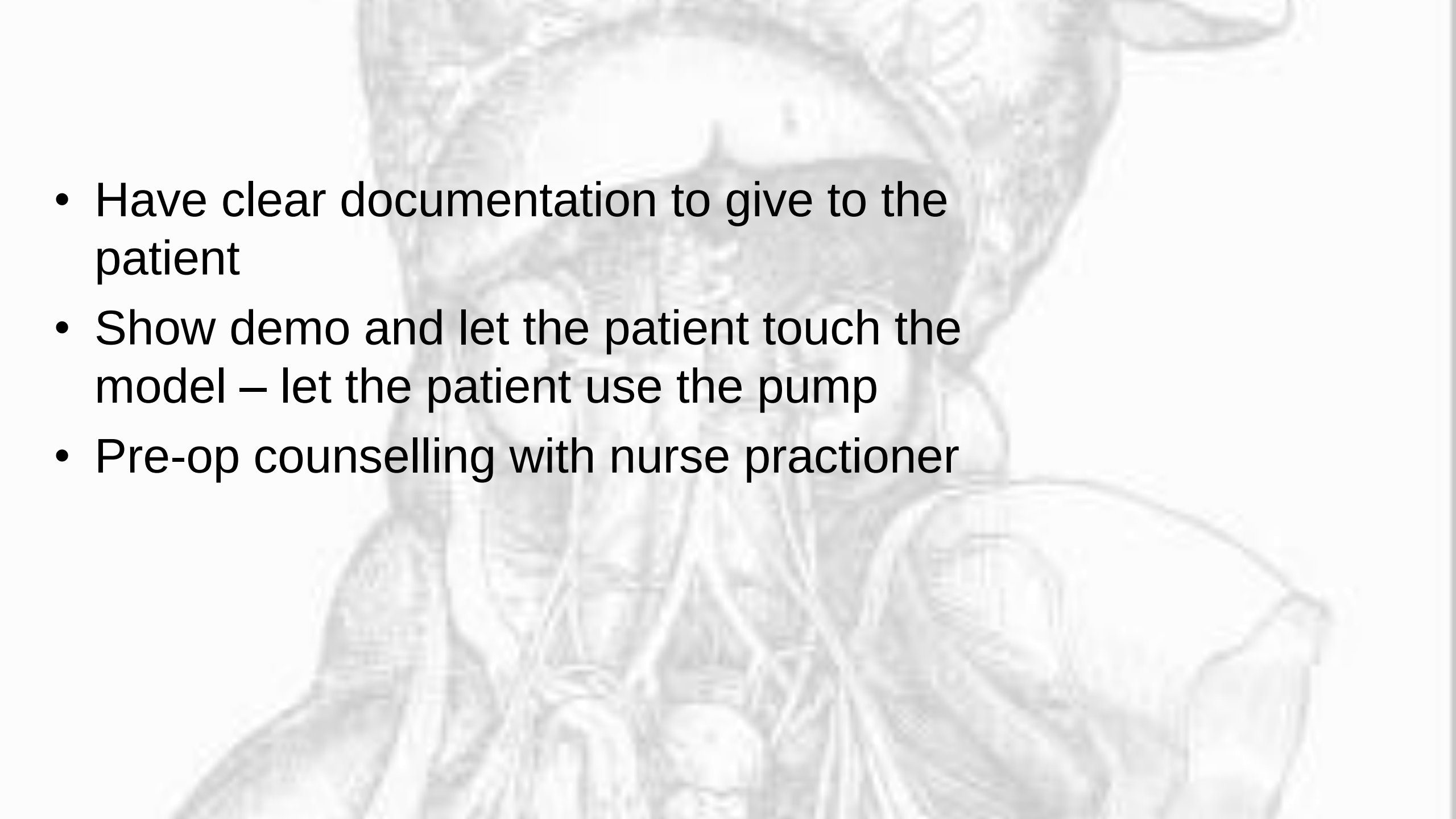
EAU GUIDELINES

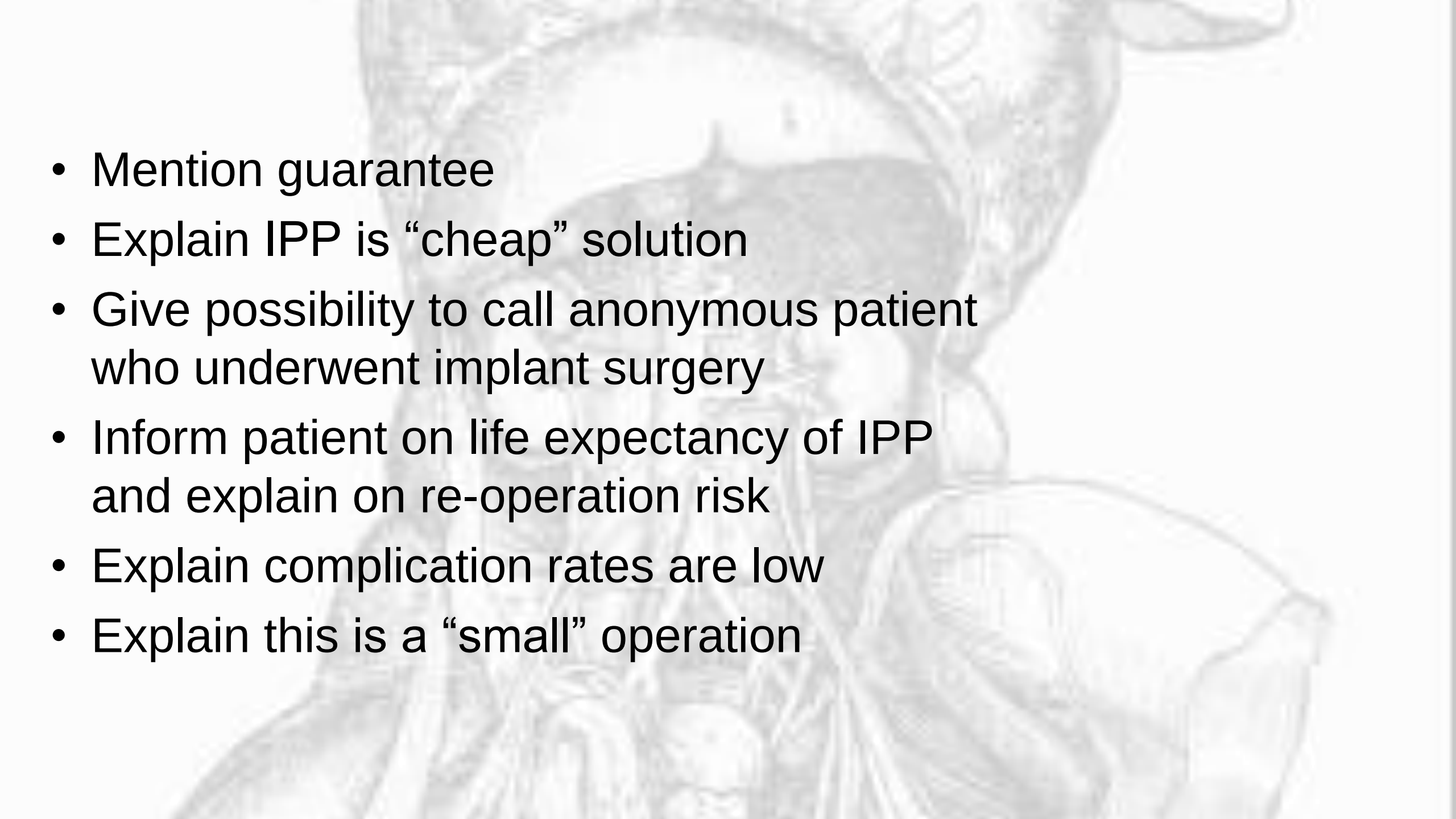
- Penile implant indicated in patients who fail pharmacotherapy or who want a permanent solution
- Most patients will prefer 3-piece IPP because erections are more natural
- Satisfaction is high if APPROPRIATE consultation

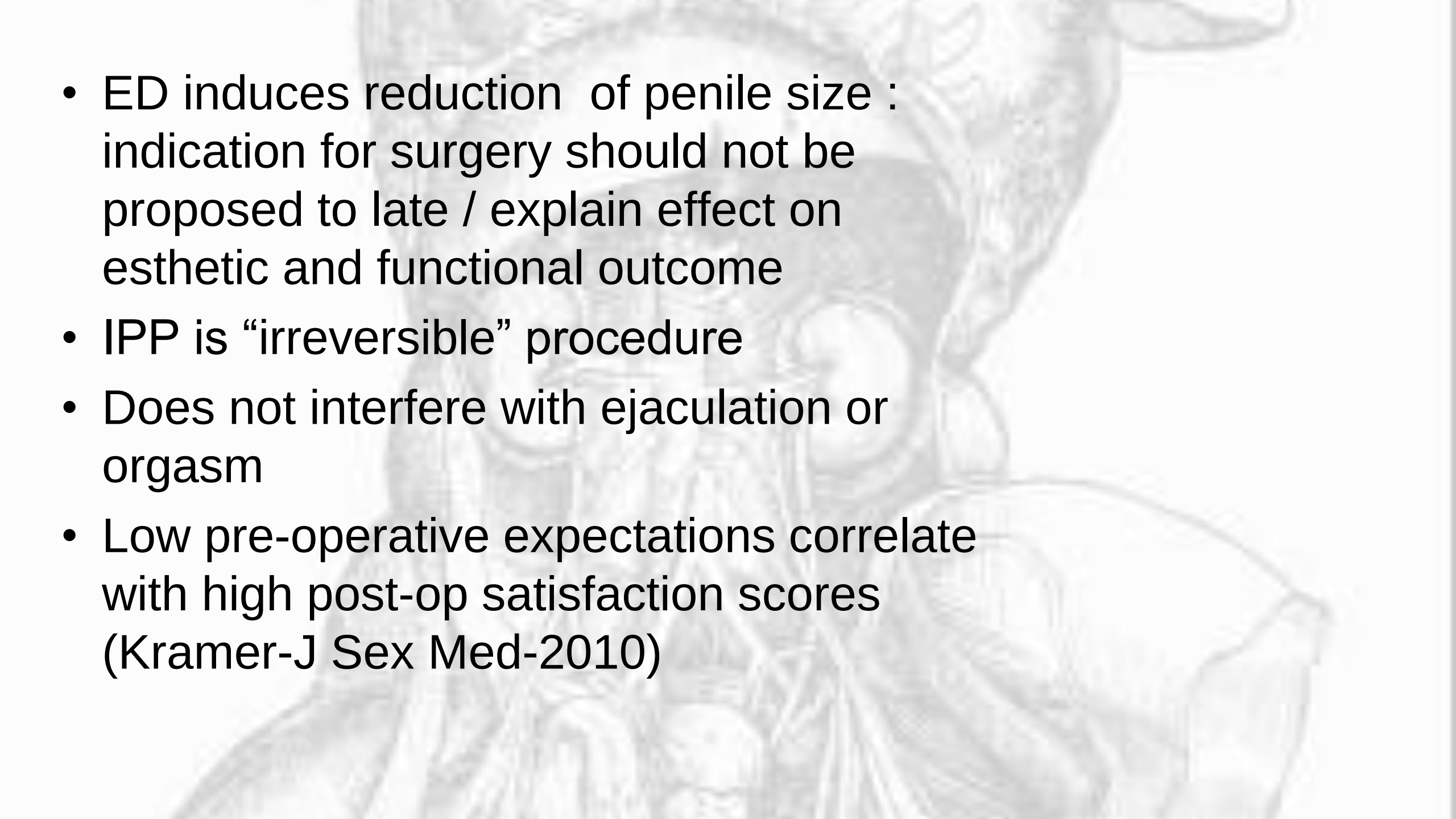
- 
- KEY TO SUCCESS = WELL TRAINED AND WELL ORGANISED TEAM
 - MULTI-DISCIPLINARY TEAM
 - UROLOGIST
 - cardiologist
 - endocrinologist
 - sexuologist
 - PARA-MEDICAL SUPPORT
 - Penile implant/prosthetic nurse practitioner
 - Dietist
 - OR assistants
 - Ward nurses

STANDARDISED WORK UP

- Integration of the couple !!!
- IIEF (SHIM)- SEP
- Hormonal evaluation
- DOPPLER
- Nocturnal penile tumescence measurements

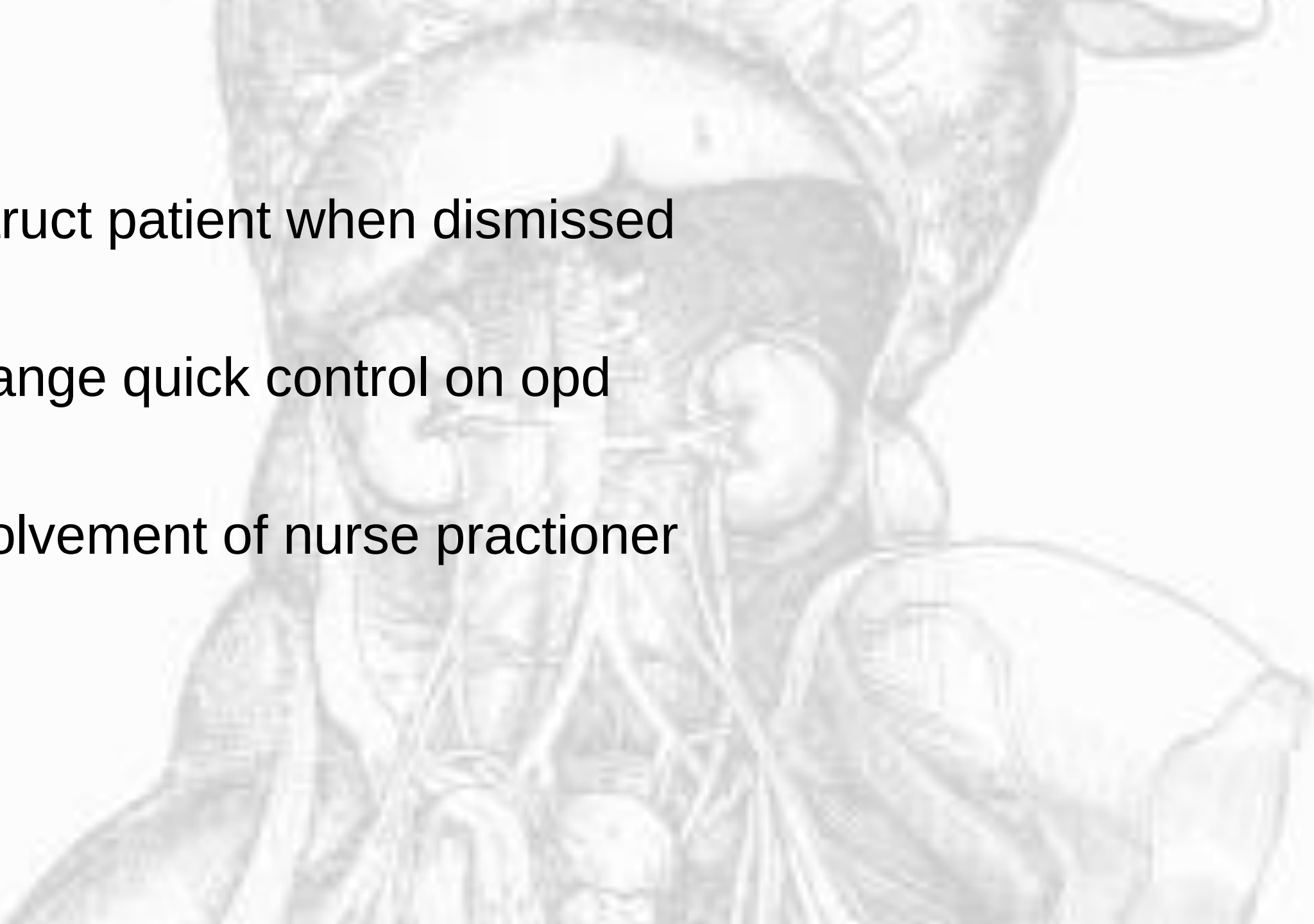
- 
- Have clear documentation to give to the patient
 - Show demo and let the patient touch the model – let the patient use the pump
 - Pre-op counselling with nurse practitioner

- 
- Mention guarantee
 - Explain IPP is “cheap” solution
 - Give possibility to call anonymous patient who underwent implant surgery
 - Inform patient on life expectancy of IPP and explain on re-operation risk
 - Explain complication rates are low
 - Explain this is a “small” operation

- 
- ED induces reduction of penile size : indication for surgery should not be proposed to late / explain effect on esthetic and functional outcome
 - IPP is “irreversible” procedure
 - Does not interfere with ejaculation or orgasm
 - Low pre-operative expectations correlate with high post-op satisfaction scores (Kramer-J Sex Med-2010)

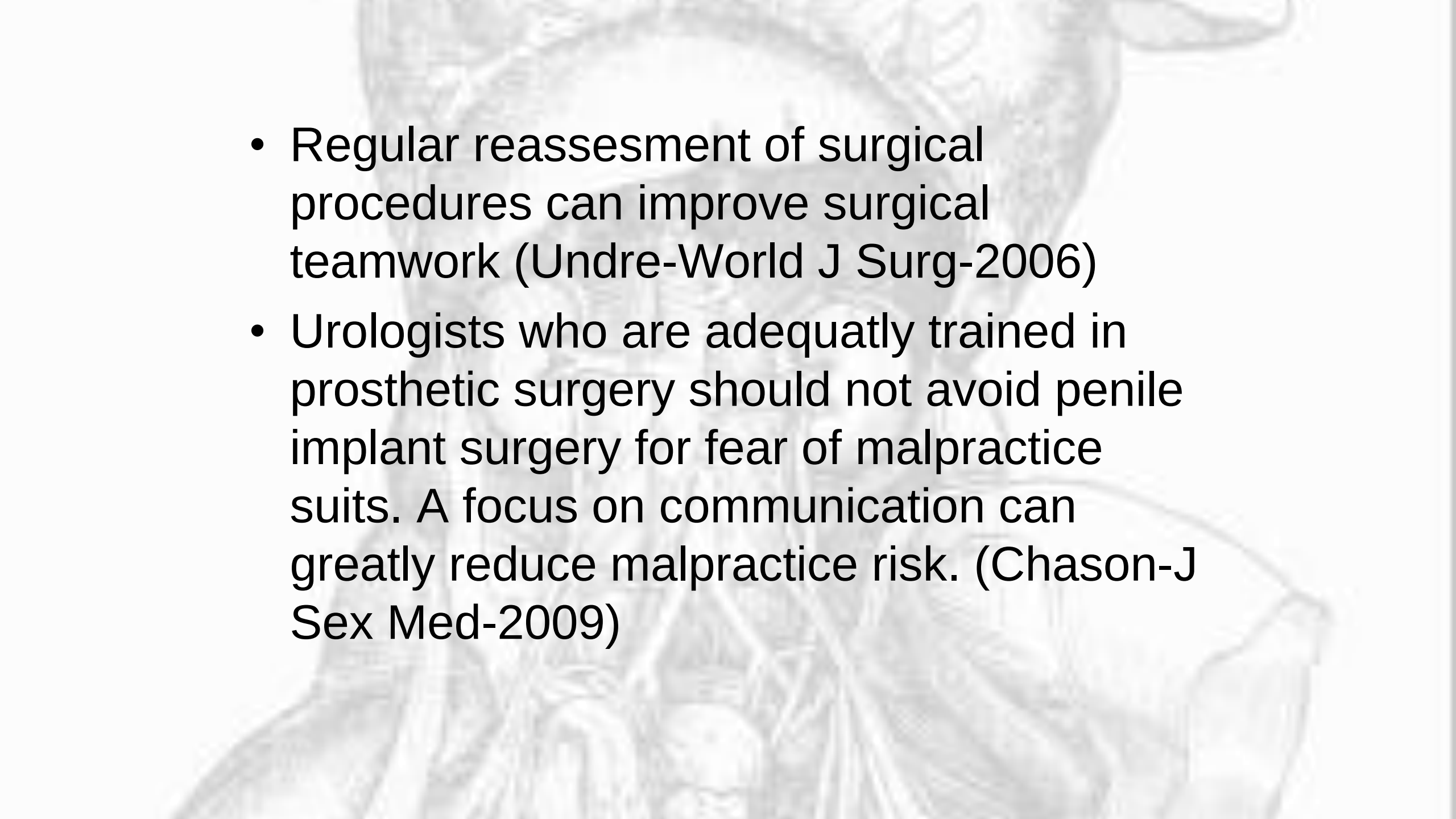
STANDARDISED PERI-OPERATIVE PROCEDURE

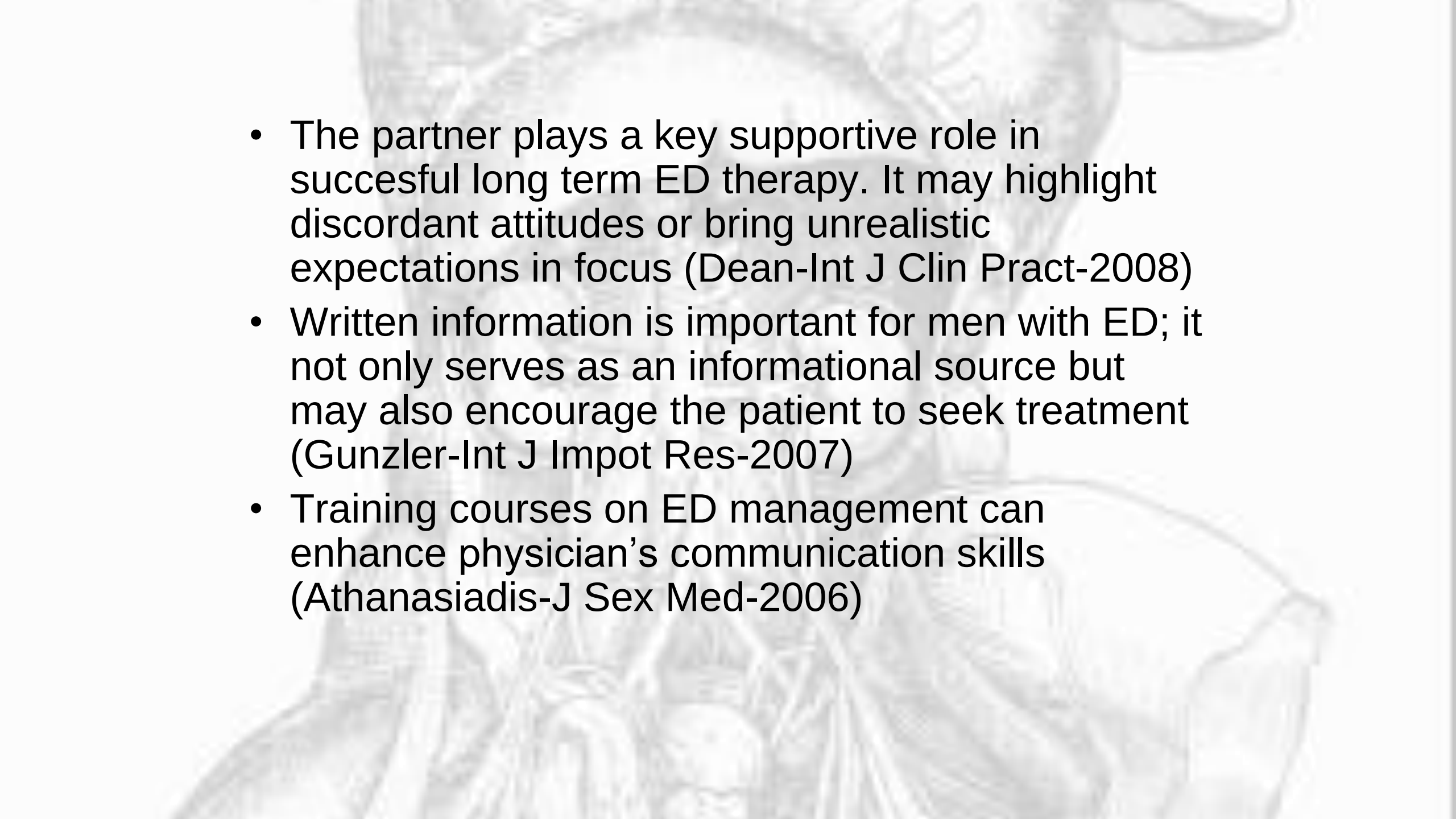
- Create “fast track” : try to improve this multi-modal intervention
 - Pre-operative : well instructed and well experienced nurses on ward (isobetadine shower-shaving-antibiotics-etc.)
 - Well trained surgical team (assemble prosthesis)
 - Clear post-op instructions on ward (TUC-ice-bandage-deflation-etc.)

- 
- Instruct patient when dismissed
 - Arrange quick control on opd
 - Involvement of nurse practitioner

LITTERATURE

- Superior outcome correlates with high volume implanters (Lotan-Urology-2003)
- Yearly sessions with a recognised expert can improve surgical outcome (King-J Sex Med-2011)
- Devellopping a fast track improves surgical program outcome (Kehlet-Am J Surg-2002)

- 
- Regular reassessment of surgical procedures can improve surgical teamwork (Undre-World J Surg-2006)
 - Urologists who are adequately trained in prosthetic surgery should not avoid penile implant surgery for fear of malpractice suits. A focus on communication can greatly reduce malpractice risk. (Chason-J Sex Med-2009)

- 
- The partner plays a key supportive role in successful long term ED therapy. It may highlight discordant attitudes or bring unrealistic expectations in focus (Dean-Int J Clin Pract-2008)
 - Written information is important for men with ED; it not only serves as an informational source but may also encourage the patient to seek treatment (Gunzler-Int J Impot Res-2007)
 - Training courses on ED management can enhance physician's communication skills (Athanasiadis-J Sex Med-2006)

WORDS OF WISDOM

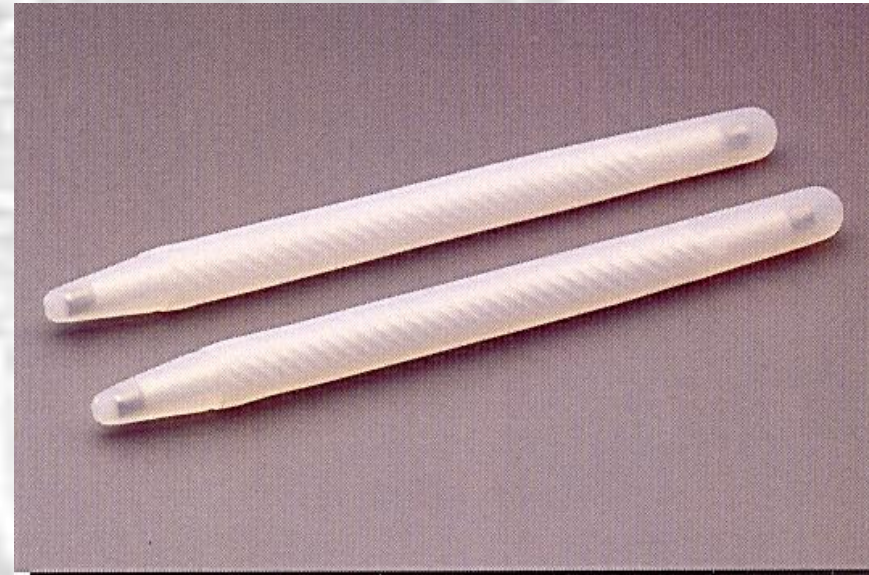
- PRIMUM NON NOCERE
- THE BEST IS THE ENEMY OF GOOD
- IN RESTRICTION THE MASTER SHOWS FIRST
- MENTAL CONDITION SHOULD NOT BE TREATED SURGICALLY

PROTHESIS TYPES



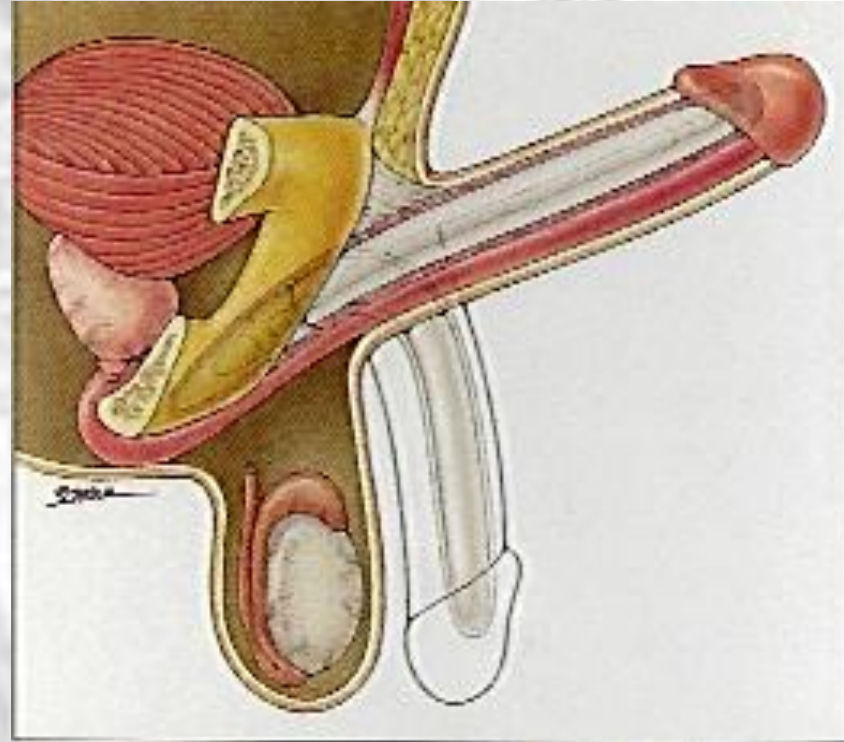
RIGID PROSTHESIS

- CONSTANT RIGIDITY
- SIMPLE PROCEDURE
- RARE TECHICAL PROBLEMS
- **CON**
 - ESTHETIC
 - EROSION



MALLEABELE PROTHESIS

- AMS
- COLOPLAST-
PORGES



INFLATABLE PROTHESIS



"Kiss me again — I think it's working!"



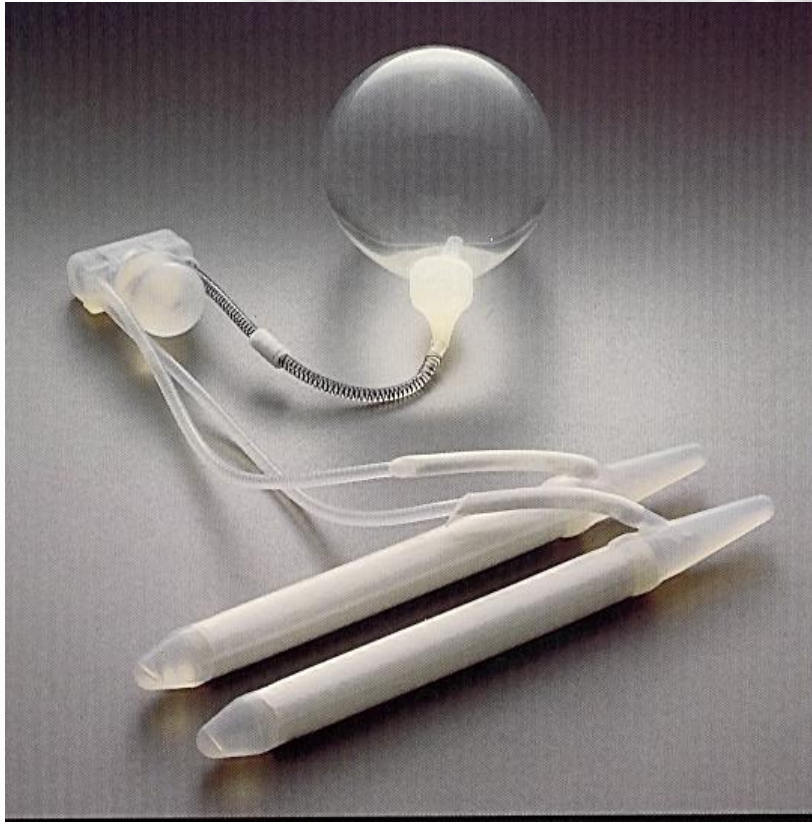
BE AFRAID BE VERY AFRAID

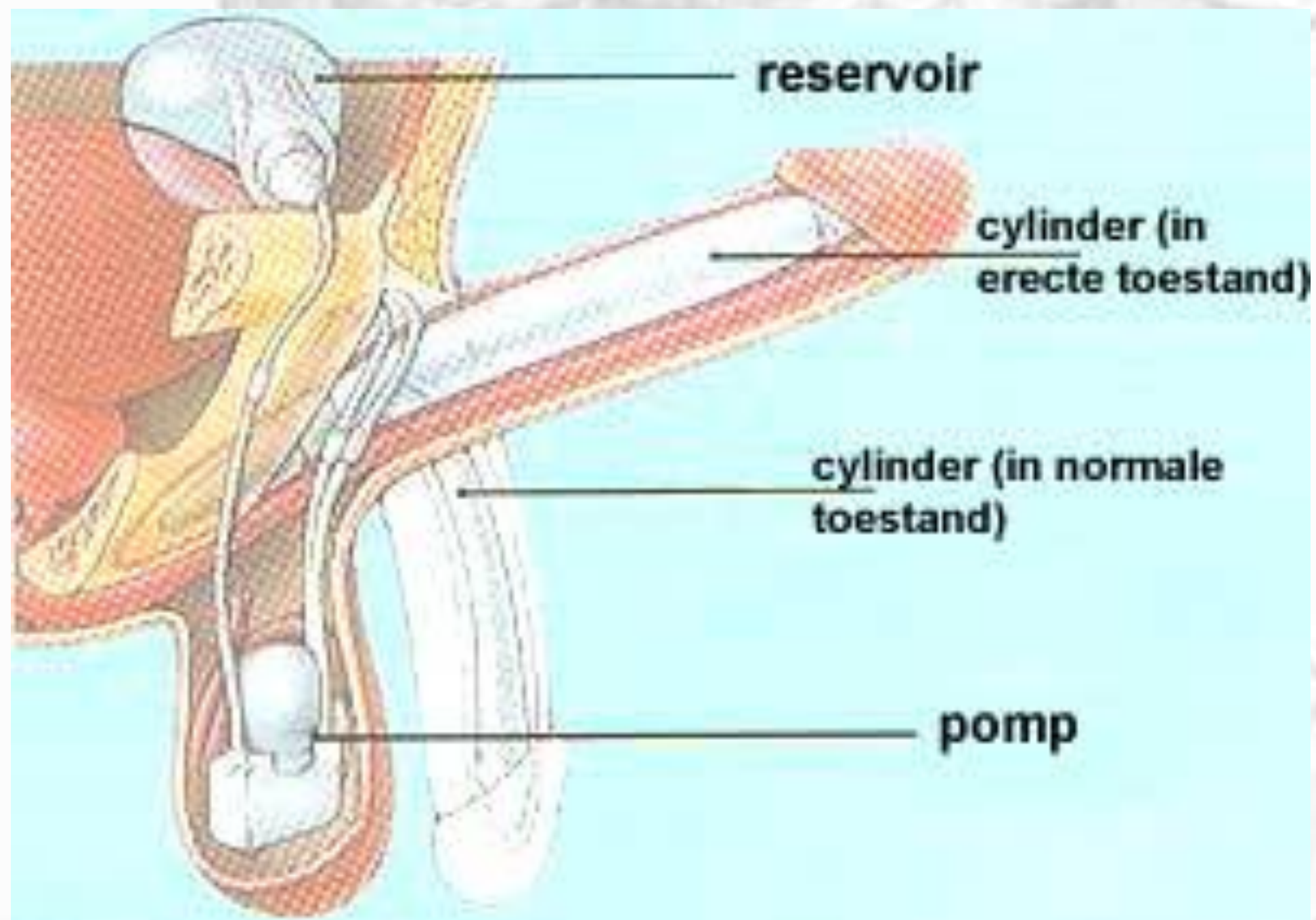
Use of IPP

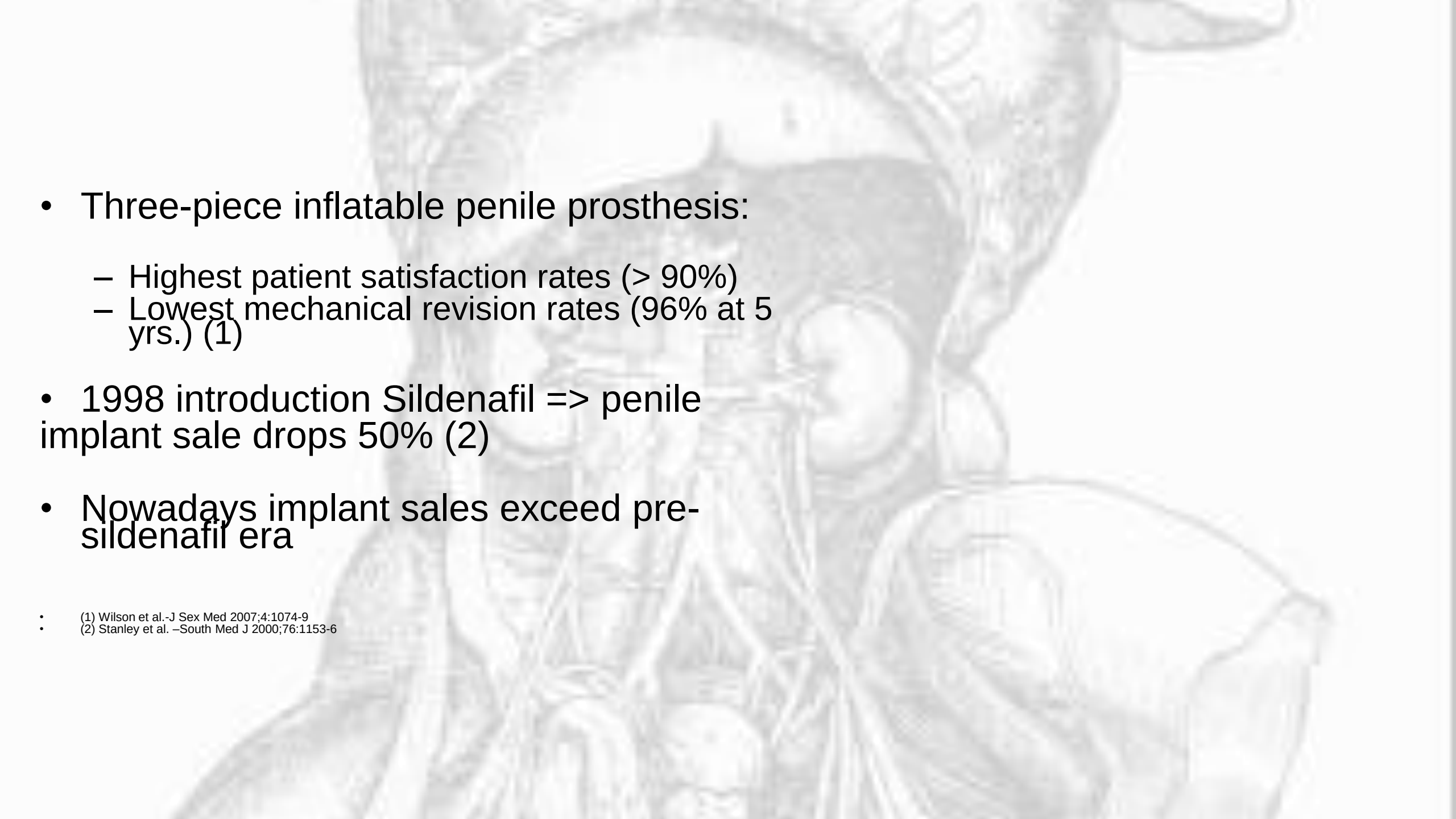
(Movie courtesy of Coloplast, publicly accesible @ youtube.com)



THREE PIECE INFLAT. PROSTH.



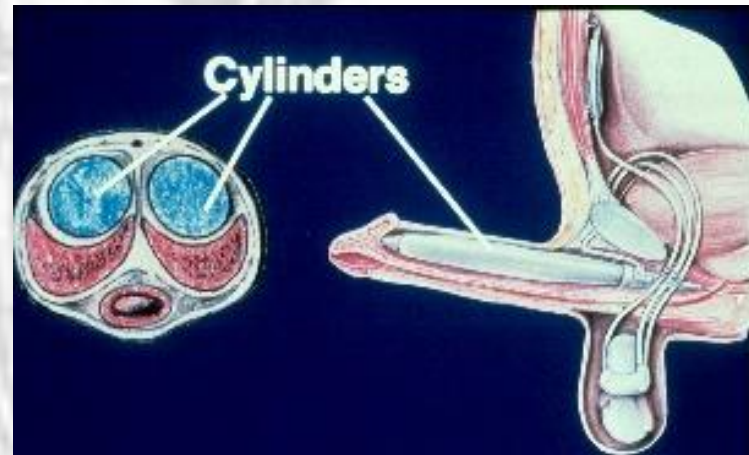


- 
- Three-piece inflatable penile prosthesis:
 - Highest patient satisfaction rates (> 90%)
 - Lowest mechanical revision rates (96% at 5 yrs.) (1)
 - 1998 introduction Sildenafil => penile implant sale drops 50% (2)
 - Nowadays implant sales exceed pre-sildenafil era

• (1) Wilson et al.-J Sex Med 2007;4:1074-9
• (2) Stanley et al. -South Med J 2000;76:1153-6

THREE PIECE INFLAT. PROTH.

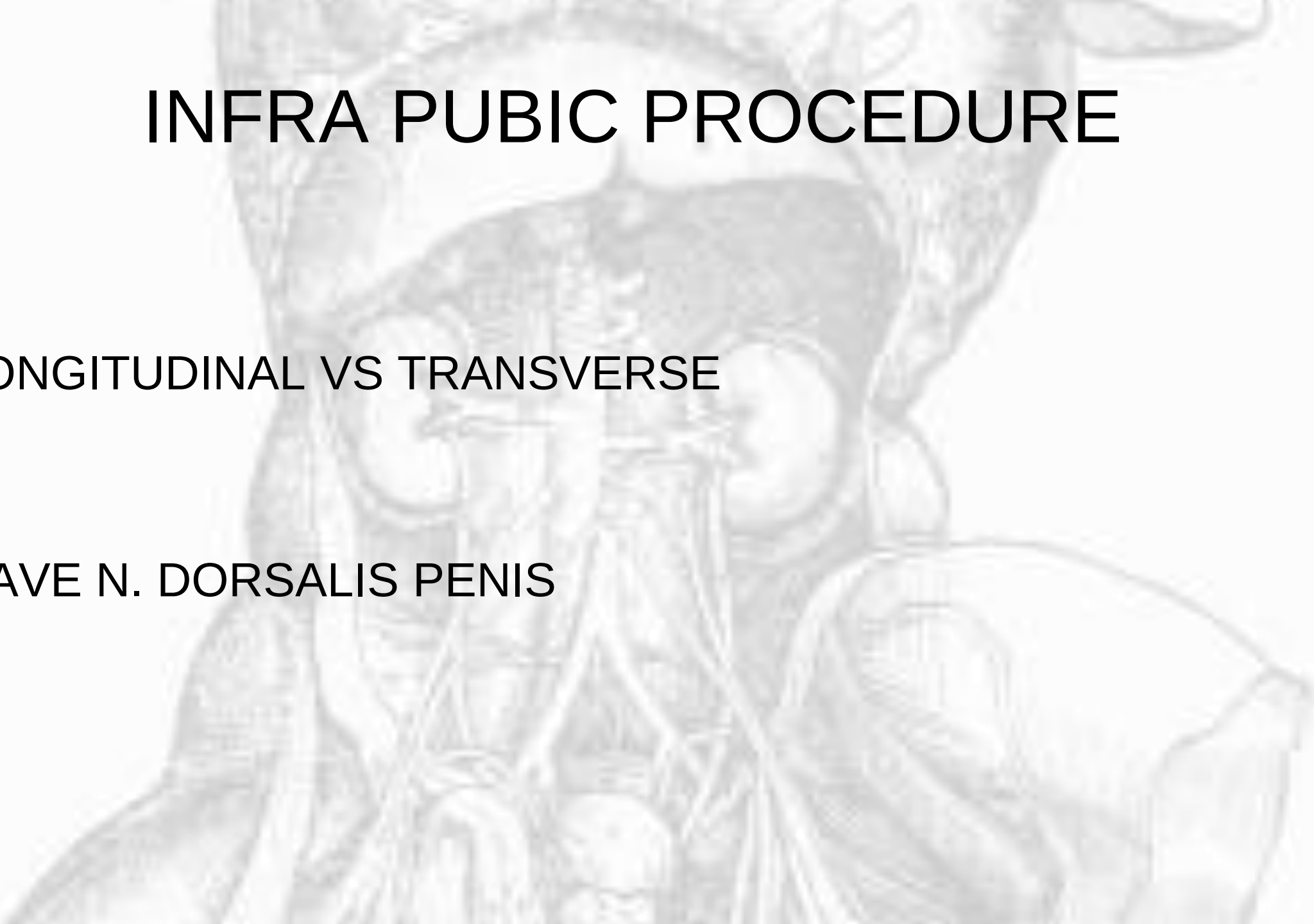
- CYLINDERS + PUMP + RESERVOIR
- AMS
 - 700 CX-CXR
 - ULTREX (EXPANSION)
 - LGX
- COLOPLAST-PORGES
 - TITAN
 - TITAN NB



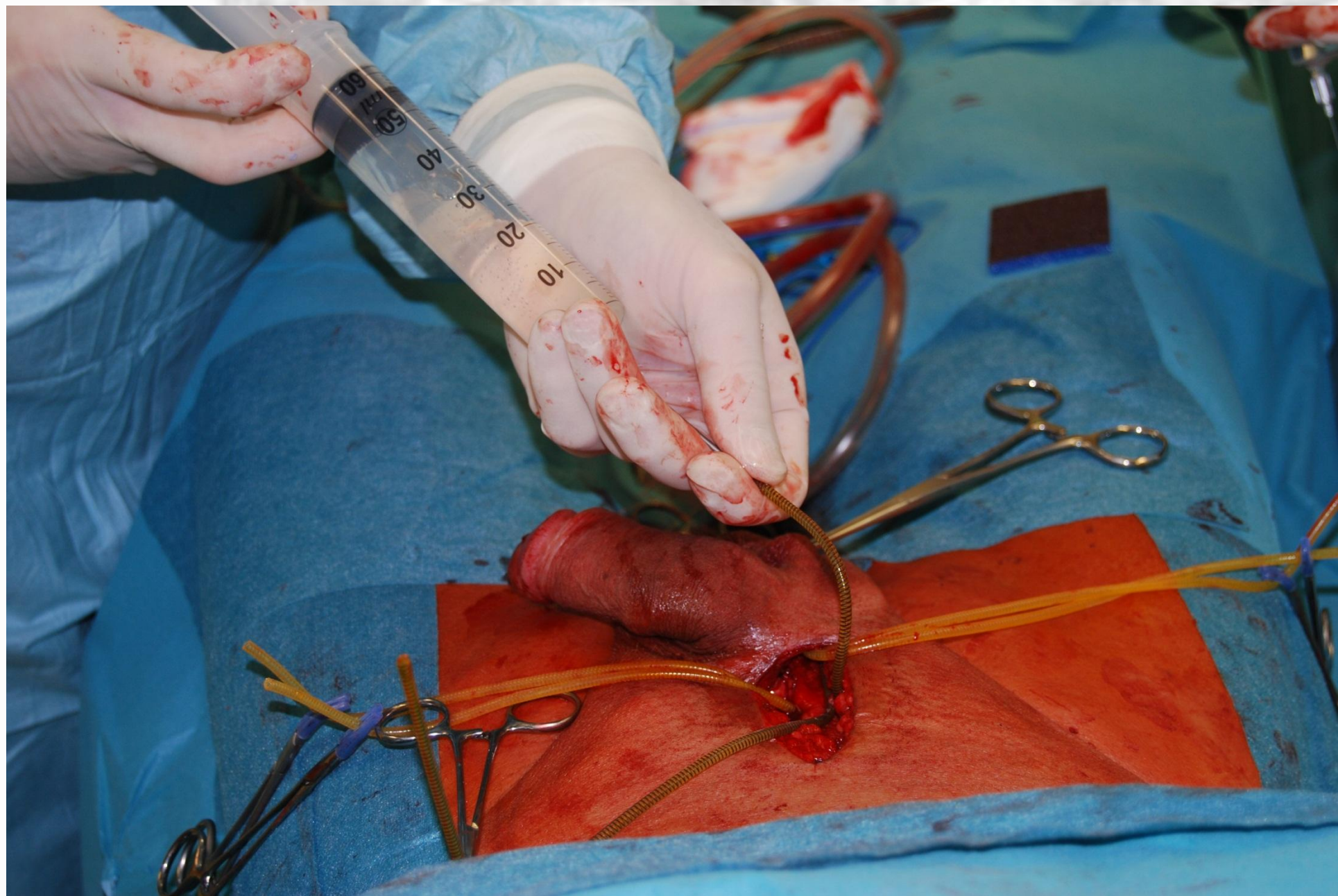
PRE-OPERATIVE

- LAB URINE ECG (THORAX)
- ISOBETADINE BADING
- FIRST PATIENT
- SHAVE PRE-OP
- SCRUB UNDER NARCOSIS
- BROAD SPECTRUM AB

INFRA PUBIC PROCEDURE

An anatomical illustration of the male pelvic region, showing the prostate gland, ureters, and vas deferens. The illustration is in a light gray tone, serving as a background for the text.

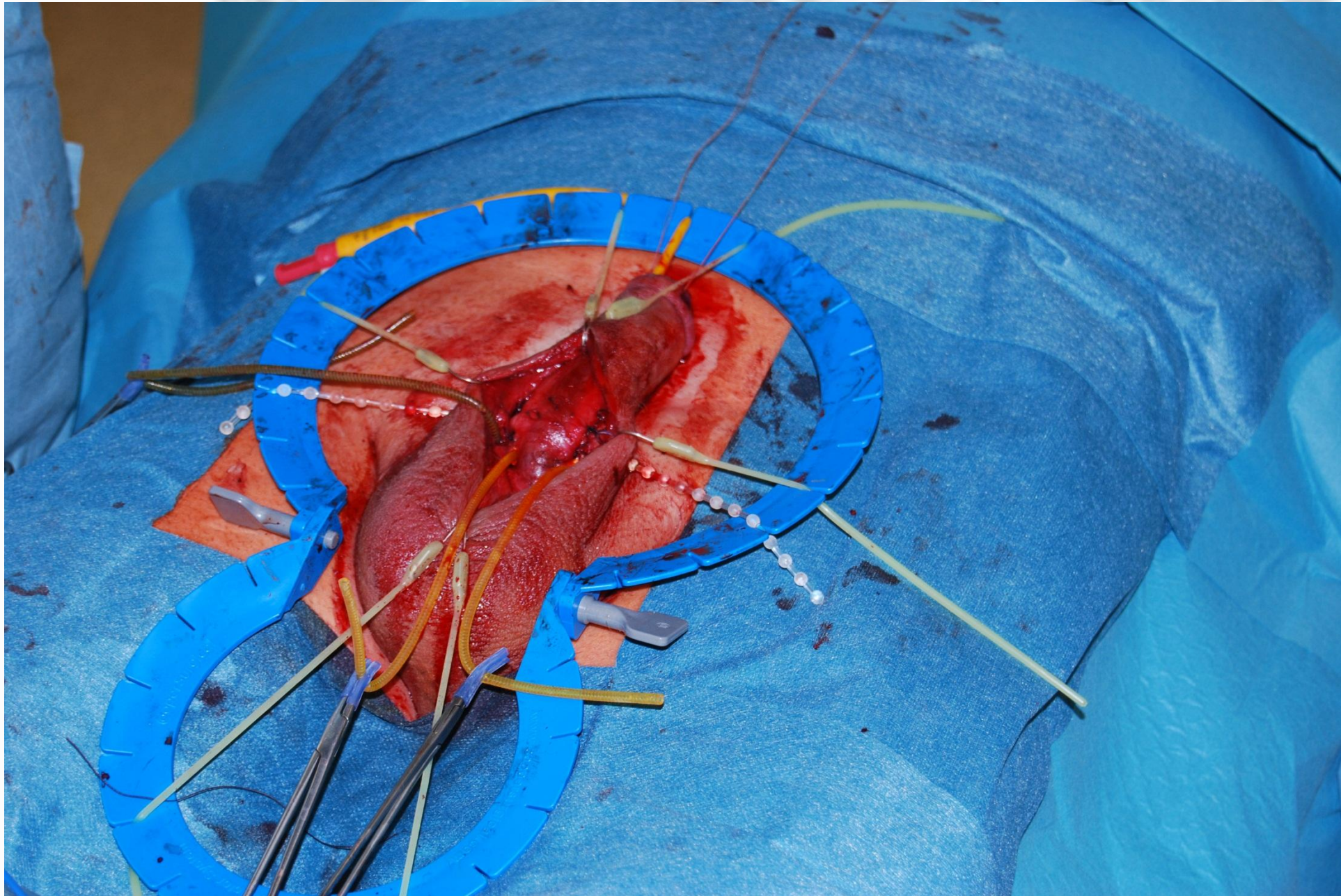
- LONGITUDINAL VS TRANSVERSE
- CAVE N. DORSALIS PENIS



PENOSCROTAL PROCEDURE

- GOOD EXPOSURE
CORPORA
- OBESE PATIENTS





POST-OP

- SEX AFTER 6 WEEKS
- ACTIVATE ASAP

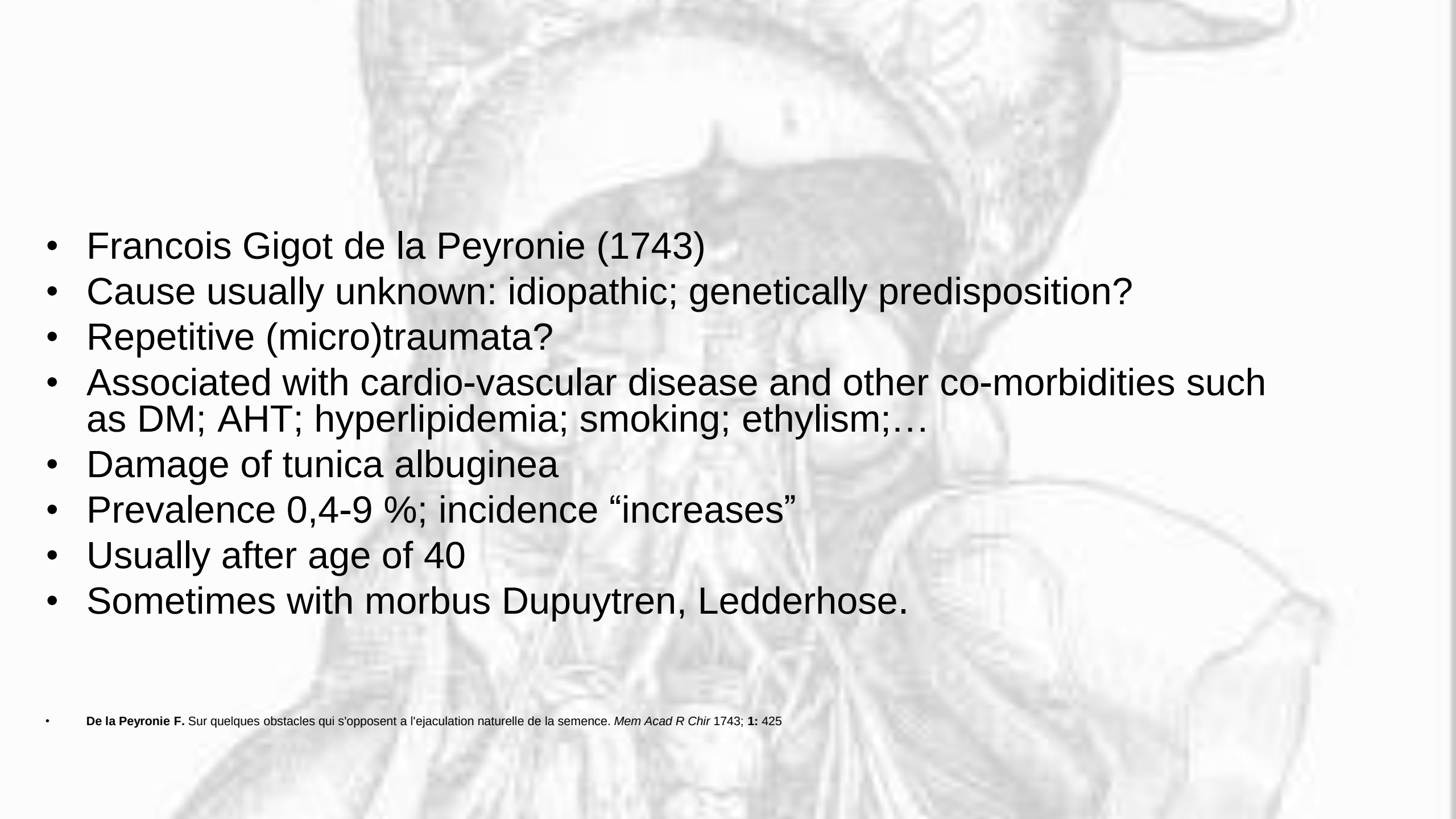


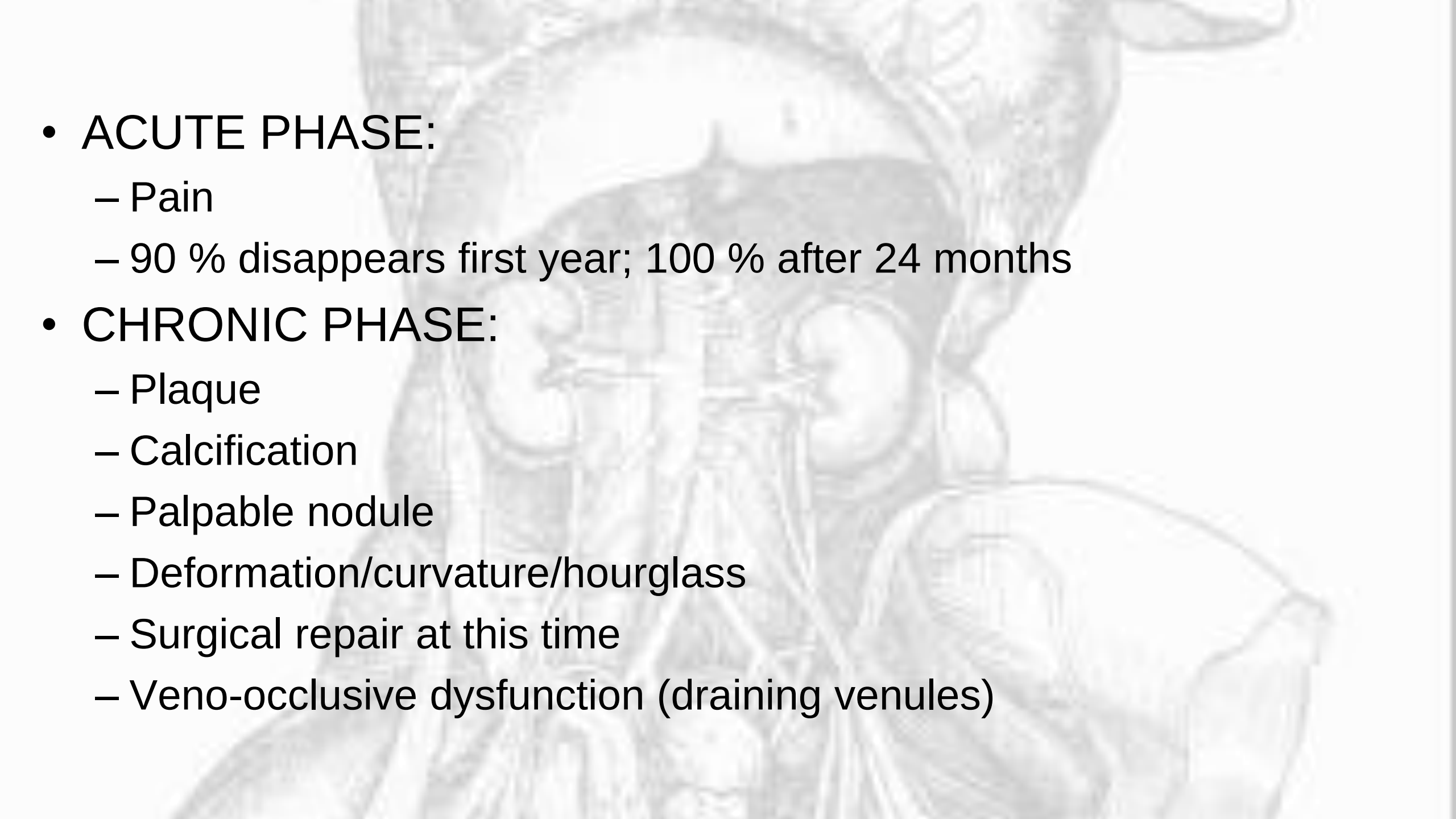
INFECTION

- PSEUDOCAPSULE
- EXPLANTATION
- LOW GRADE LATE COMPLICATION
- PRIMAIR 2%
- SECUND. 13%
- DIABETES
- SALVAGE PROCEDURE

An anatomical illustration of the male reproductive system, specifically the penis and testicles, rendered in a light gray, semi-transparent style. The illustration shows the internal structures, including the urethra, vas deferens, and the corpus cavernosum. The text 'ZIEKTE VAN PEYRONIE' is overlaid in the center. The background is white.

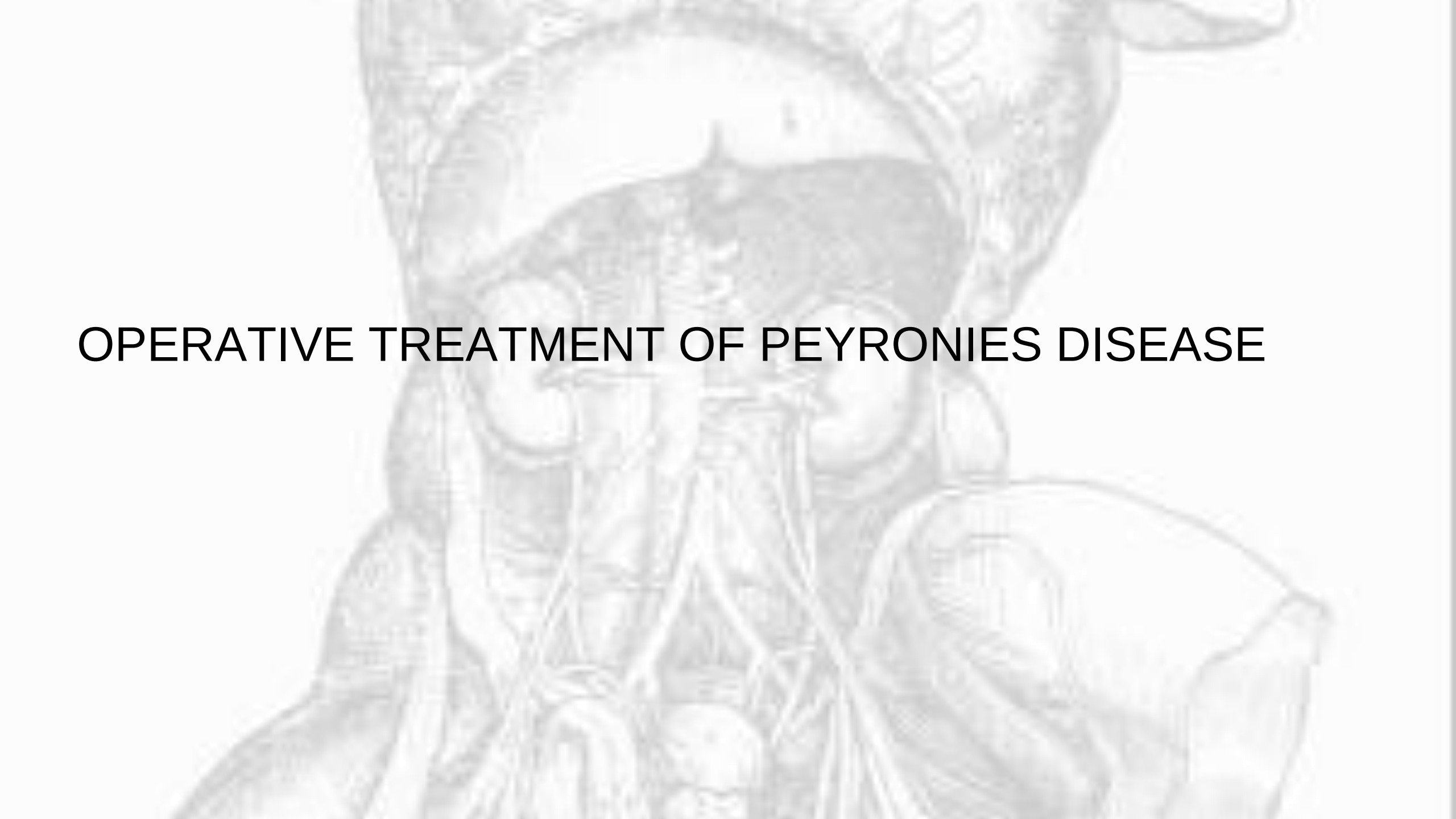
ZIEKTE VAN PEYRONIE

- 
- Francois Gigot de la Peyronie (1743)
 - Cause usually unknown: idiopathic; genetically predisposition?
 - Repetitive (micro)traumata?
 - Associated with cardio-vascular disease and other co-morbidities such as DM; AHT; hyperlipidemia; smoking; ethylism;...
 - Damage of tunica albuginea
 - Prevalence 0,4-9 %; incidence “increases”
 - Usually after age of 40
 - Sometimes with morbus Dupuytren, Ledderhose.

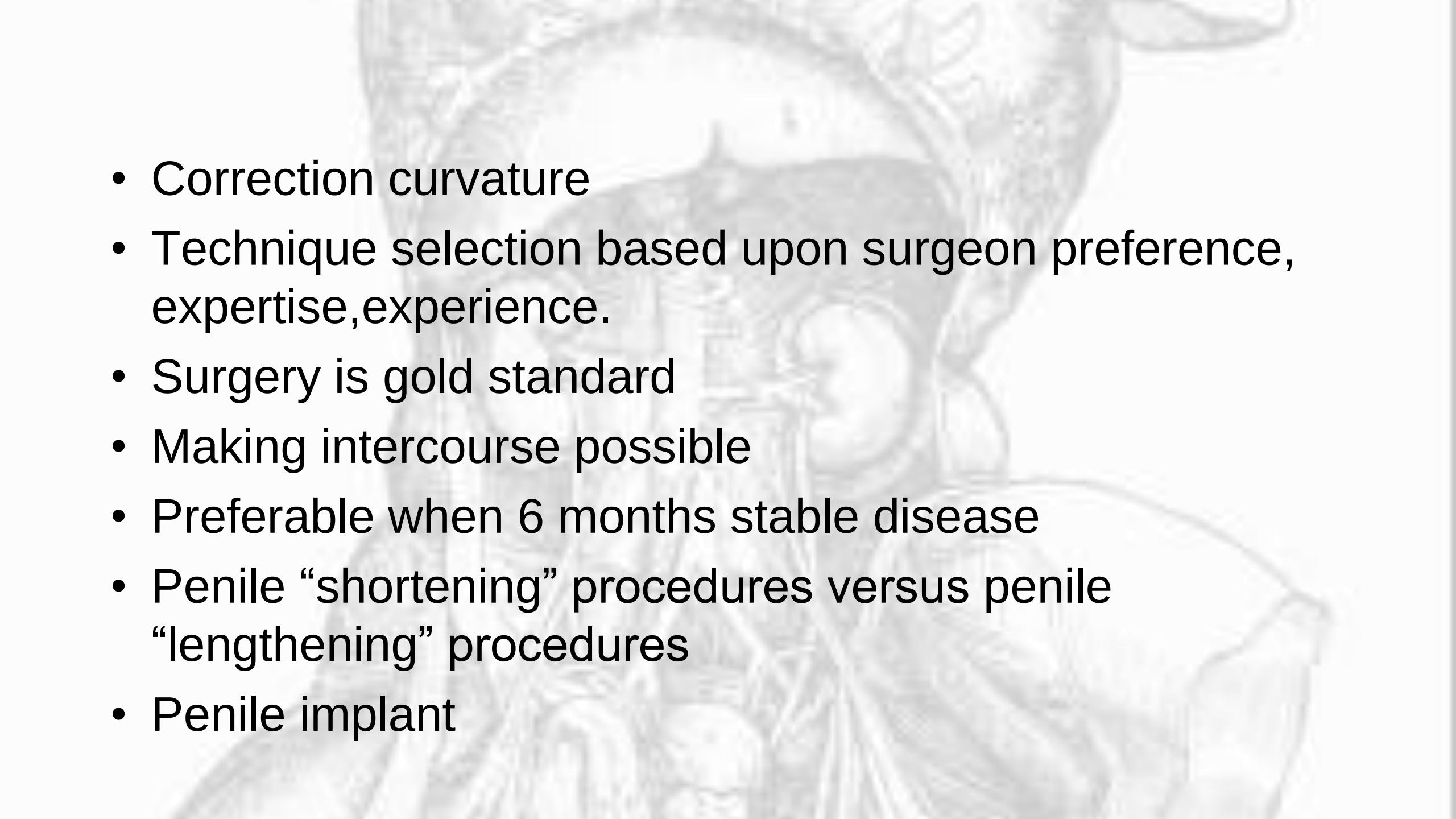
- 
- **ACUTE PHASE:**
 - Pain
 - 90 % disappears first year; 100 % after 24 months
 - **CHRONIC PHASE:**
 - Plaque
 - Calcification
 - Palpable nodule
 - Deformation/curvature/hourglass
 - Surgical repair at this time
 - Veno-occlusive dysfunction (draining venules)

Non-operative treatment

- **Conservative** treatment of Peyronie's disease is primarily focused on patients in the **early stages** of disease.
- Several options have been suggested, including oral pharmacotherapy (vitamin E, potassium para-aminobenzoate, tamoxifen, colchicine, acetyl esters of carnitine, pentoxifylline), intra-lesional injection therapy (steroids, verapamil, clostridial collagenase, interferon) and other topical treatments (verapamil, iontophoresis, extracorporeal shock wave therapy, traction devices, vacuum devices).

An anatomical illustration of the male reproductive system, showing the penis, testis, vas deferens, and associated structures. The illustration is in a classic, detailed style, likely from a medical textbook. The text "OPERATIVE TREATMENT OF PEYRONIES DISEASE" is overlaid on the image.

OPERATIVE TREATMENT OF PEYRONIES DISEASE

- 
- Correction curvature
 - Technique selection based upon surgeon preference, expertise, experience.
 - Surgery is gold standard
 - Making intercourse possible
 - Preferable when 6 months stable disease
 - Penile “shortening” procedures versus penile “lengthening” procedures
 - Penile implant

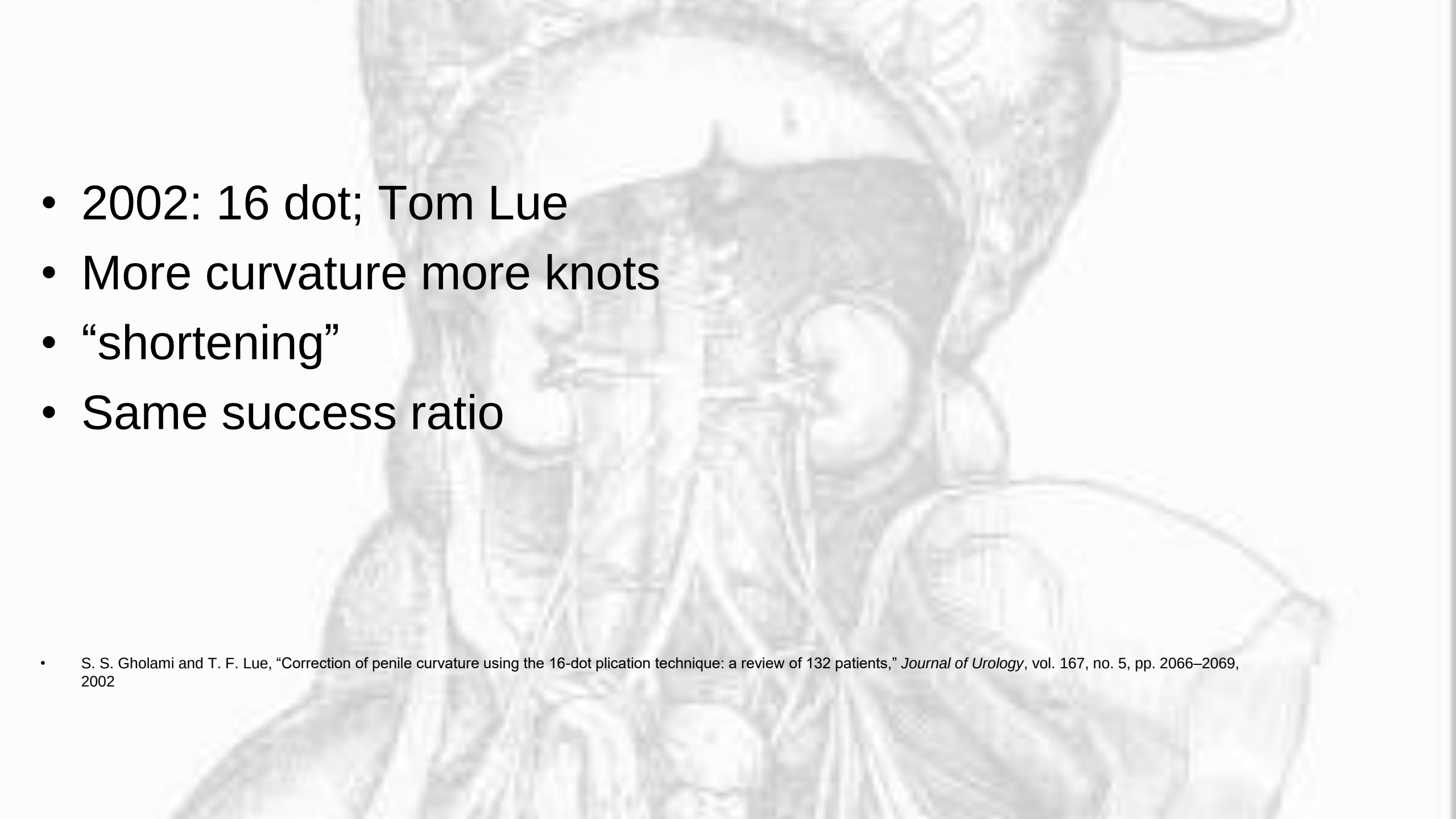
- PENILE LENGTHENING:

- Graft (autologous – v. saphena; tutoplast - bovine; SIS - porcine; etc.)
- Cave veno-occlusive dysfunction
- Shrinking of graft (Kadioglu)
- Nerve damage (neuro-vascular bundle) with diminished glandular sensation

- T. F. Lue, "Venous patch graft for peyronie's disease. Part I: technique," *Journal of Urology*, vol. 160, no. 6 I, pp. 2047–2049, 1998
- A. Kadioglu, O. Sanli, T. Akman, M. Cakan, B. Erol, and F. Mamadov, "Surgical treatment of Peyronie's disease: a single center experience with 145 patients," *European Urology*, vol. 53, no. 2, pp. 432–440, 2008
- E. C. Serefoglu and W. J. G. Hellstrom, "Treatment of Peyronie's disease: 2012 update," *Current Urology Reports*, vol. 12, pp. 444–452, 2011
- F. Montorsi, A. Salonia, T. Maga et al., "Evidence based assessment of long-term results of plaque incision and vein grafting for Peyronie's disease," *Journal of Urology*, vol. 163, no. 6, pp. 1704–1708, 2000

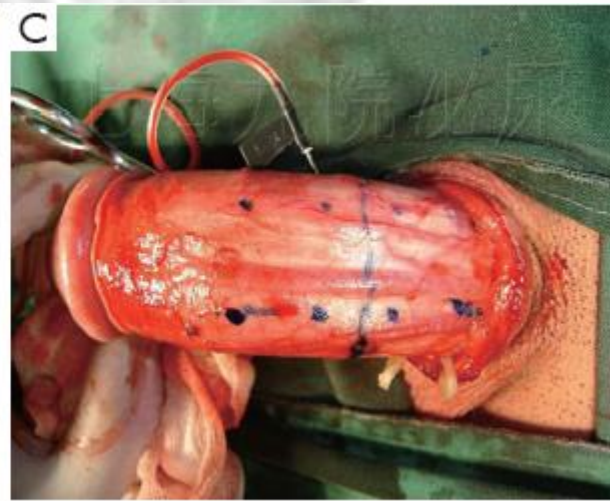
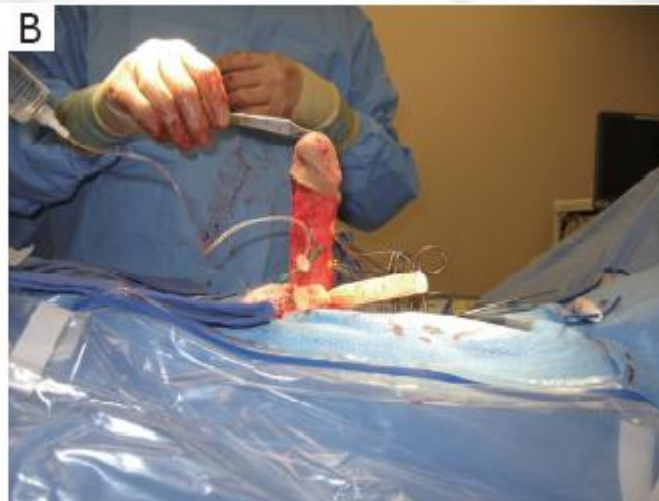
- **PENILE SHORTENING:**
 - Nesbitt procedure (1965)
 - Initially contra lateral elliptical excision (5-10 mm; transverse; closure)
 - Modified Nesbitt: plicaturation; non absorbable sutures
 - Absorbable sutures ? (vicryl)
 - Circumcision
 - High satisfaction (related to pre-op counselling) (Licht / Langston / Kim)
 - Not shortening! Penis will have length of “shortest side”
 - Any reduction in penile length does not lead to difficulties in sexual intercourse (Cantoro)

- R. M. Nesbit, “Congenital curvature of the phallus: report of three cases with description of corrective operation,” *The Journal of Urology*, vol. 93, pp. 230–232, 1965
- J. Rehman, A. Benet, L. S. Minsky, and A. Melman, “Results of surgical treatment for abnormal penile curvature: Peyronie’s disease and congenital deviation by modified Nesbit plication (tunical shaving and plication),” *Journal of Urology*, vol. 157, no. 4, pp. 1288–1291, 1997
- E. Essed and F. H. Schroeder, “New surgical treatment for Peyronie disease,” *Urology*, vol. 25, no. 6, pp. 582–587, 1985
- A. Basiri, R. Sarhanghejad, S. M. Ghahestanizadeh, and M. H. Radfar, “Comparing absorbable and nonabsorbable sutures in corporeal plication for the treatment of congenital penile curvature,” *Journal of Urology*, vol. 8, no. 4, pp. 302–306, 2011
- M. R. Licht and R. W. Lewis, “Modified Nesbit procedure for the treatment of Peyronie’s disease: a comparative outcome analysis,” *Journal of Urology*, vol. 158, no. 2, pp. 460–463, 1997
- J. P. Langston and C. C. Carson, “Peyronie disease: plication or grafting,” *Urologic Clinics of North America*, vol. 38, no. 2, pp. 207–216, 2011
- D. H. Kim, T. F. Lesser, and S. R. Aboseif, “Subjective patient-reported experiences after surgery for Peyronie’s disease: corporeal plication versus plaque incision with vein graft,” *Urology*, vol. 71, no. 4, pp. 698–702, 2008
- Cantoro et al, “Penile plication for Peyronie’s disease: our results with mean follow-up of 103 months on 89 patients. *Int J Impot Res* 2014 Jul-Aug;26(4): 156-9

- 
- 2002: 16 dot; Tom Lue
 - More curvature more knots
 - “shortening”
 - Same success ratio

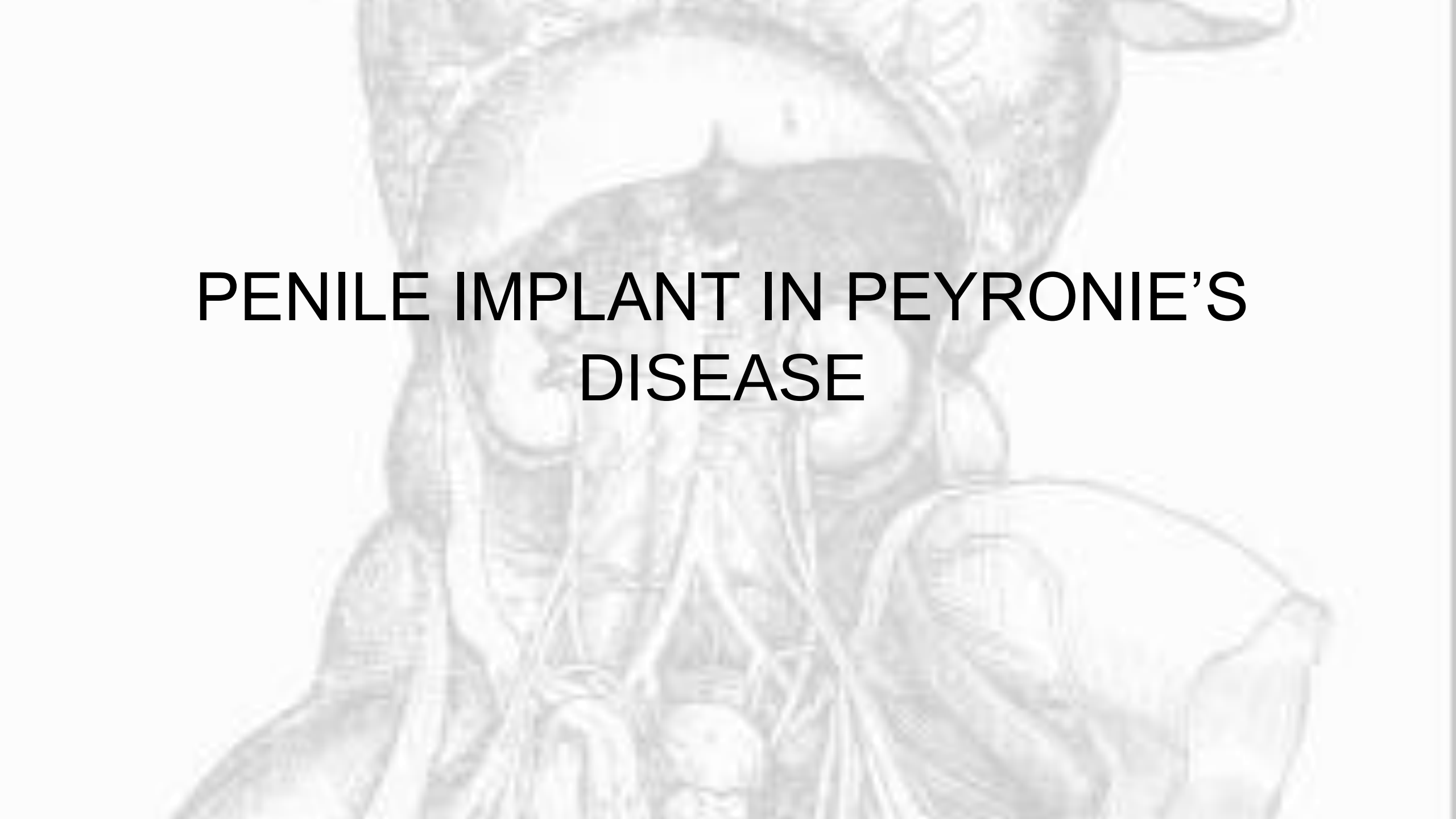
- S. S. Gholami and T. F. Lue, “Correction of penile curvature using the 16-dot plication technique: a review of 132 patients,” *Journal of Urology*, vol. 167, no. 5, pp. 2066–2069, 2002

Lue 16-dot

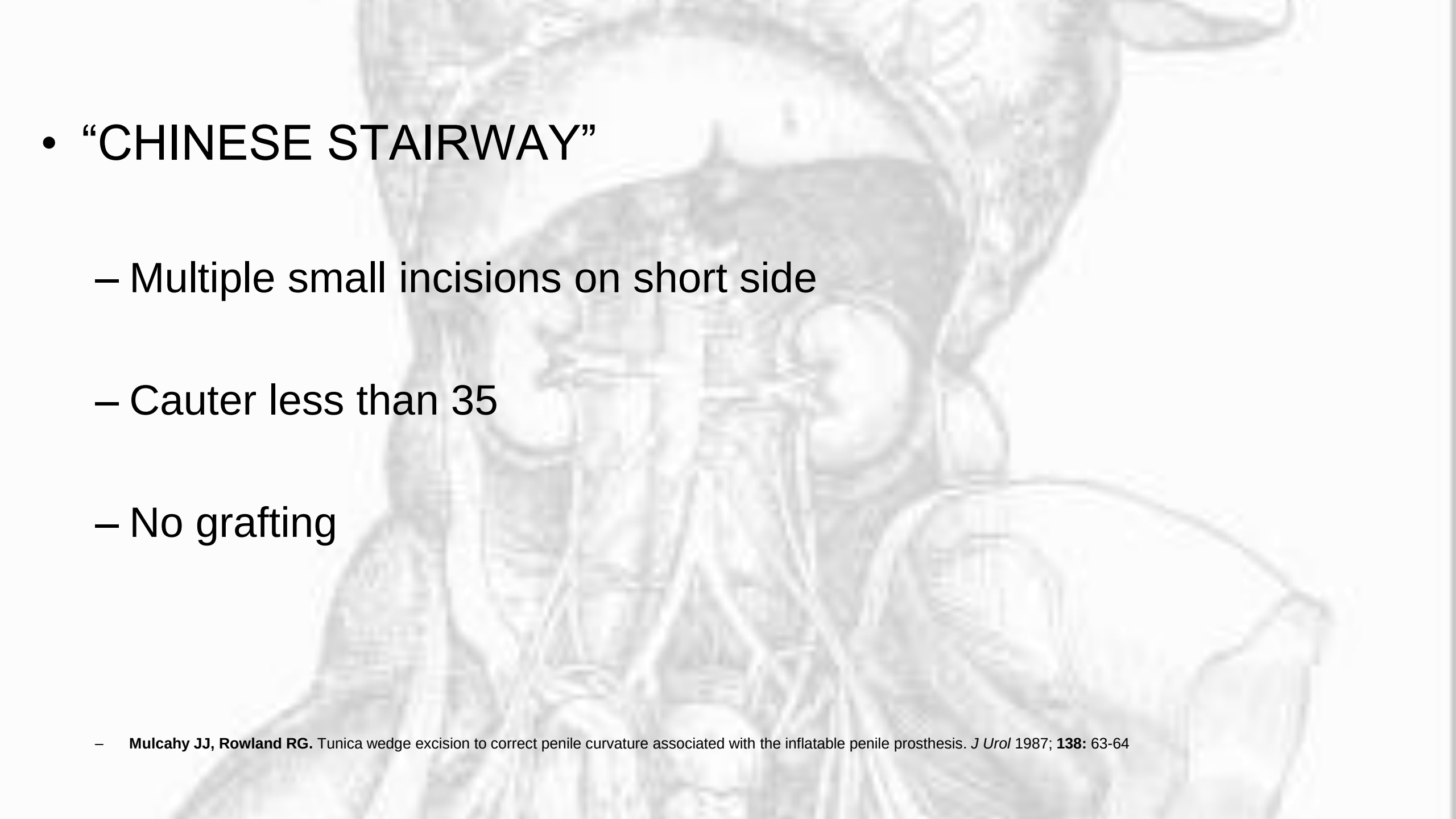


Surgical treatment

	Tunical shortening procedures		Tunical lengthening procedures
	Nesbit	Plication	Grafts
Penile shortening	4.7-30.8%	41-90%	0-40%
Penile straightening	79-100%	58-100%	74-100%
Persistent or recurrent curvature	4-26.9%	7.7-10.6%	0-16.7%
Post-operative erectile dysfunction	0-13%	0-22.9%	0-15%
Penile hypoesthesia	2-21%	0-21.4%	0-16.7%
Technical modifications	1	At least 3	Many types of grafts and techniques used



PENILE IMPLANT IN PEYRONIE'S DISEASE

- 
- “CHINESE STAIRWAY”
 - Multiple small incisions on short side
 - Cauter less than 35
 - No grafting

– **Mulcahy JJ, Rowland RG.** Tunica wedge excision to correct penile curvature associated with the inflatable penile prosthesis. *J Urol* 1987; **138**: 63-64

- **MODELING PROCEDURE:**

- Steve Wilson
- Fully inflated cylinders
- Break plaque
- Protection of pump
- Cave sutures corporotomies
- Bend penis 90 seconds
- Remove clamps and inflate 2-3 times extra
- Same procedure in some cases
- Pull strings cylinders for adequate placement
- Over time cylinders will act as tissue expanders (rapid activation)
- Cave urethral damage

IPP + remodelling



- The method of prosthesis implantation with post implantation manual modelling allows the surgeon to repeat manipulation to achieve maximal correction but runs the risk of possible urethral injury
- IPP implantation with grafting can be indicated for severe persistent curvature $>30^\circ$ with severe indentation or penile shortening
- Steve Wilson maneuver

- INTRA CORPORAL CORRECTION

- Normal dilation (Brooks)
- Rossello dilators
- Uramix cavernotomes
- Hooked knife “scratch technique”
- Otis urethrotome



LATE ONSET HYPOGONADISM

HYPOGONADISM

TESTICULAR = PRIMARY
HYPERGONADOTROPIC
HYPOGONADISM

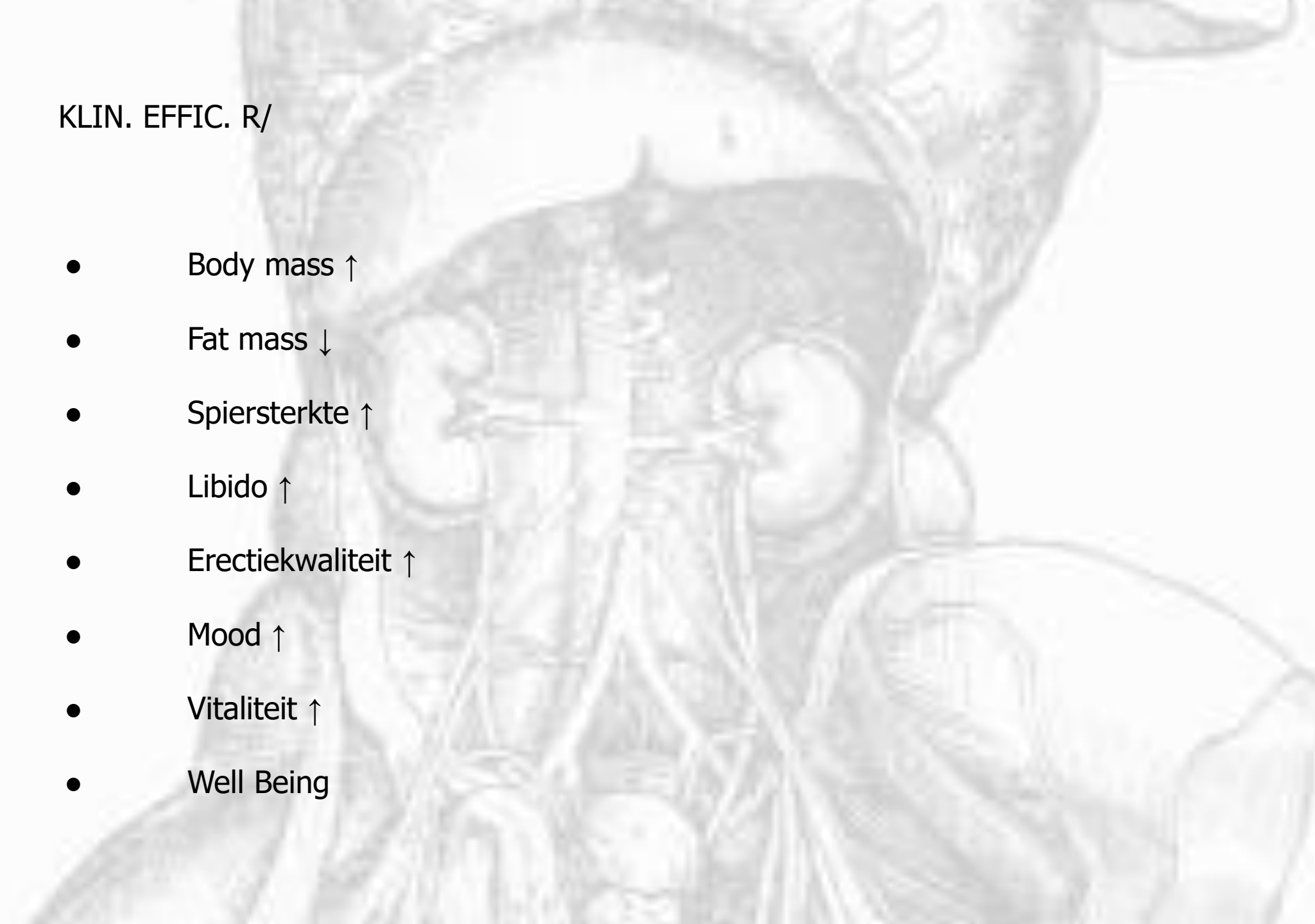
HYPOTH. –HYPOF.FAILURE =
SECOND. HYPOGONADOTROPIC
HYPOGONADISM

- 
- Andropauze
 - Male Climacterium
 - Penopauze
 - ADAM
 - PADAM
 - TDS
 - LOH

Late Onset Hypogonadisme

Een biochemisch syndroom dat geassocieerd is aan verouderen en gekarakteriseerd wordt door een deficiëntie van androgenen in het bloedserum met of zonder een afgenomen erfelijke gevoeligheid voor androgenen.

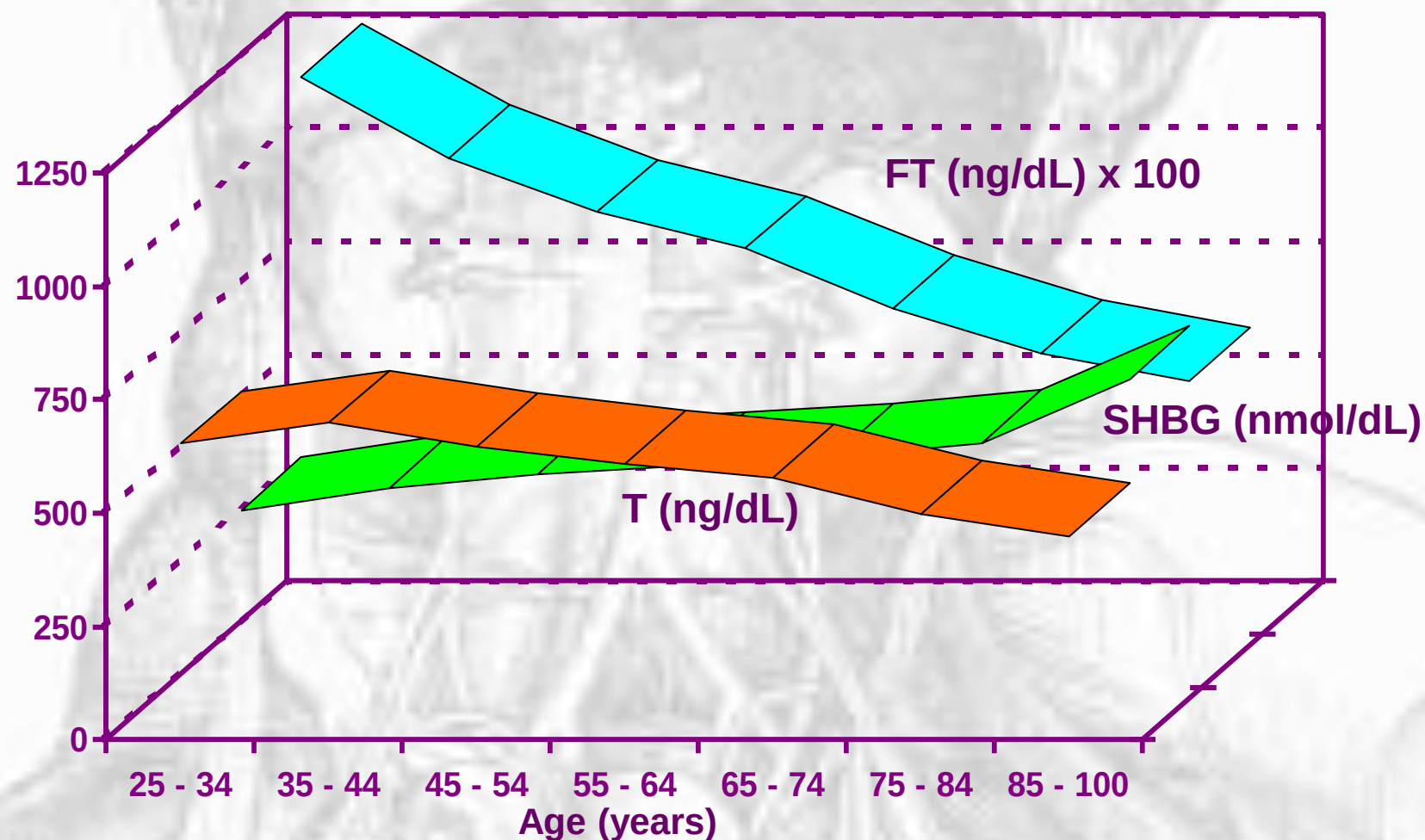
Dit kan resulteren in veranderingen in de levenskwaliteit en een aantal lichamelijke functies en organen negatief beïnvloeden.



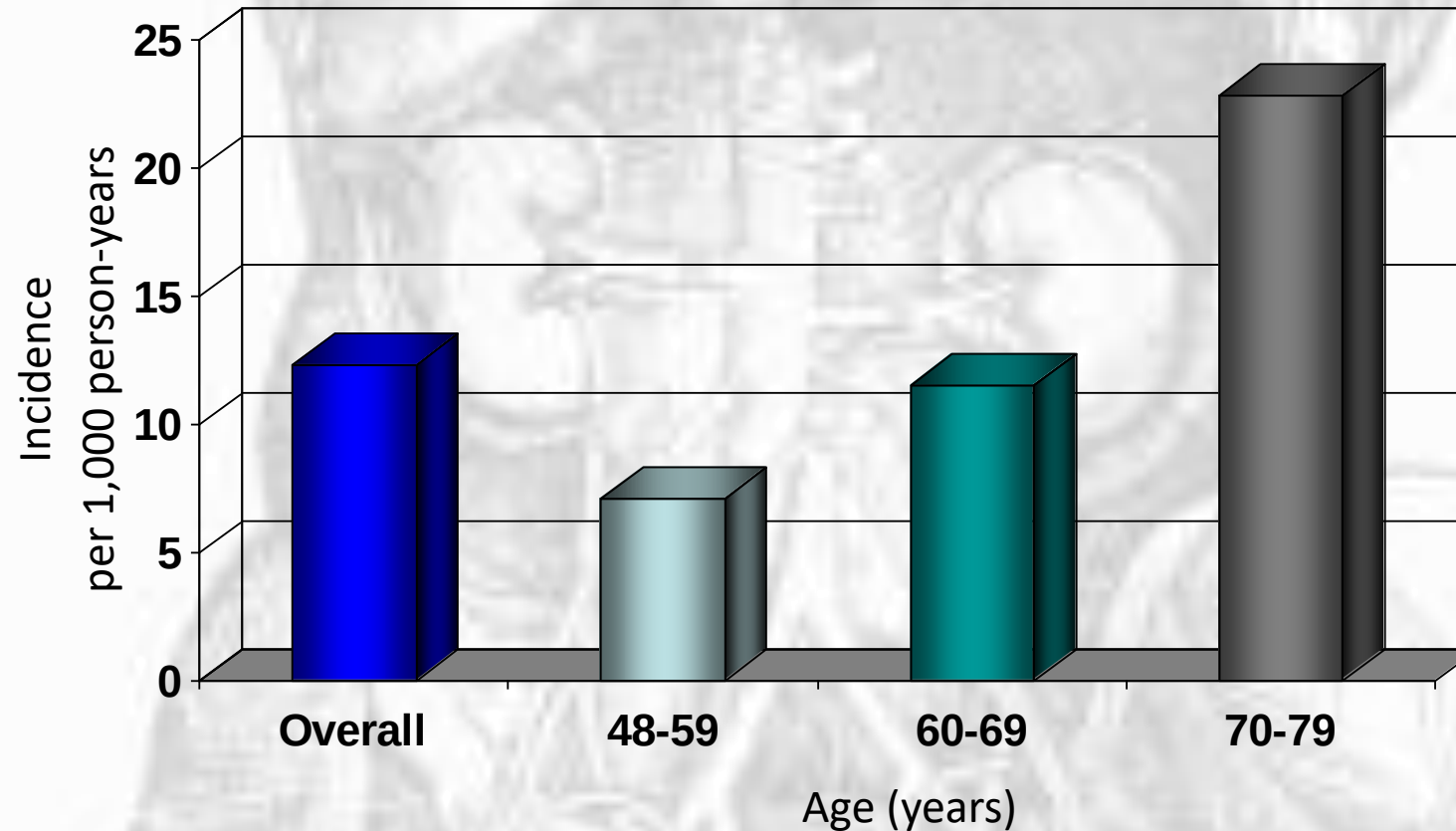
KLIN. EFFIC. R/

- Body mass ↑
- Fat mass ↓
- Spiersterkte ↑
- Libido ↑
- Erectiekwaliteit ↑
- Mood ↑
- Vitaliteit ↑
- Well Being

Leeftijdsgebonden veranderingen in Serum Concentraties van Totaal (T) en Vrij (FT) Testosterone, en Sex Hormoon Gebonden Globuline (SHBG)



Hypogonadism incidence and age (US data)



- Expected 481,000 new cases p.a. in US men 40-69 yrs

Hypogonadism and CV risk factors

- Low testosterone levels in men frequently co-exist with
 - Type 2 diabetes mellitus
 - Erectile dysfunction
 - Abdominal obesity
 - Other CV risk factors
- Component of the metabolic syndrome?

Behandeling

- I.M.
 - meest gebruikt
 - serumpiek- & dalwaarden
 - Sustanon (250 mg./3we) = mengpreparaat
 - Nebido
- P.O.
 - dosis makkelijk aan te passen
 - 3-4 x/dag na de maaltijd
 - Undestor
 - First pass
- Transdermaal
 - Recenter beschikbaar
 - Betere dosering & minder hepatotoxiciteit
 - Testimigel, Androderm, TestodermTTS, Andractim, Androgel

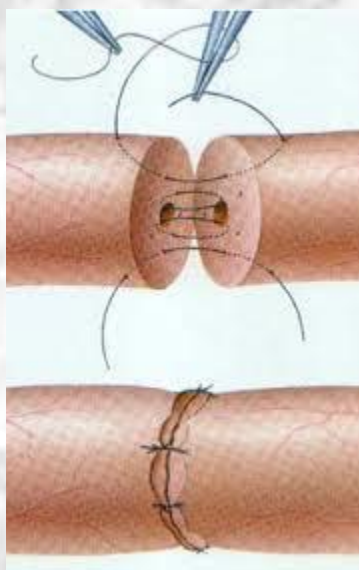
A detailed anatomical illustration of the female reproductive system, showing the uterus, fallopian tubes, ovaries, and associated blood vessels. The illustration is rendered in a light, sketchy style, with the central organs in a darker shade of gray. The text "FERTILITEITSPROBLEMEN" is overlaid in the center of the image.

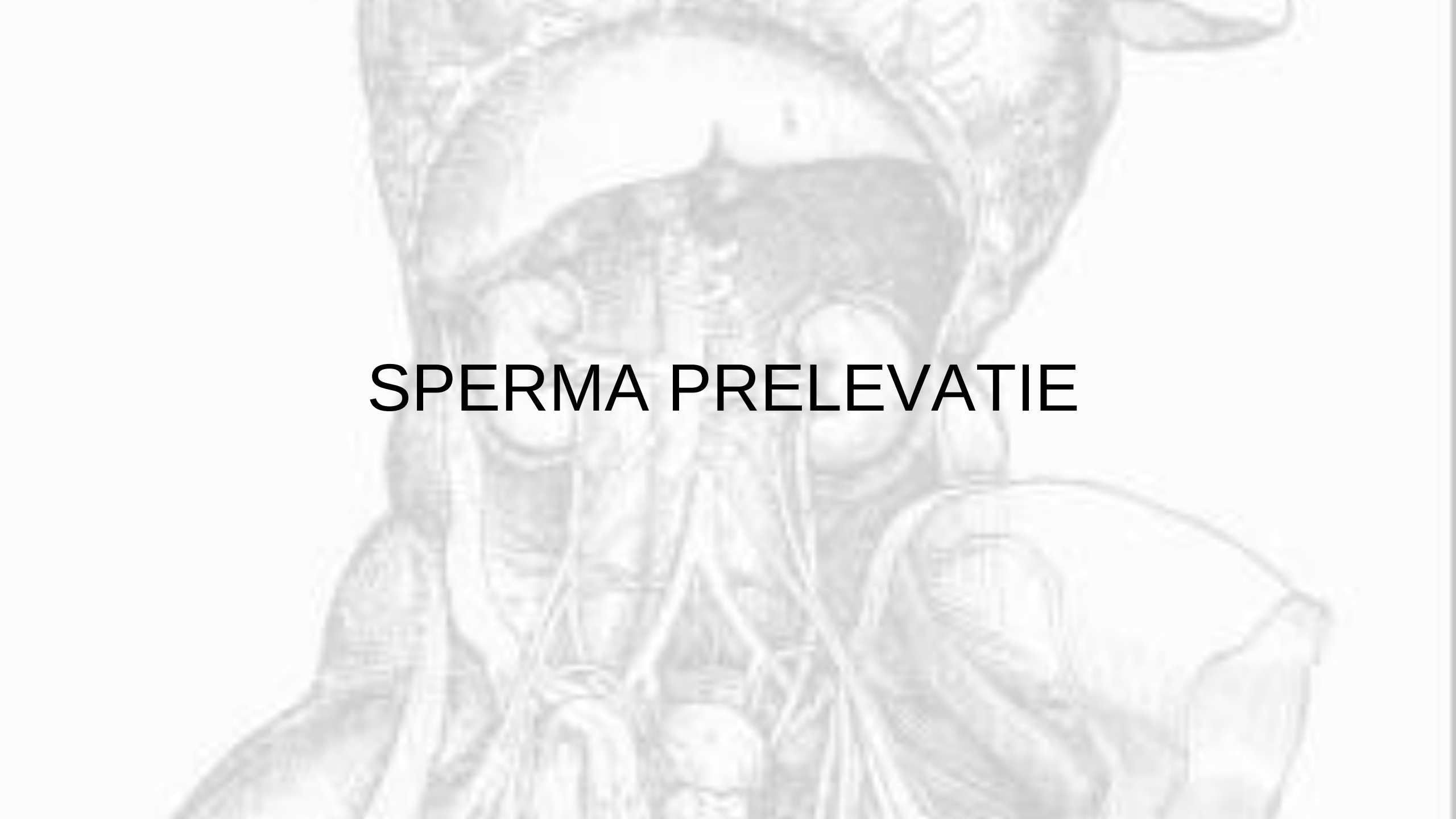
FERTILITEITSPROBLEMEN



VASO VASO STOMIE

- 
- Hersteloperatie na vroegere vasectomie
 - Spijtoptanten
 - Steeds meer: nieuwe relaties,...
 - Succesratio technisch vs spermioogram vs zwangerschap
 - Cut off 10 jaar
 - Vaso-epididymostomie
 - Evt. met TESE
 - Evt. geassisteerde procreatie (biobank, labo!)
 - Microscoop vs loupebril (vs robot)
 - Prolene 7/0; extra- en intralumineel
 - Stent vs geen stent
 - Nederlandse patiënten



A detailed anatomical illustration of the male reproductive system, showing the testes, vas deferens, ureters, and associated structures. The illustration is in a classic medical style, with clear labels and shading to indicate depth and structure. The text "SPERMA PRELEVATIE" is overlaid on the central part of the image.

SPERMA PRELEVATIE

- Tese
- Tesa
- Mese
- Mesa
- Geassisteerde procreatie



PUBLIC RELATIONS

- Coach up nurses
- Information sessions for patients
- Teaching sessions for GP's
- Live surgery
- www.ageingmaleclinic.be
- www.kliniekvoorurologie.be

And the
fascination
never ends!!

